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Active Directory Infrastructure & Administrative Security Controls

Executive Summary

Network Security – Module 2:
Administrative Network Security Implementation Lab

This lab demonstrates the implementation of core administrative network security controls within a virtualized enterprise environment.
Using VMware as the hypervisor, a three-machine domain infrastructure was built consisting of a Windows Server 2019 Domain Controller (DC01), an administrative workstation (ADMIN01), and a client machine (CL01), all operating within an isolated host-only network under the domain **crack.local**.

The lab was structured across five implementation phases

- **Phase 1,** Active Directory was configured with a proper organizational structure including departmental OUs (IT, HR, Finance), role-based user accounts, and appropriate administrative privileges enforcing the principle of least privilege.
- **Phase 2** established a Security Baseline through Group Policy, enforcing domain-wide password complexity requirements, account lockout thresholds after five failed attempts, and audit logging for login events — all verified through Windows Event Viewer logs (Event IDs 4624 and 4625).
- **Phase 3** deployed Lansweeper for network asset discovery, successfully identifying all domain-joined devices across the 192.168.10.0/24 subnet.
- **Phase 4** implemented Privileged Access Management using ManageEngine Password Manager Pro, centralizing the storage of administrative credentials in an encrypted vault to eliminate insecure password practices.
- **Phase 5** utilized SpyAgent on the client machine to demonstrate user activity monitoring, capturing keystrokes, application usage, and web browsing activity to support insider threat detection.

Collectively, these implementations reflect a layered administrative security approach — combining identity management, policy enforcement, asset visibility, credential protection, and behavioral monitoring — to strengthen an organization's overall security posture.

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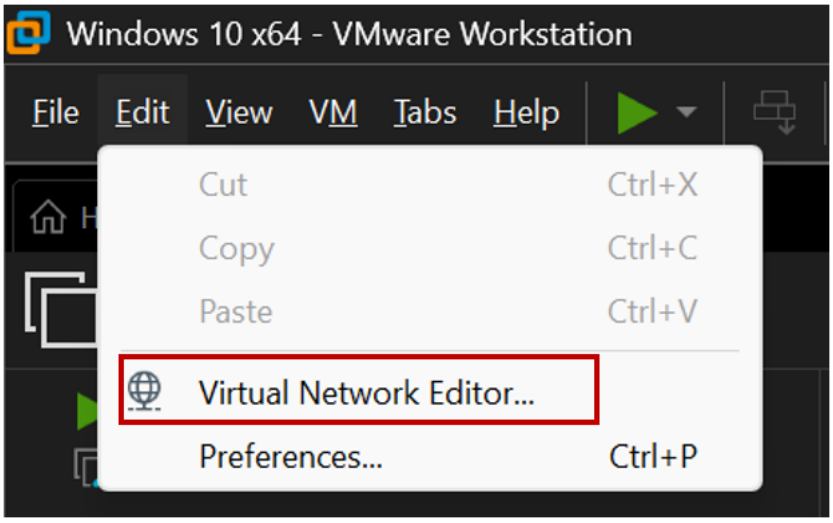
1. VMware Virtual Network Configuration

Before starting the virtual machines, you must configure the virtual network adapters in VMware Workstation. This creates an isolated internal network for the domain and a NAT network for internet access.

1.1 Opening the Virtual Network Editor

To open the Virtual Network Editor:

1. Open VMware Workstation
2. Click the Edit menu in the top menu bar
3. Select "Virtual Network Editor…"

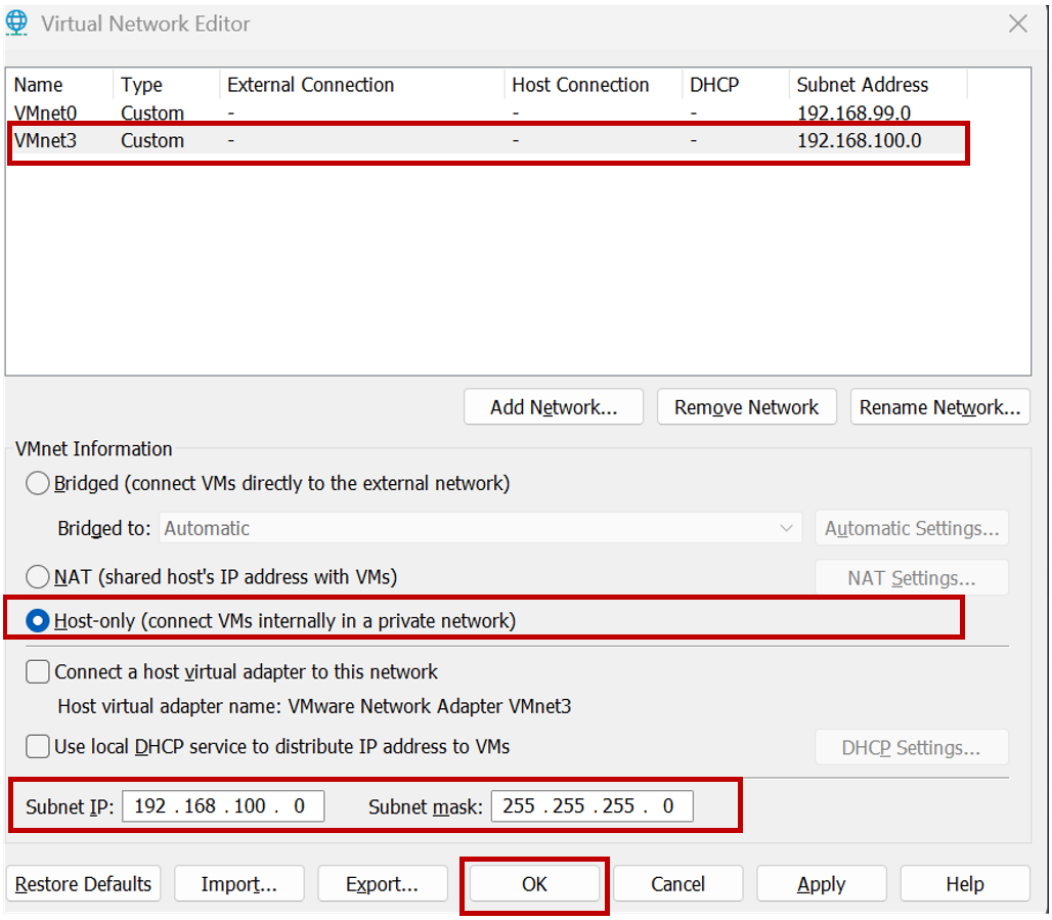


1.2 Creating VMnet3 – Host-Only Internal Network

VMnet3 is the internal network used exclusively for domain communication between the Server, ADM01, and CLI01. DHCP is disabled because we assign static IPs manually.

Configuration Settings:

- Network Name: VMnet3
- Type: Host-only (connect VMs internally in a private network)
- Subnet IP: 192.168.100.0
- Subnet Mask: 255.255.255.0
- DHCP: Disabled (uncheck "Use local DHCP service")
- External Connection: None

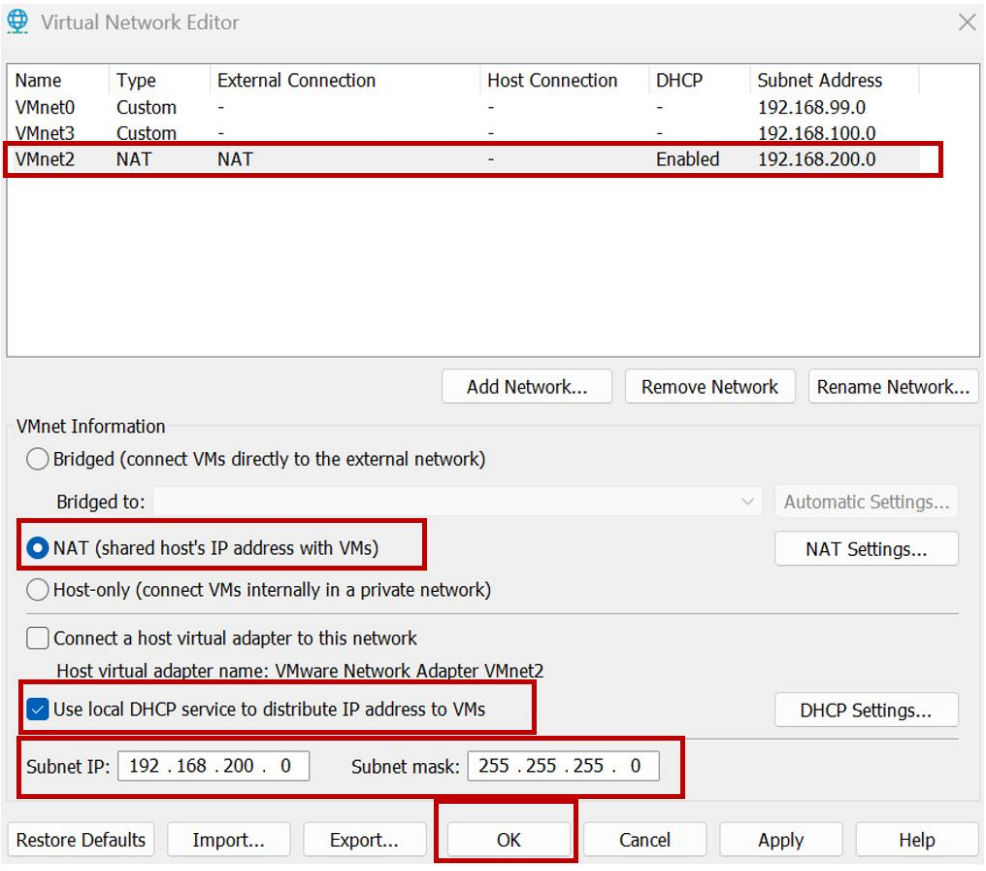


1.3 Creating VMnet2 – NAT Network (Internet)

VMnet2 provides internet connectivity to the VMs by sharing the host machine's IP address via NAT. DHCP is enabled on this network.

Configuration Settings:

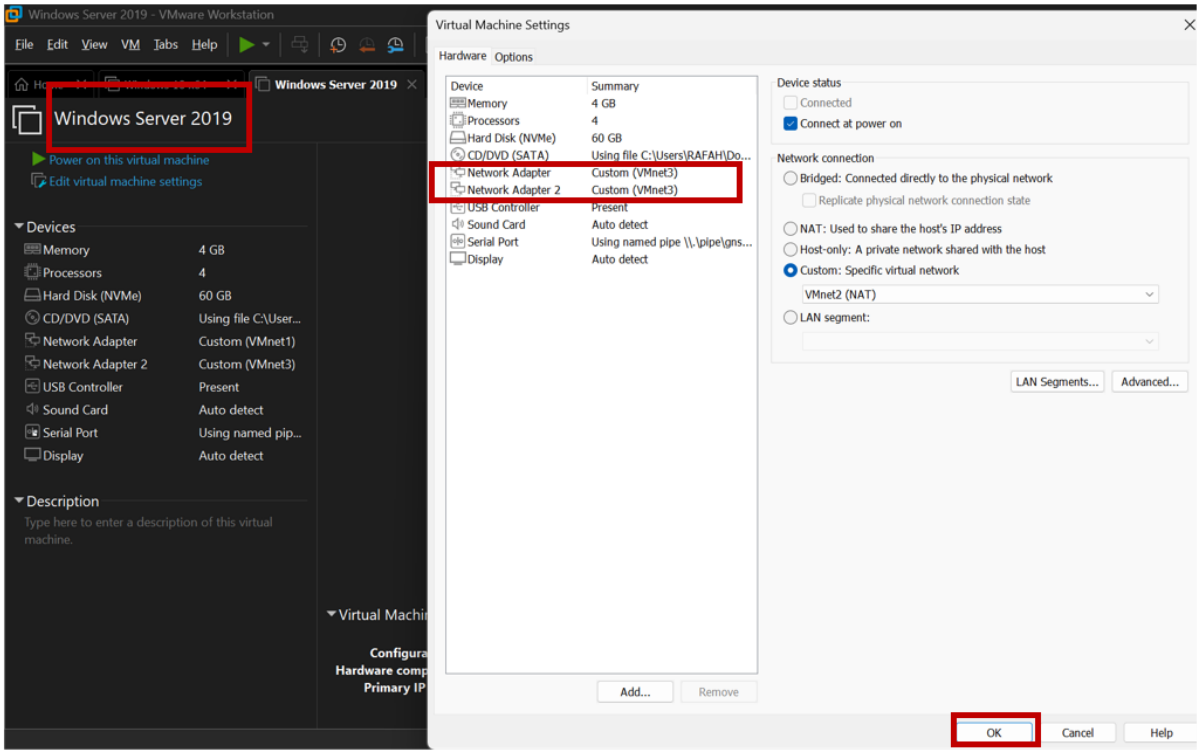
- Network Name: VMnet2
- Type: NAT (shared host's IP address with VMs)
- Subnet IP: 192.168.200.0
- Subnet Mask: 255.255.255.0
- DHCP: Enabled
- External Connection: NAT



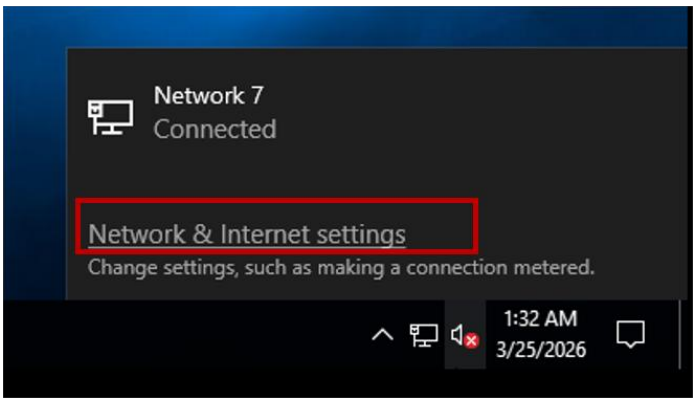
1.4 Assigning Network Adapters to Windows Server 2019 VM

The Windows Server 2019 virtual machine needs two network adapters — one connected to VMnet3 (internal domain) and one to VMnet2 (internet/NAT). Right-click the VM and select "Edit virtual machine settings" then configure both adapters:

- Network Adapter 1: Custom – VMnet1 (or VMnet3) — Internal domain network
- Network Adapter 2: Custom – VMnet3 — mapped accordingly



Once network adapters are configured, power on the Windows Server 2019 VM. Right-click the network icon in the system tray and open "Network & Internet settings".

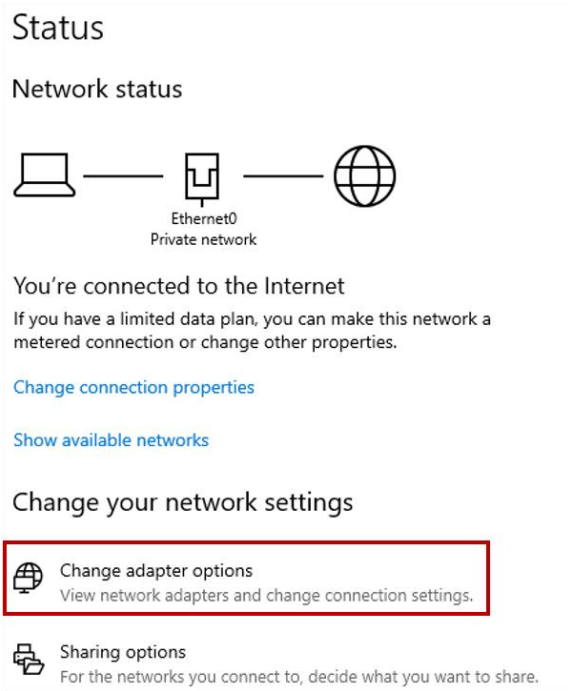


2. Windows Server 2019 – Network Setup

After booting into Windows Server 2019, configure the network adapters with static IP addresses so the server can function as a domain controller.

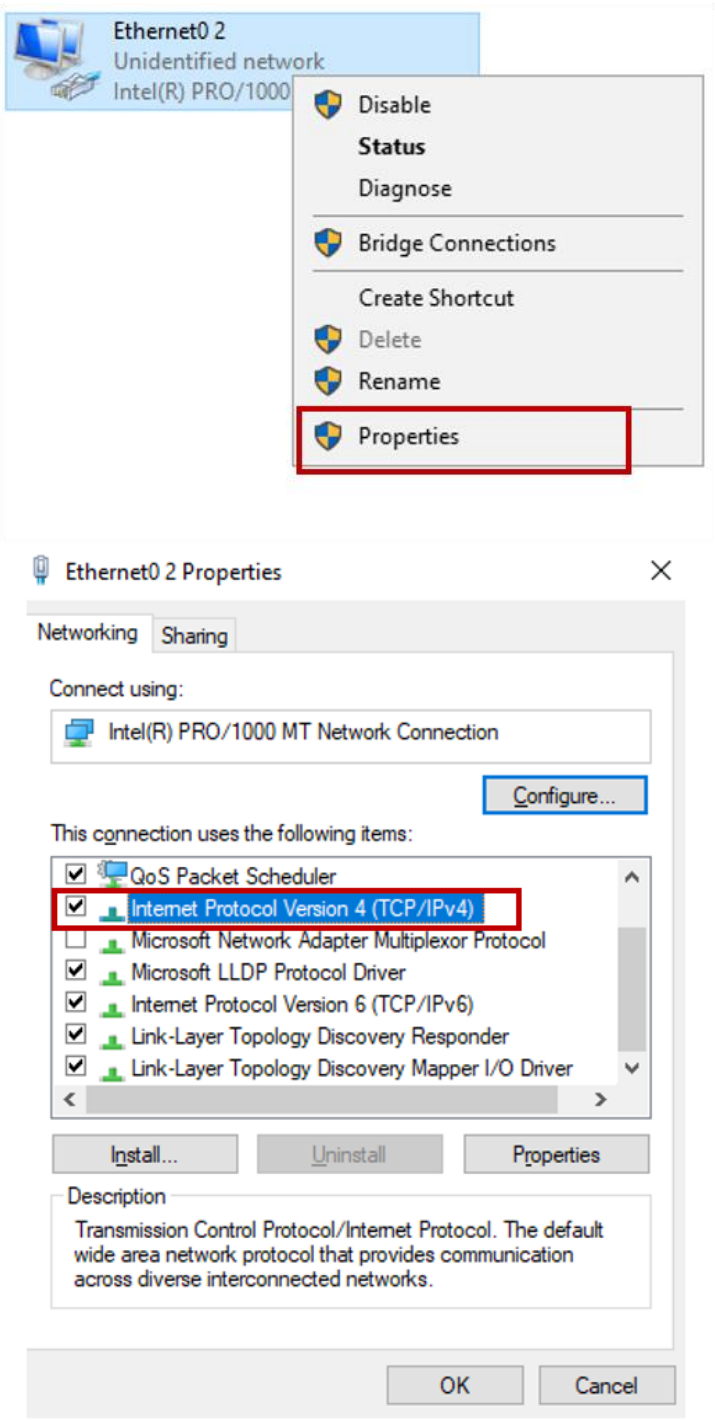
2.1 Accessing Change Adapter Options

1. Click the network icon in the system tray
2. Select "Network & Internet settings"
3. Click "Change adapter options"



2.2 Configuring Ethernet0 2 – Static IP

Right-click the Ethernet adapter connected to VMnet3 and select Properties. Then select "Internet Protocol Version 4 (TCP/IPv4)" and click Properties.

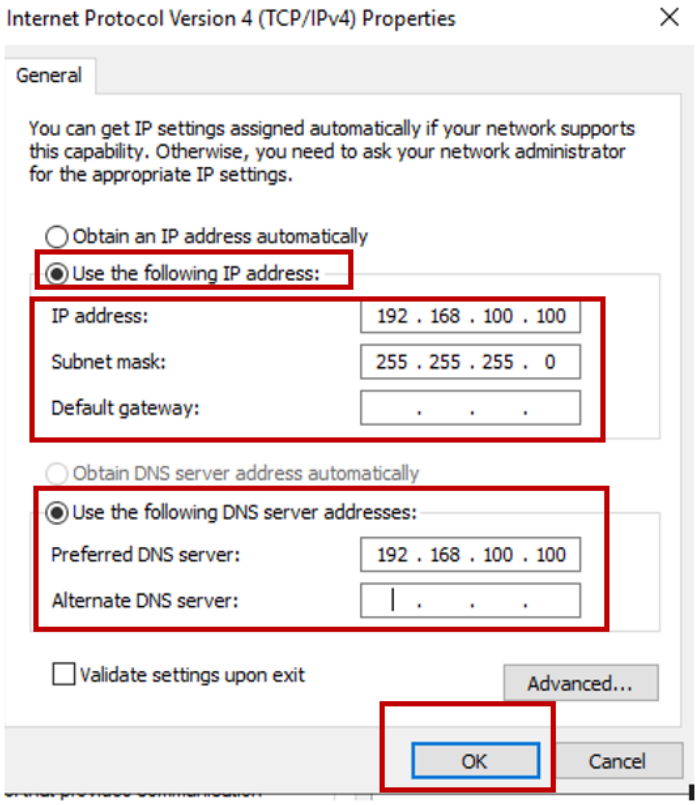


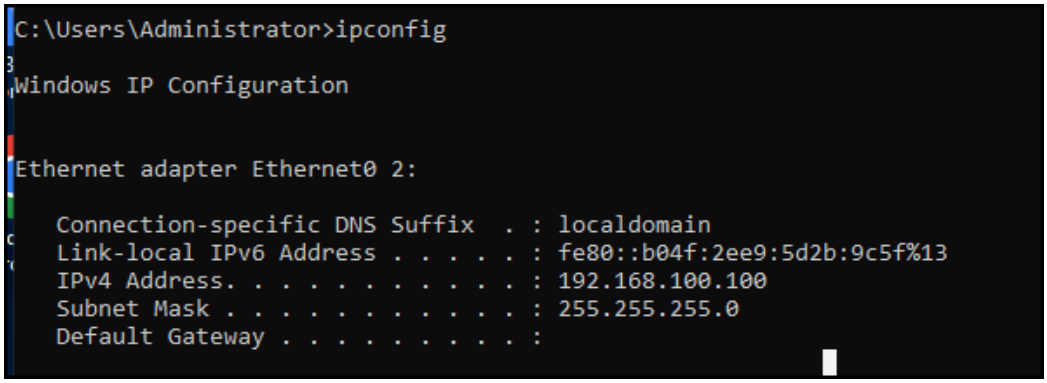
IPv4 Properties – Internal Adapter (Ethernet0 2):

- IP Address: 192.168.100.100
- Subnet Mask: 255.255.255.0
- Default Gateway: (leave blank)
- Preferred DNS Server: 192.168.100.100 (points to itself – configured after AD DS install)

4. Click OK to apply

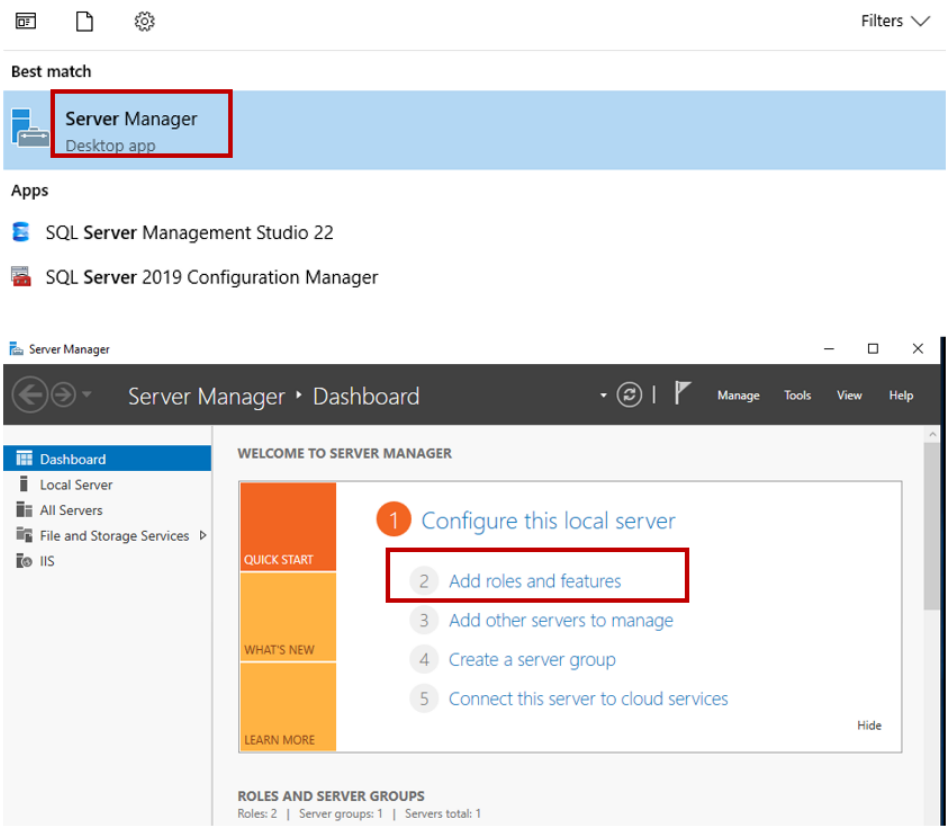
After clicking OK, open Command Prompt and run ipconfig to verify the IP address is correctly assigned.





2.3 Opening Server Manager

Search for "Server Manager" and open it. From the Dashboard, you will use "Add roles and features" to install Active Directory Domain Services.

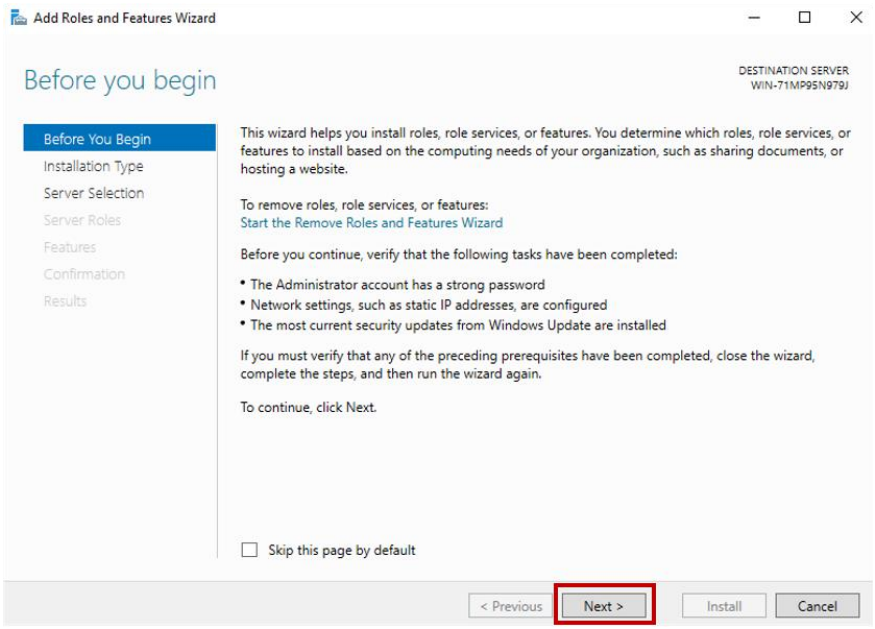


3. Active Directory Domain Services Installation

Active Directory Domain Services (AD DS) is installed using the Add Roles and Features Wizard in Server Manager.

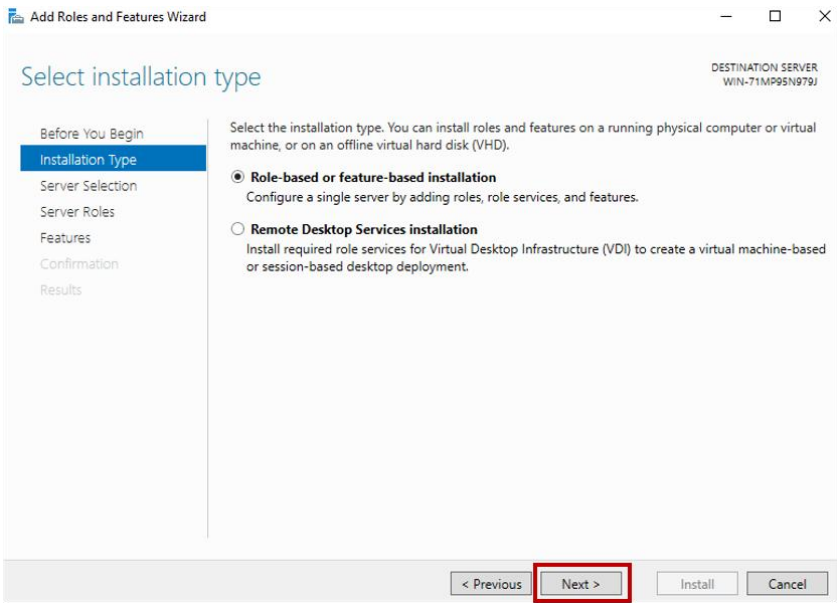
3.1 Before You Begin

Click "Add roles and features" from the Server Manager Dashboard. On the "Before You Begin" page, verify prerequisites and click Next.



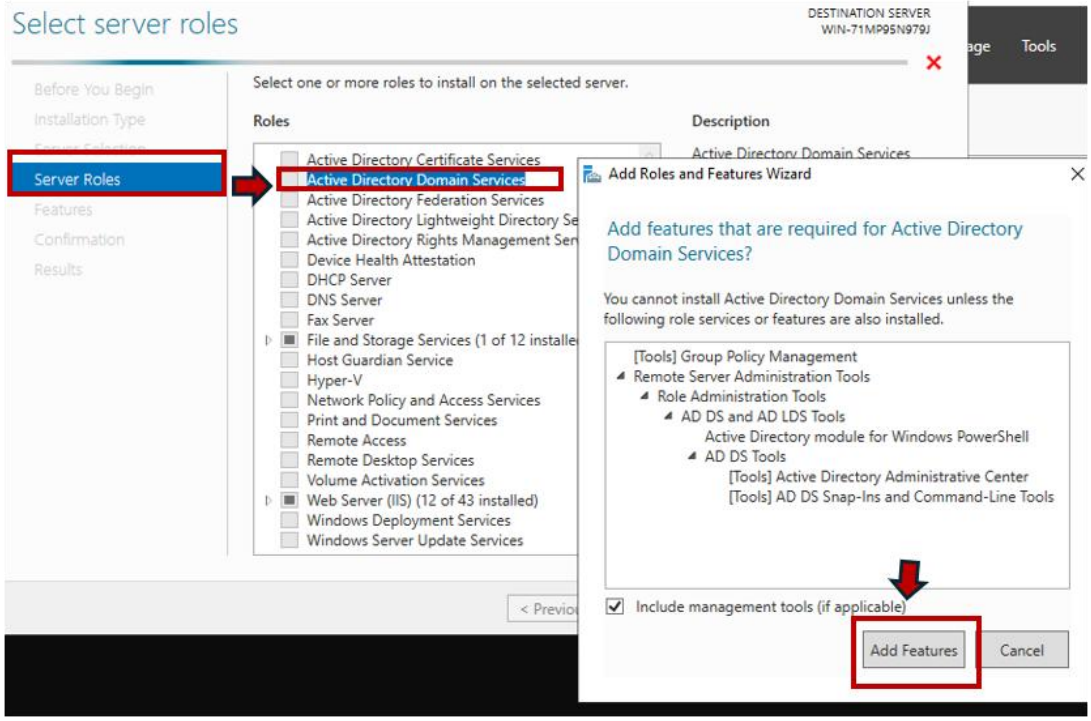
3.2 Select Installation Type

Select "Role-based or feature-based installation" to install AD DS on the local server. Click Next.

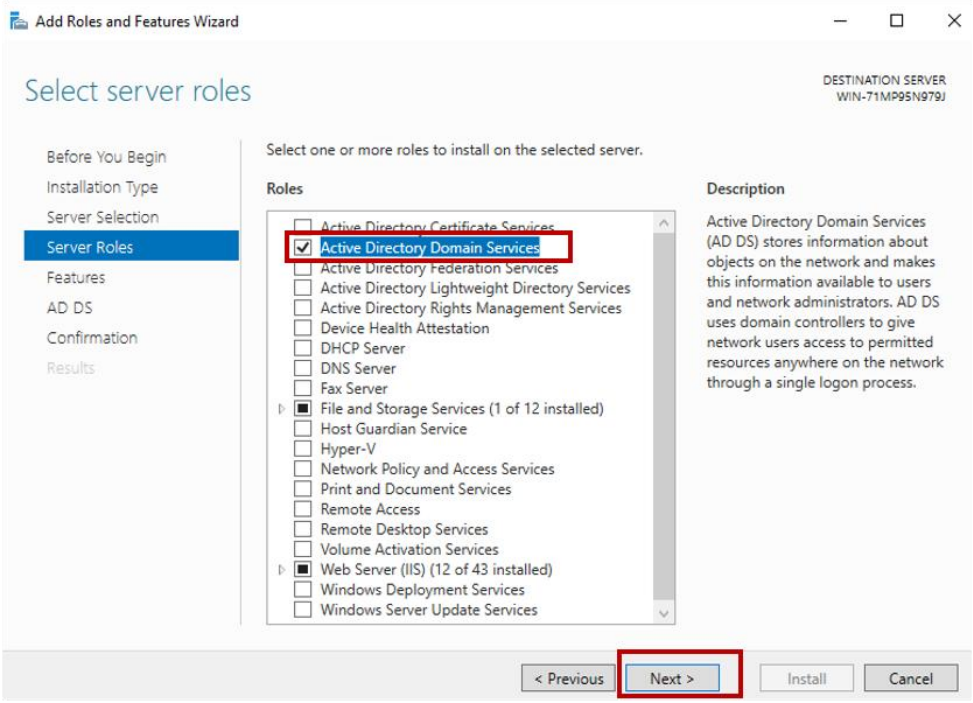


3.3 Select Server Roles – Active Directory Domain Services

On the Server Roles page, check "Active Directory Domain Services". A dialog will appear asking to add required features — click "Add Features".

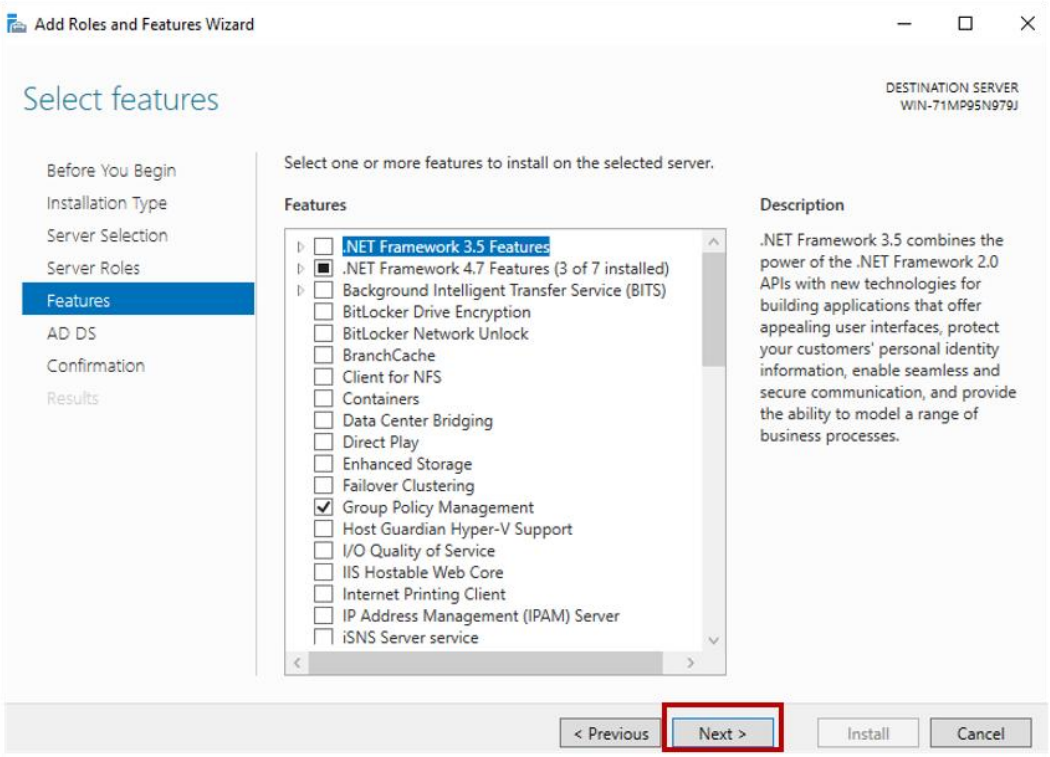


After adding features, you will see Active Directory Domain Services is checked. Click Next.



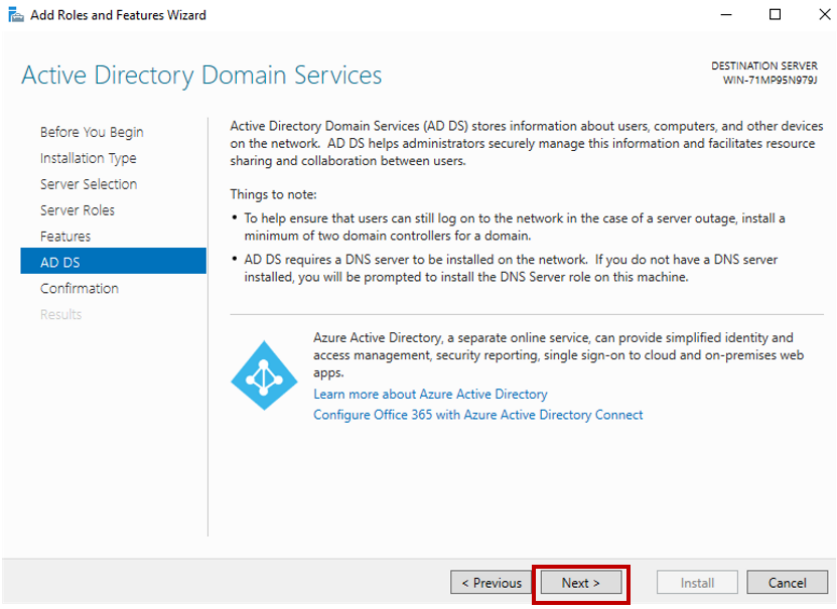
3.4 Select Features

The Features page shows Group Policy Management is pre-checked. No additional features need to be selected. Click Next.



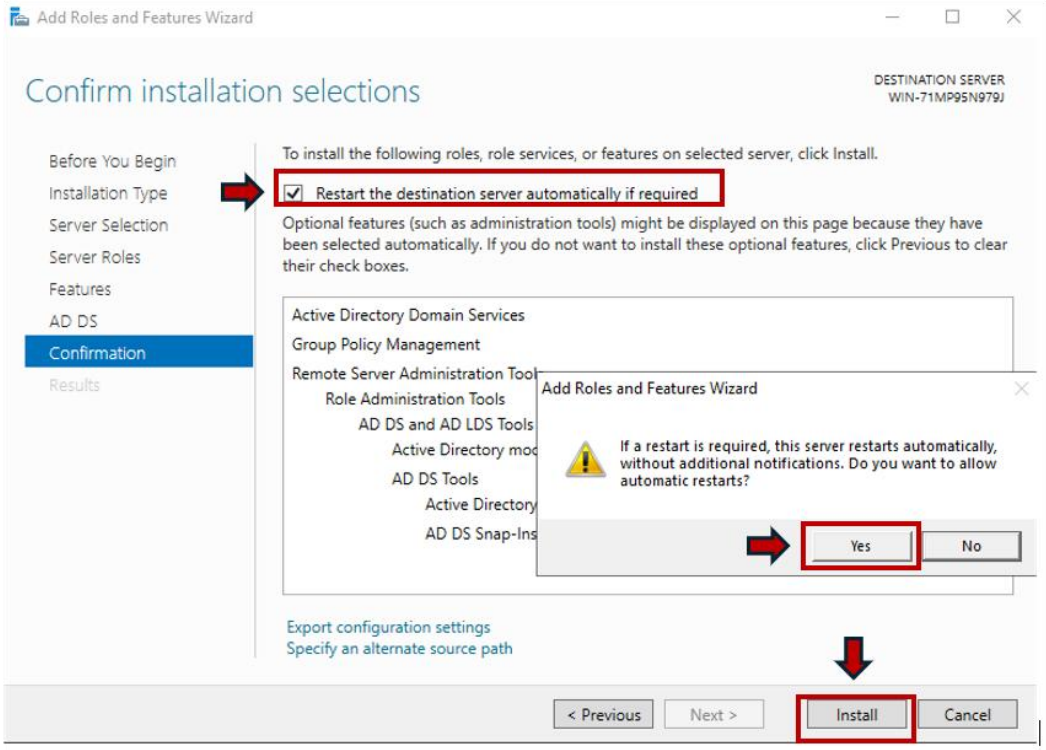
3.5 AD DS Information Page

Read the information about AD DS requirements. Note that AD DS requires a DNS server. Click Next.



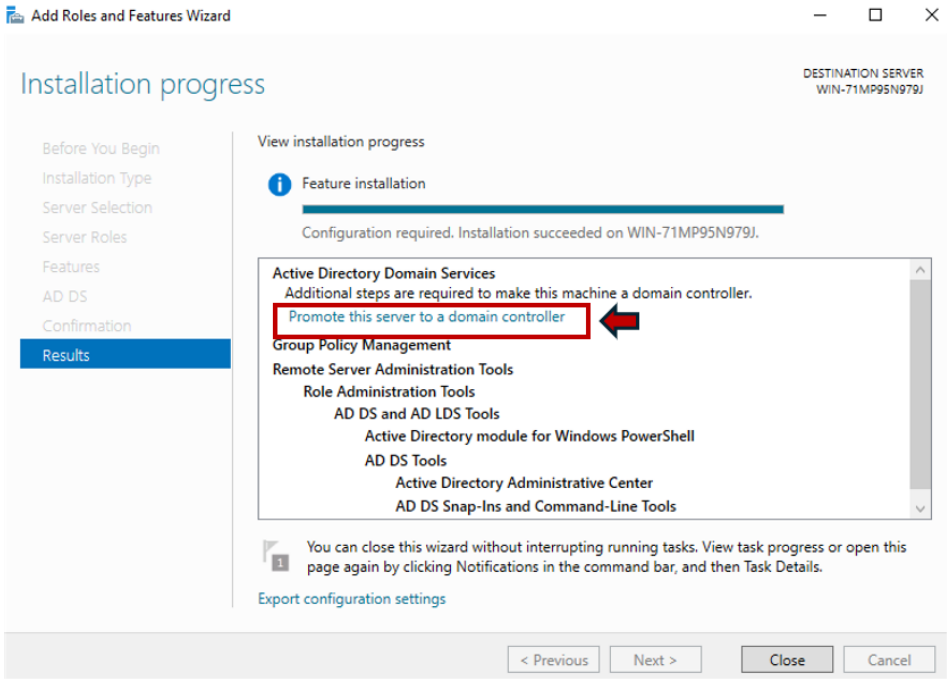
3.6 Confirm Installation – Restart & Install

On the Confirmation page, check "Restart the destination server automatically if required". When prompted, click Yes. Then click Install.



3.7 Installation Progress & Promote to Domain Controller

Once installation completes, you will see "Configuration required. Installation succeeded." Click "Promote this server to a domain controller" to launch the configuration wizard.



4. Domain Controller Promotion

After installing the AD DS role, the server must be promoted to a Domain Controller by creating a new forest.

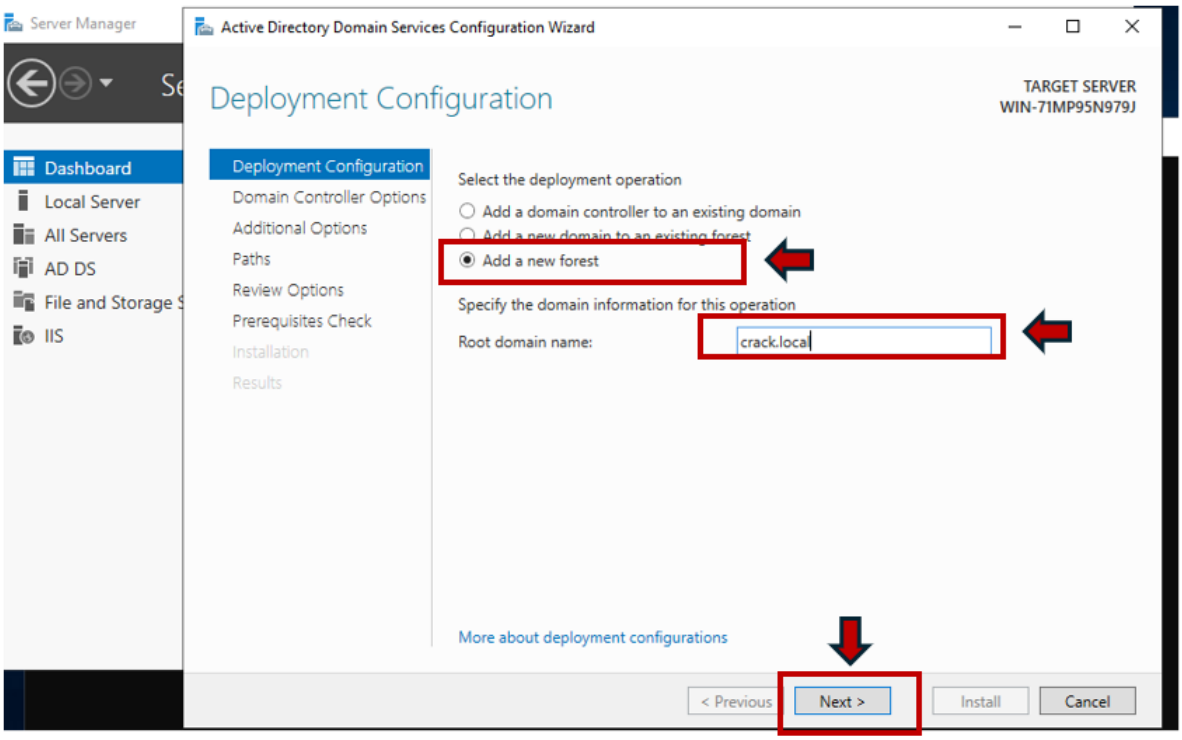
4.1 Deployment Configuration – New Forest

The AD DS Configuration Wizard opens. Select "Add a new forest" and enter the root domain name.

Settings:

- Select: Add a new forest
- Root domain name: crack.local

1. Click Next

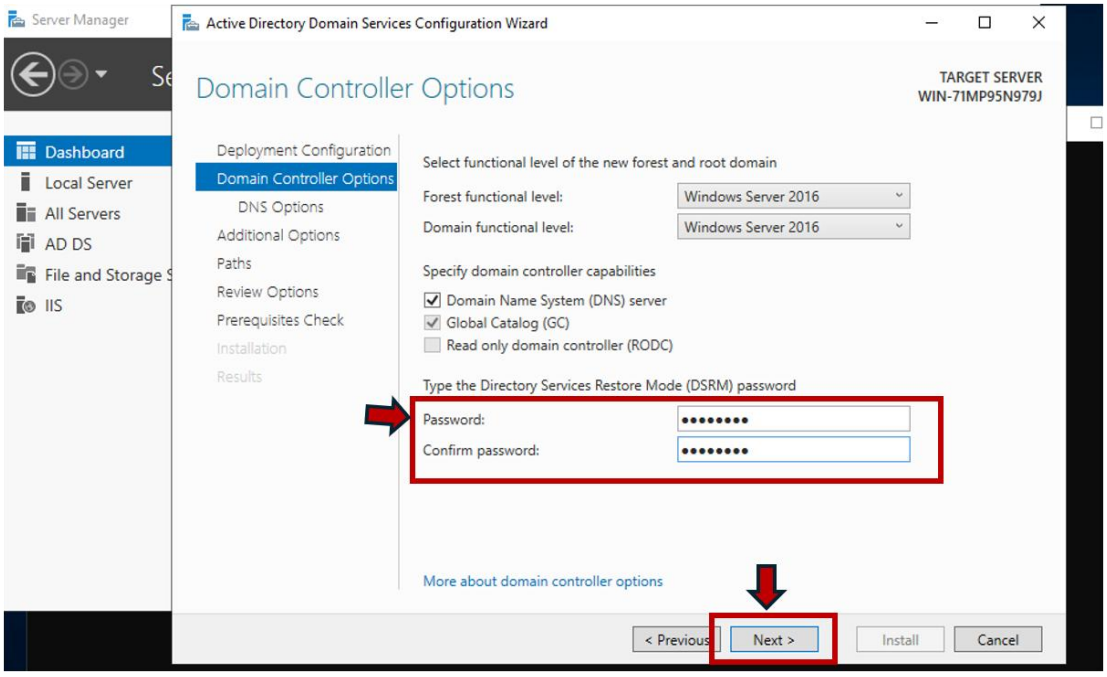


4.2 Domain Controller Options – DSRM Password

Set the functional levels and enter the DSRM (Directory Services Restore Mode) password. This password is used to recover AD if needed.

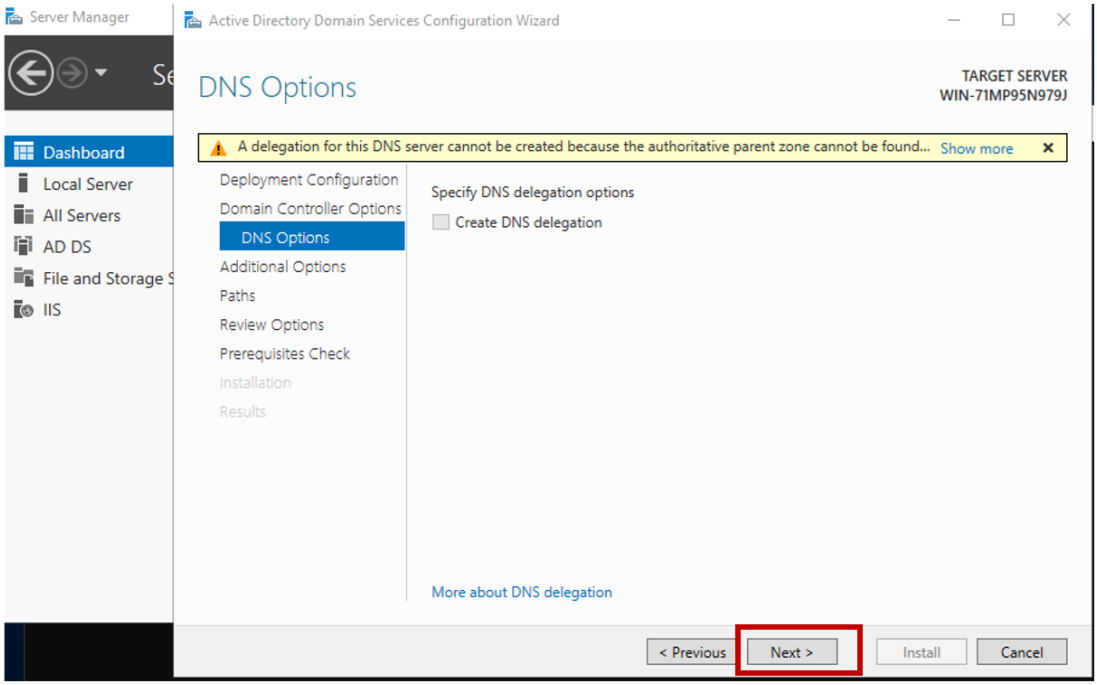
Settings:

- Forest functional level: Windows Server 2016
- Domain functional level: Windows Server 2016
- Domain Name System (DNS) server: ✓ Checked
- Global Catalog (GC): ✓ Checked
- DSRM Password: Crack123
- Confirm password: Crack123



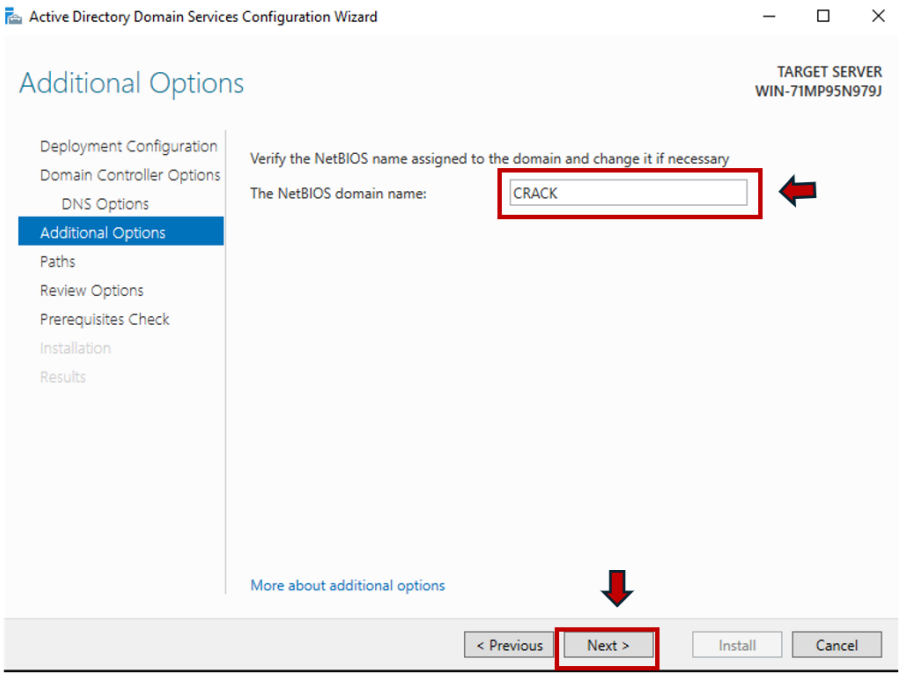
4.3 DNS Options

A warning appears that a DNS delegation cannot be created. This is normal for a new forest — there is no parent DNS zone. Leave "Create DNS delegation" unchecked and click Next.



4.4 Additional Options – NetBIOS Name

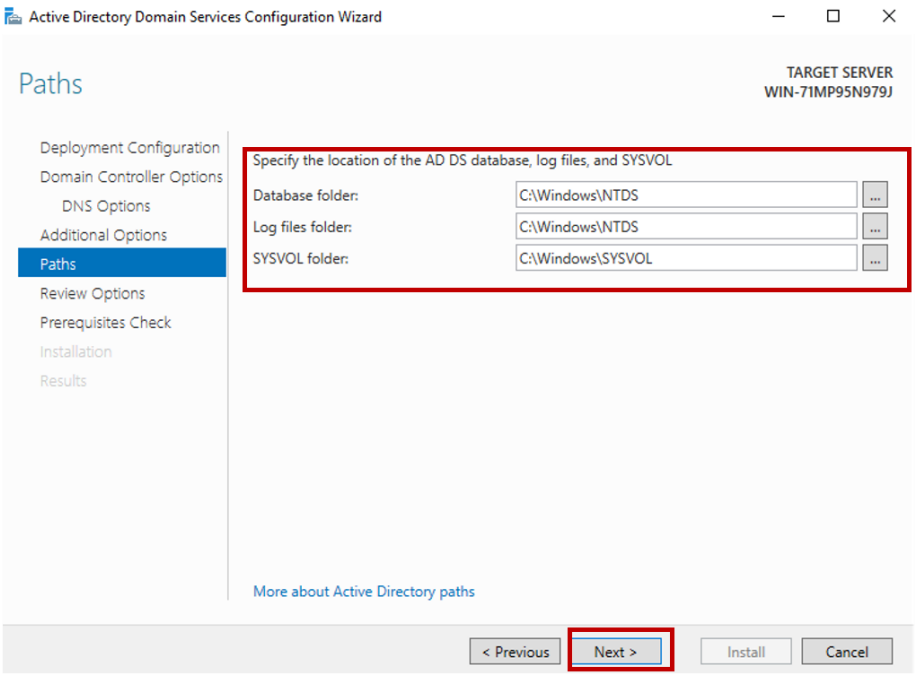
The NetBIOS domain name is automatically generated from the root domain name. Verify it shows "CRACK" and click Next.



4.5 Paths – Database, Logs, SYSVOL

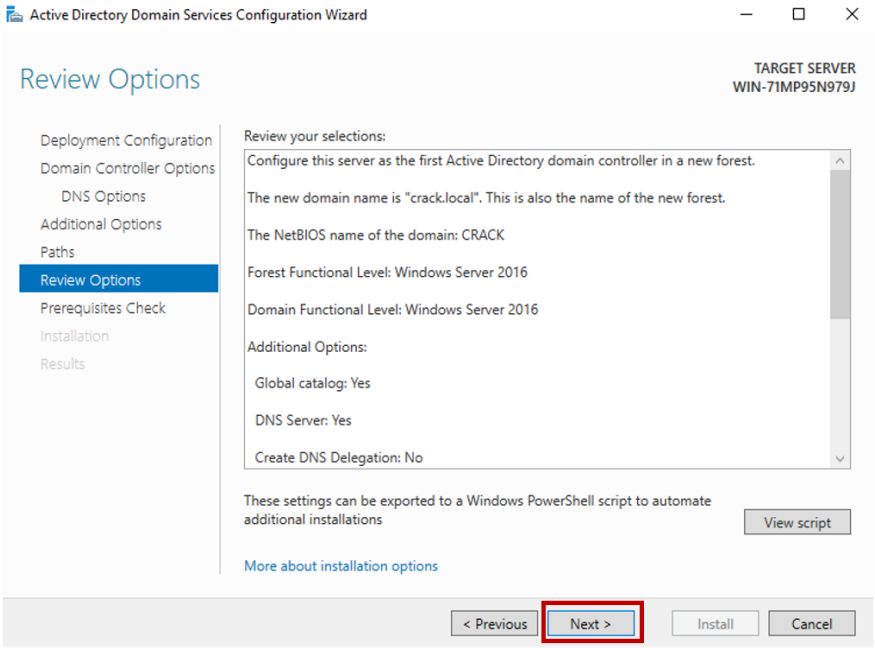
Keep the default paths for the AD DS database, log files, and SYSVOL folder and click Next.

- Database folder: C:\Windows\NTDS
- Log files folder: C:\Windows\NTDS
- SYSVOL folder: C:\Windows\SYSVOL



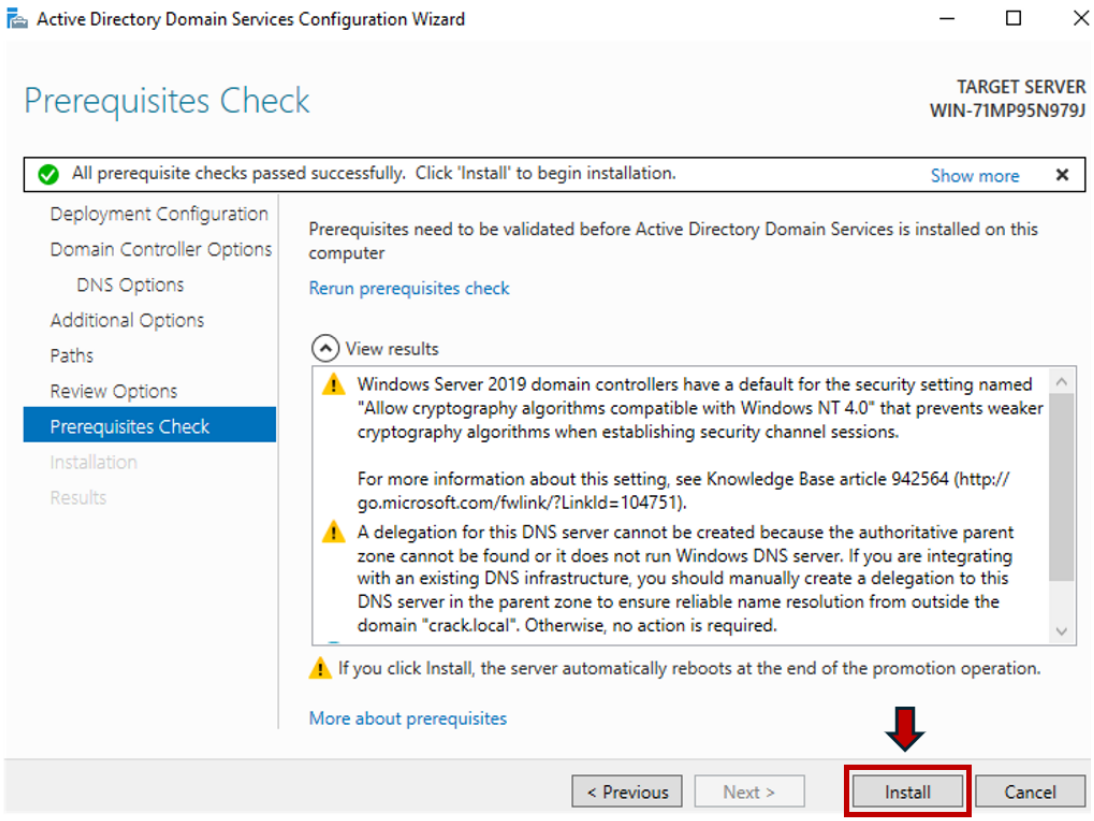
4.6 Review Options

Review the configuration summary. Confirm domain name is "crack.local", NetBIOS name is "CRACK", Forest/Domain functional level is Windows Server 2016. Click Next.



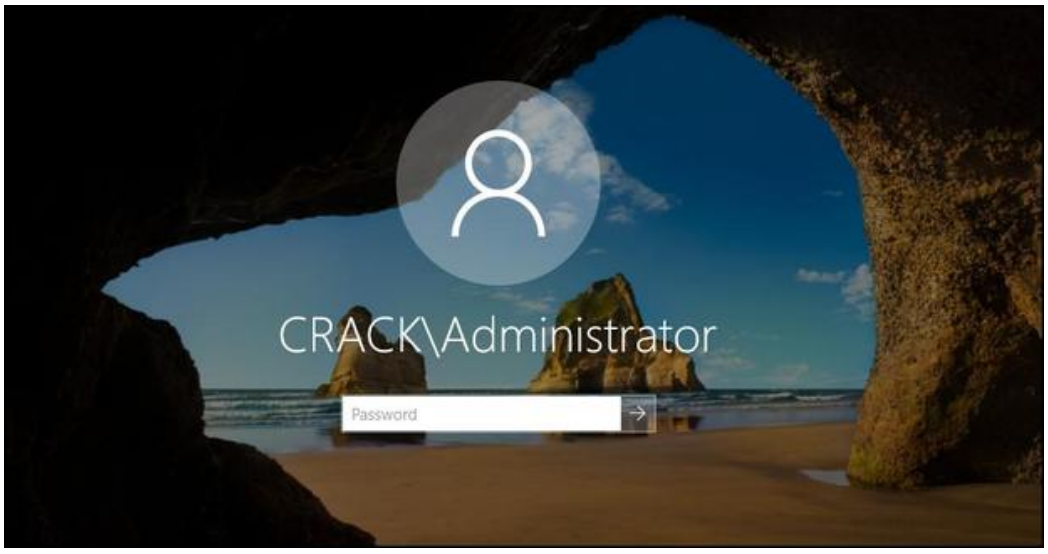
4.7 Prerequisites Check & Install

All prerequisite checks pass (warnings are acceptable). Click Install to begin domain controller promotion. The server will restart automatically when complete.



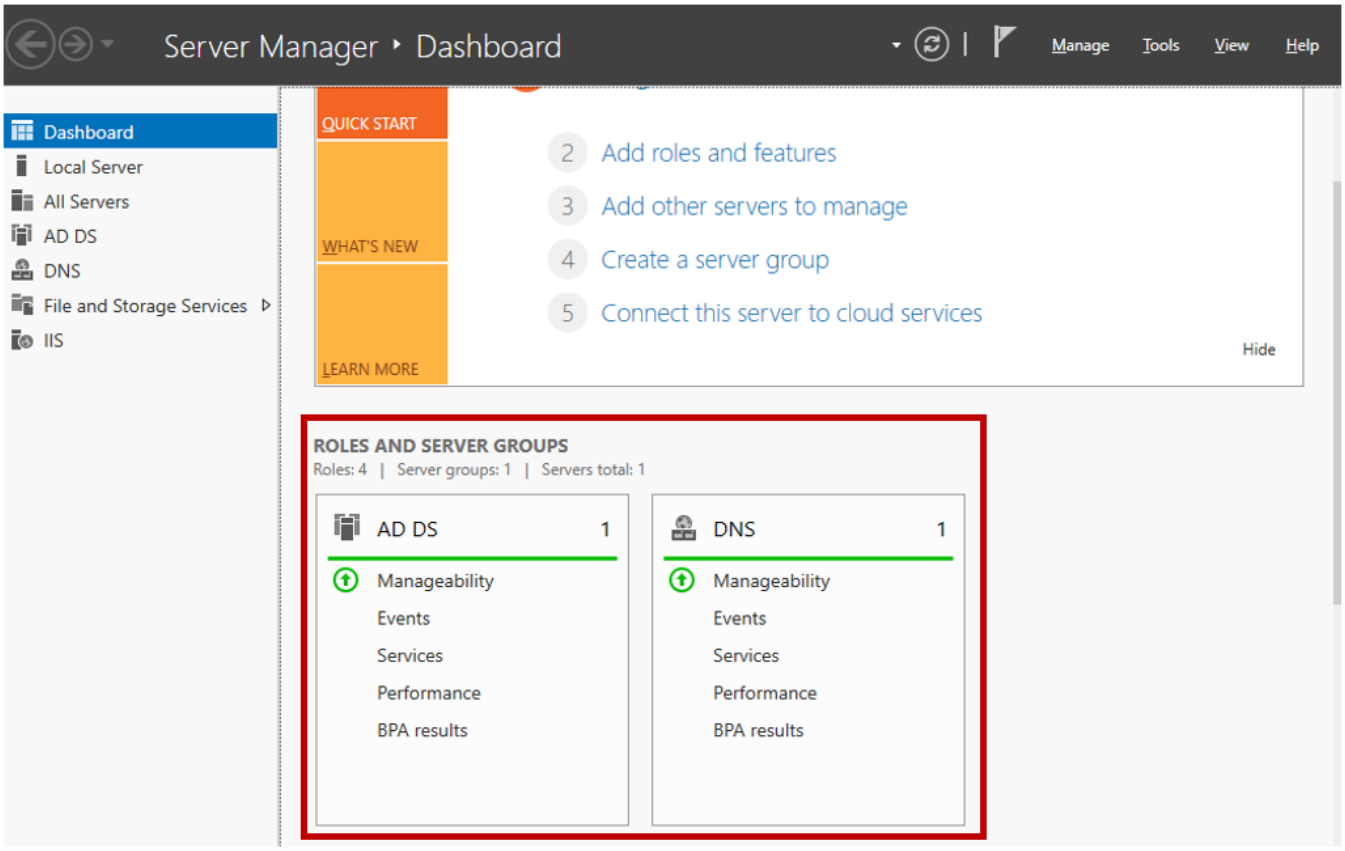
4.8 Post-Reboot Login

After the server restarts, log in with CRACK\Administrator and the password Admin123.



4.9 Verifying AD DS and DNS in Server Manager

After logging in, open Server Manager. The Dashboard now shows AD DS and DNS roles listed under Roles and Server Groups, confirming successful domain controller promotion.

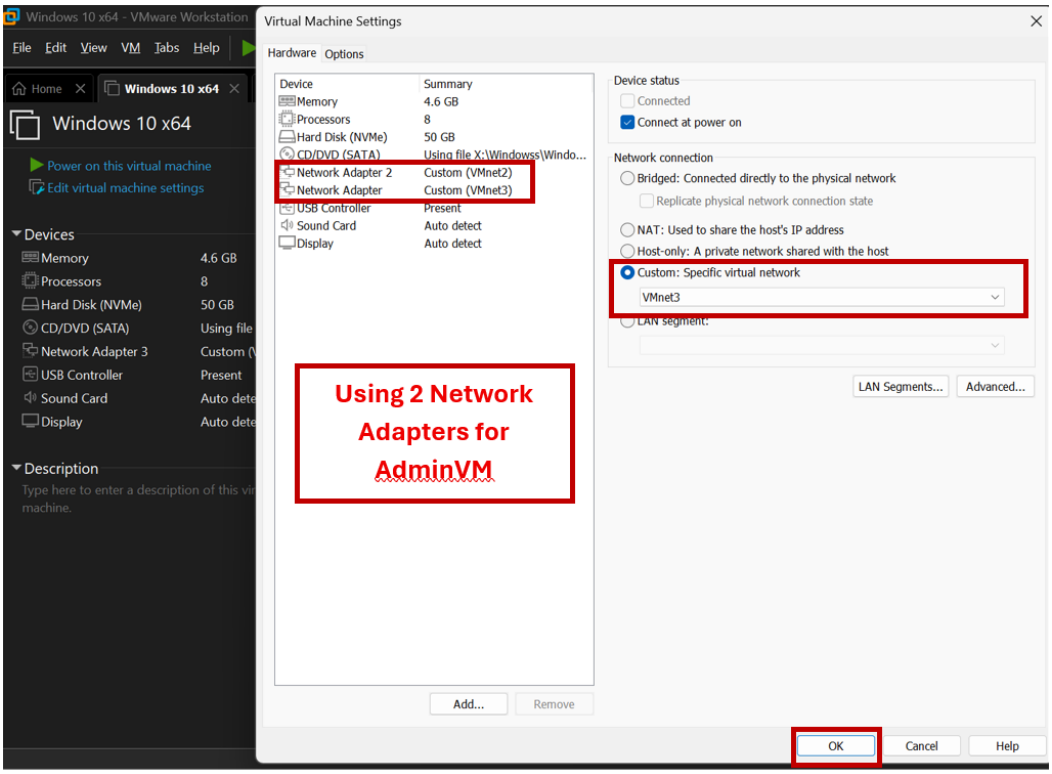


5. Windows 10 Admin VM (ADM01) – Network & Domain Join

The Windows 10 Admin VM (ADM01) will act as the administrative workstation. It needs two network adapters configured identically to the server — VMnet2 (NAT) and VMnet3 (host-only internal).

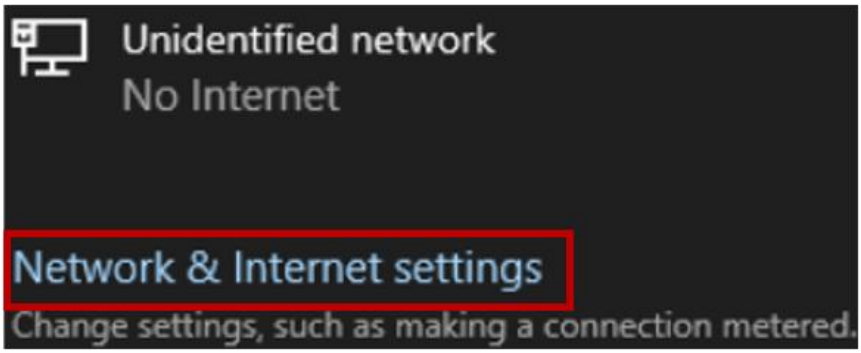
5.1 Configuring VM Network Adapters

In VMware settings for the Windows 10 VM, add two network adapters: one for VMnet2 (NAT) and one for VMnet3 (host-only). After configuration, power on the VM.



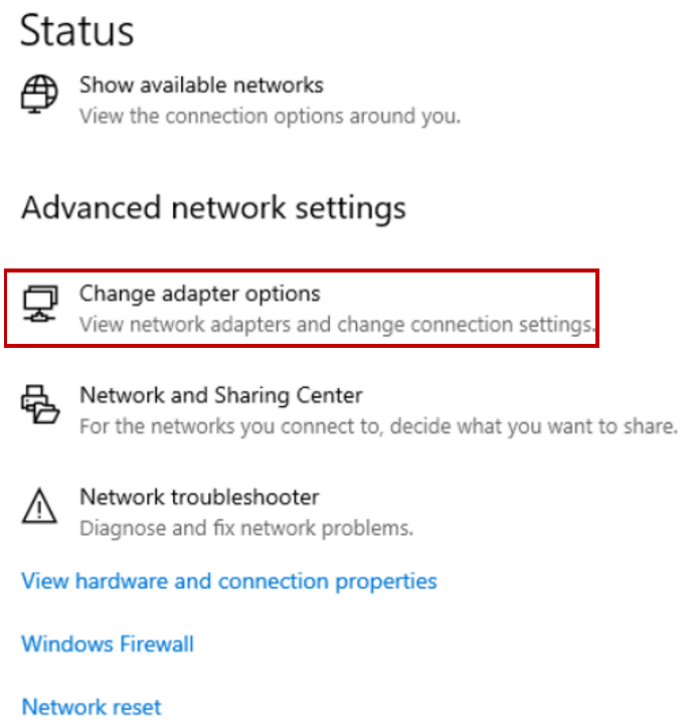
5.2 Connecting to Network & Internet Settings

After Windows 10 boots, click the network icon in the system tray. Open Network & Internet settings.



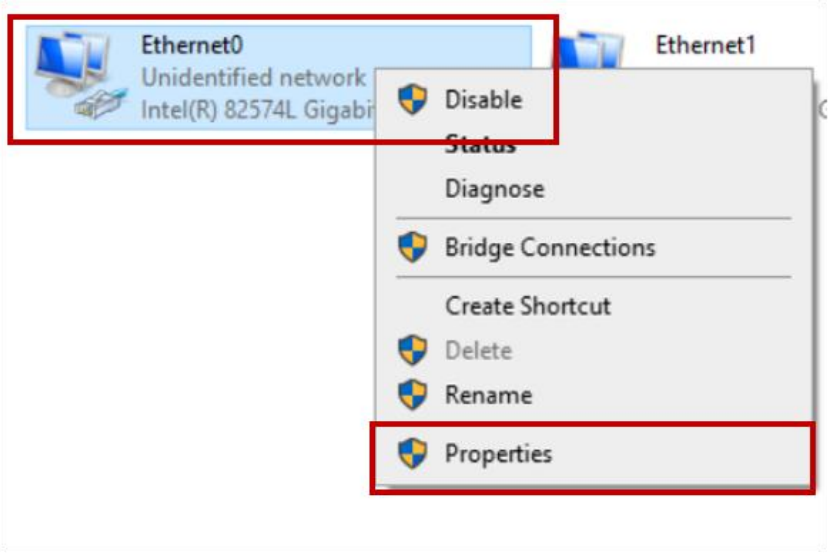
5.3 Change Adapter Options

Navigate to Change adapter options in the Network status page.



5.4 Assigning Static IP to ADM01

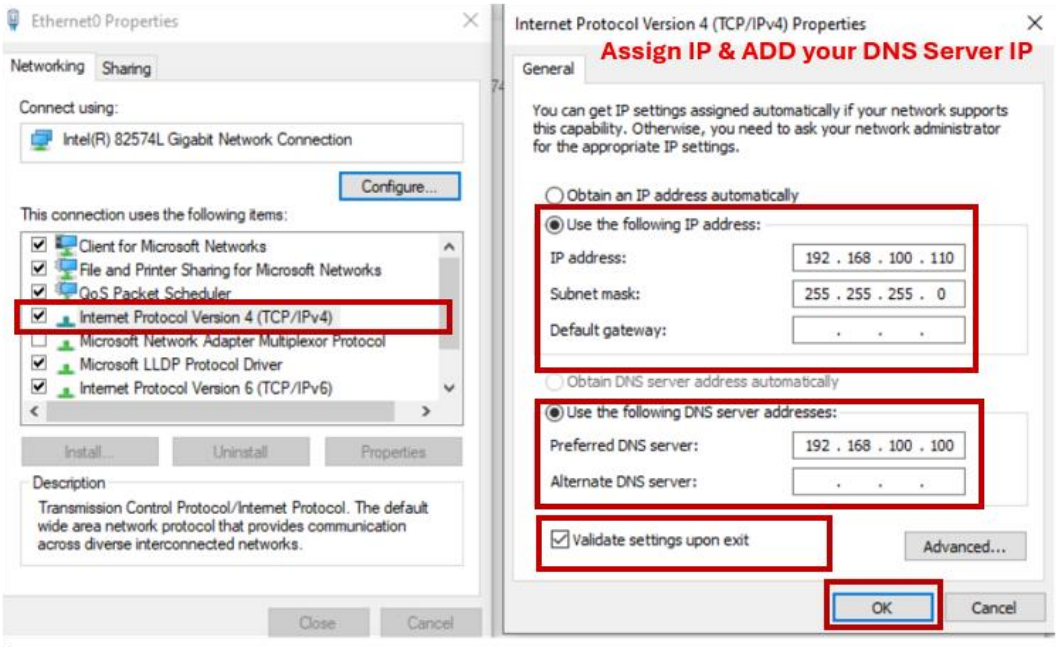
Right-click Ethernet0 (the VMnet3 adapter — shows "Unidentified network") and select Properties. Select "Internet Protocol Version 4 (TCP/IPv4)" and click Properties.



IPv4 Properties for ADM01:

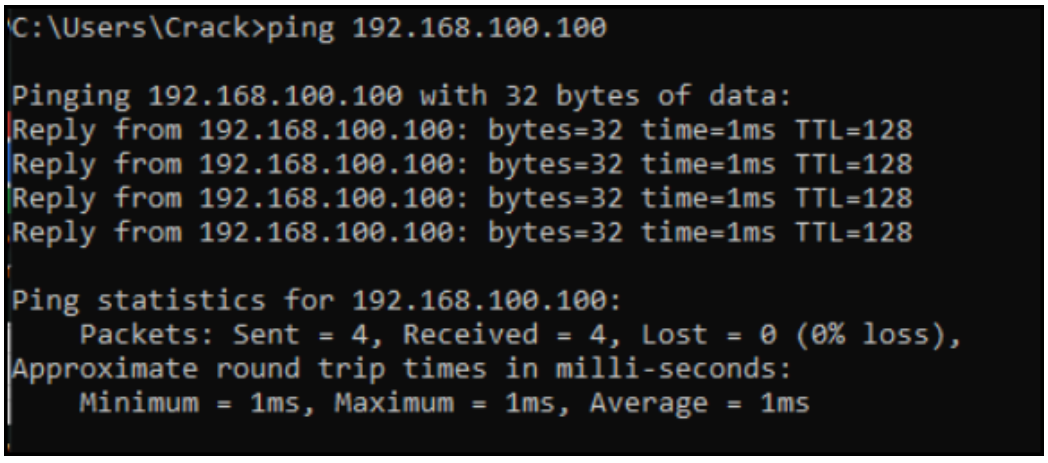
- IP Address: 192.168.100.110
- Subnet Mask: 255.255.255.0
- Default Gateway: (leave blank)
- Preferred DNS Server: 192.168.100.100 (the Domain Controller IP)

1. Check "Validate settings upon exit"
2. Click OK



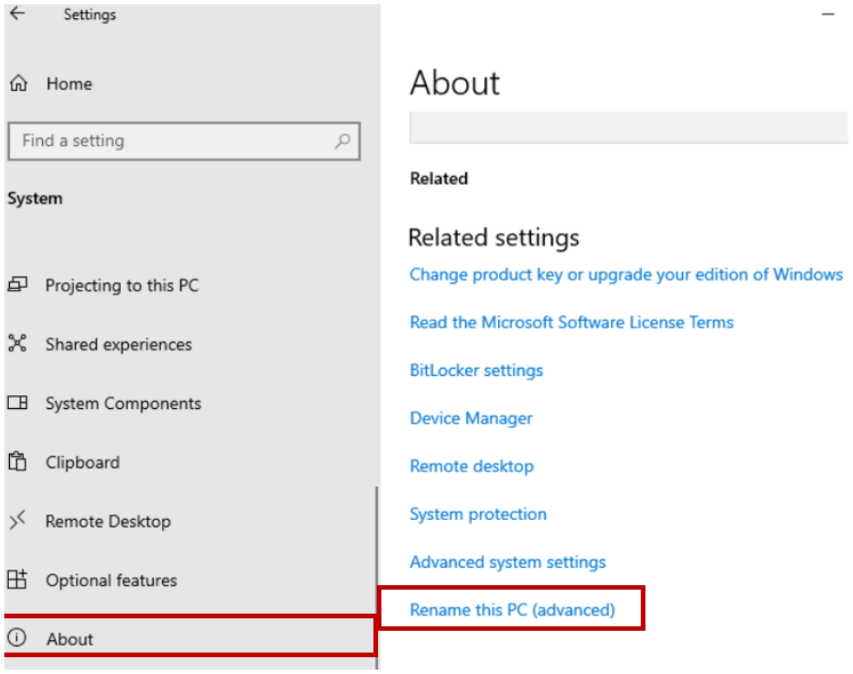
5.5 Testing Connectivity to Domain Controller

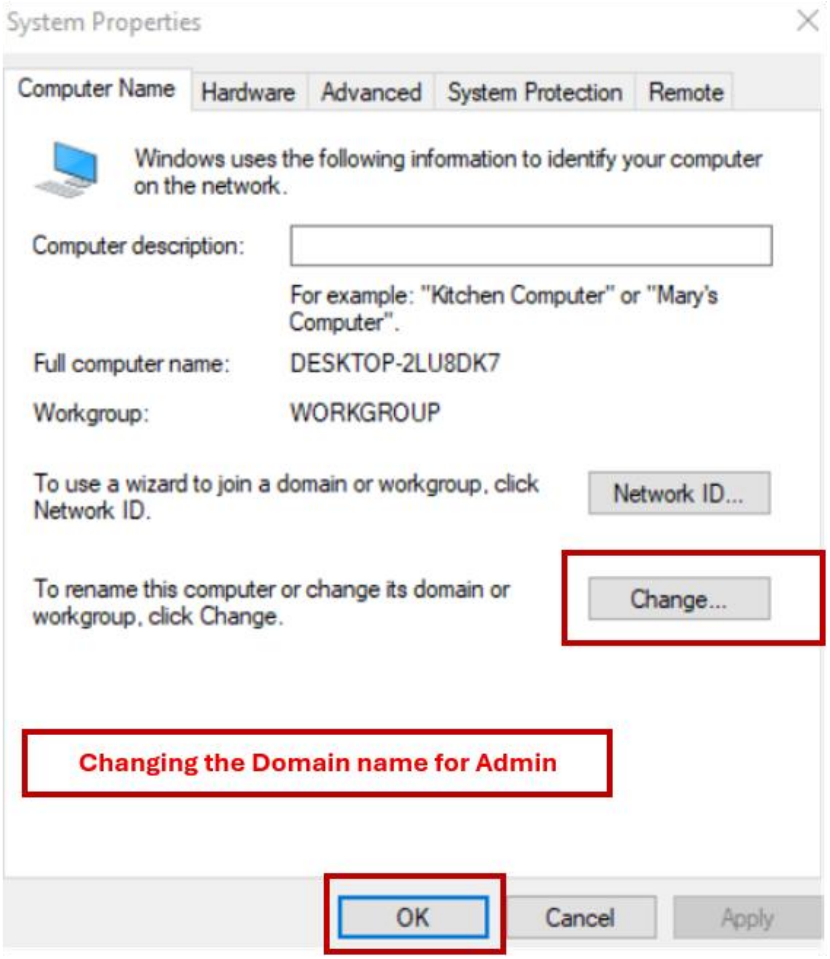
Open Command Prompt and run ping 192.168.100.100. All 4 packets should receive replies with 0% loss, confirming connectivity to the domain controller.



5.6 Renaming the PC and Joining the Domain

Navigate to Settings > System > About and click "Rename this PC (advanced)". Then click the Change… button.

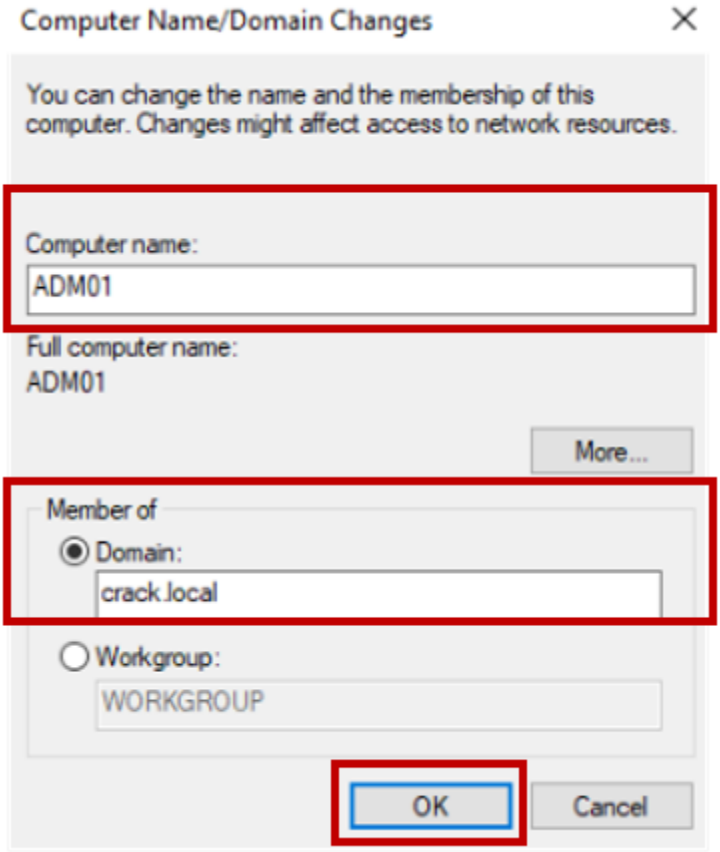




Computer Name/Domain Changes:

- Computer name: ADM01
- Member of: Domain → crack.local

3. Click OK

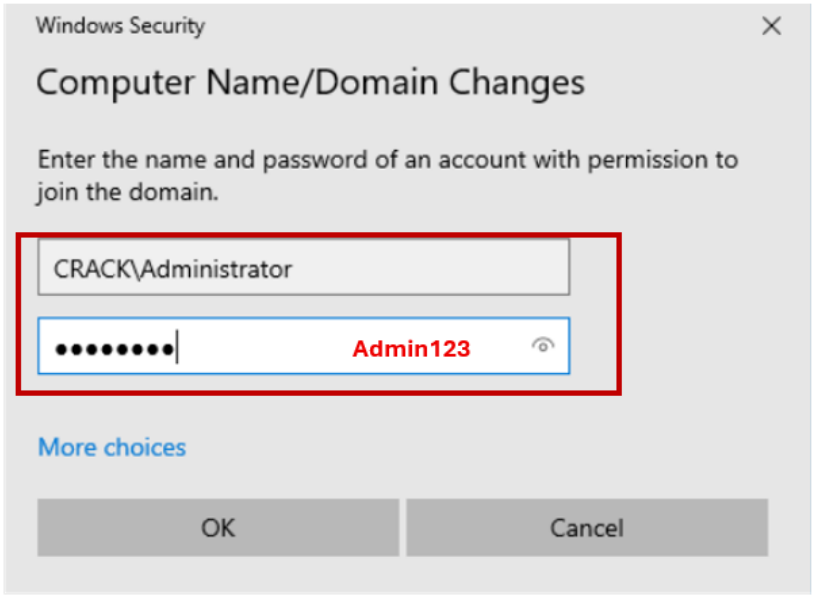


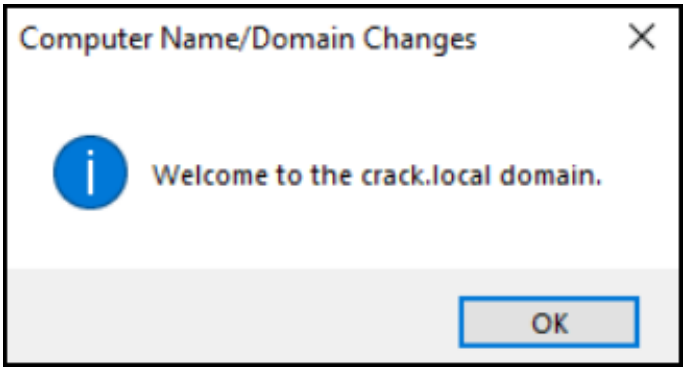
5.7 Providing Domain Credentials

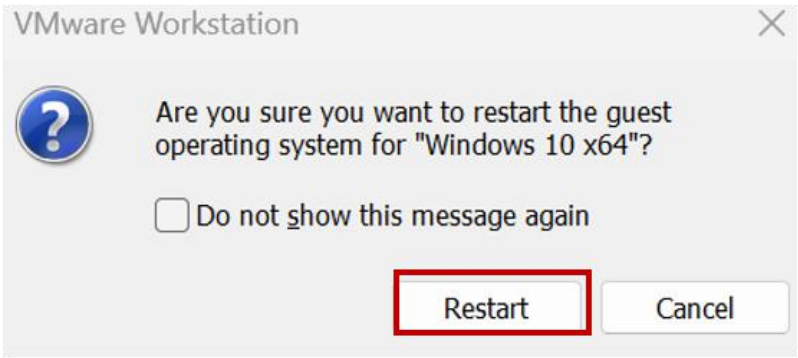
A Windows Security prompt appears asking for domain credentials to authorize joining. Enter the domain administrator credentials.

- Username: CRACK\Administrator
- Password: Admin123

A "Welcome to the crack.local domain" message confirms successful domain join. Click OK and restart the system when prompted.

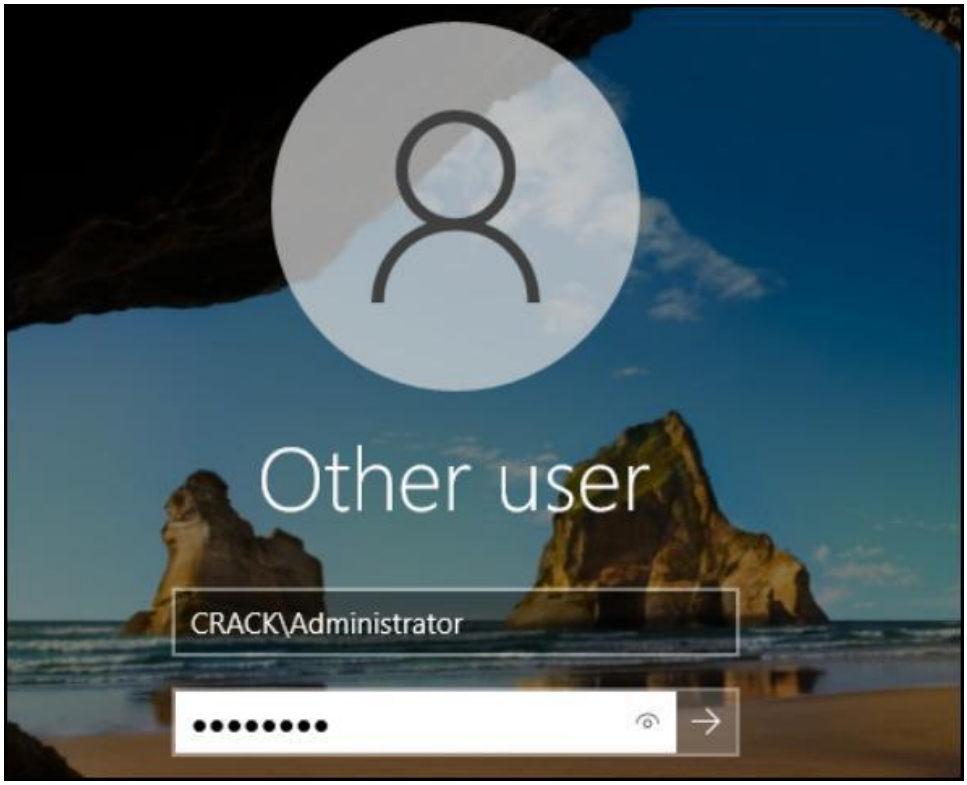






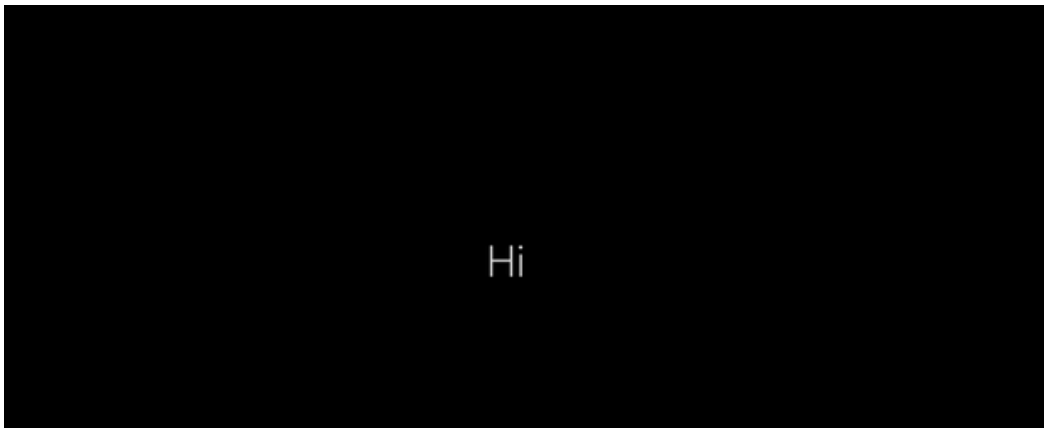
5.8 Post-Restart Domain Login

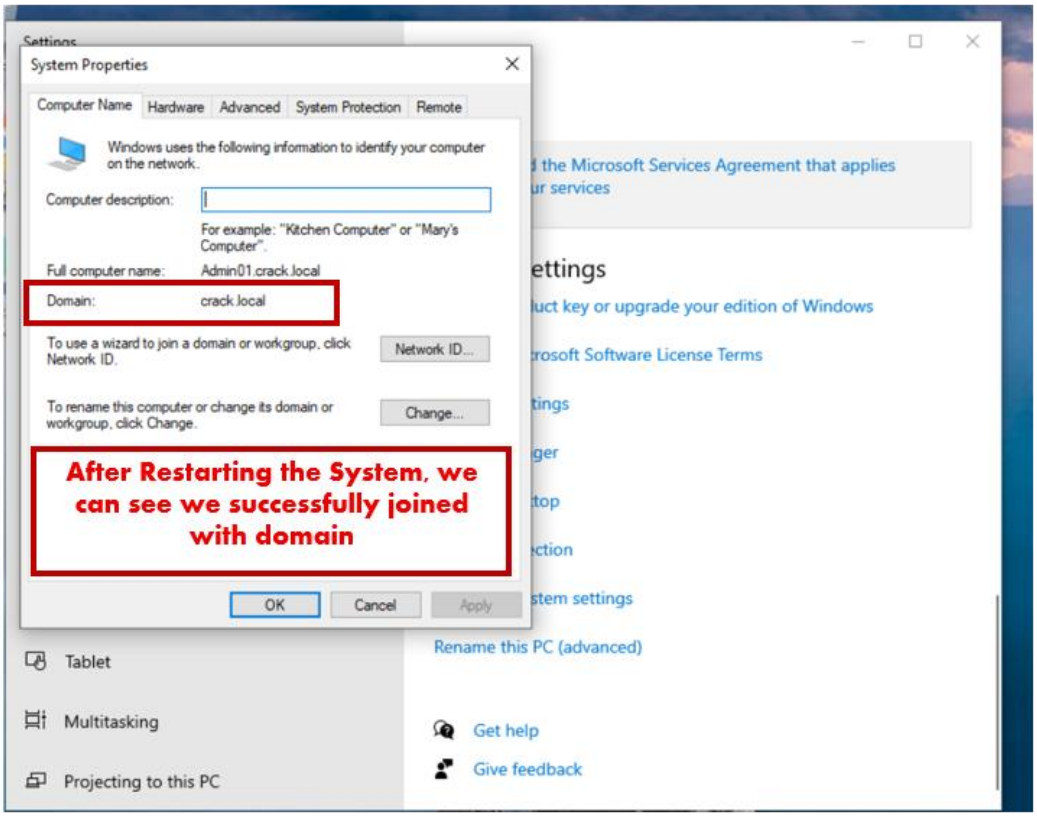
After restarting, select "Other user" on the login screen. Notice the domain "Sign in to: CRACK" is displayed. Log in with CRACK\Administrator / Admin123.



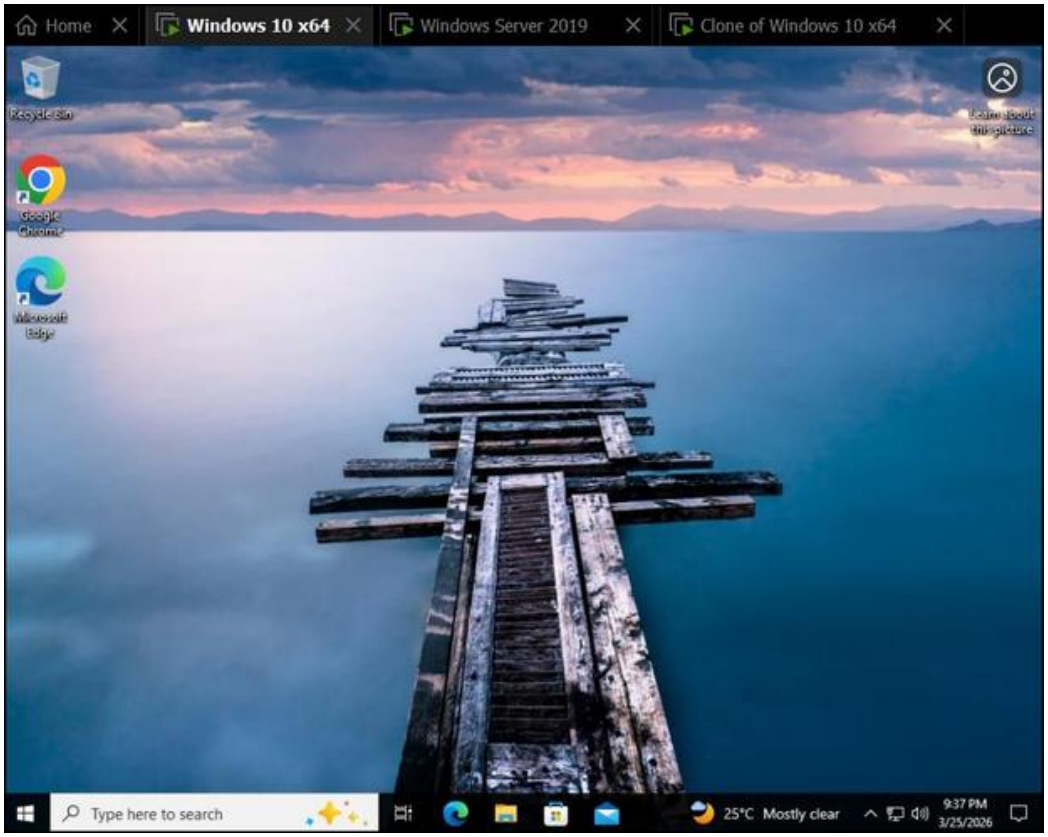
5.9 Verifying Domain Membership

After login, go to System Properties. The domain now shows "crack.local" and the full computer name shows "Admin01.crack.local", confirming successful domain membership.





ADM01 is now running as a domain-joined Windows 10 workstation. Note the three tabs in VMware: Windows 10 x64 (ADM01), Windows Server 2019, and the upcoming Clone (CLI01).

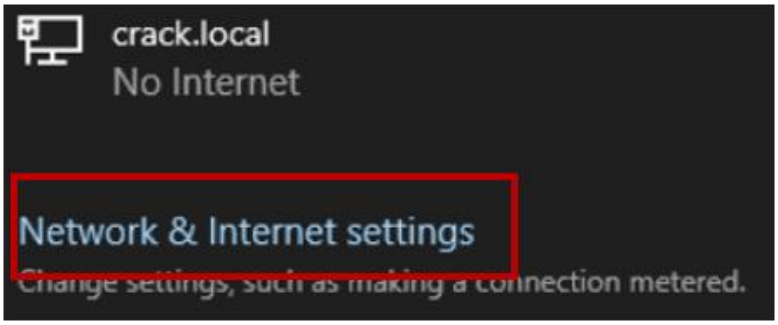


6. Windows 10 Client VM (CLI01) – Creating and Domain Join

CLI01 is created by cloning the ADM01 virtual machine. Since all the network settings (VMnet2/VMnet3 adapters, domain membership) are already configured in the clone, we only need to assign a new IP address and rename the machine.

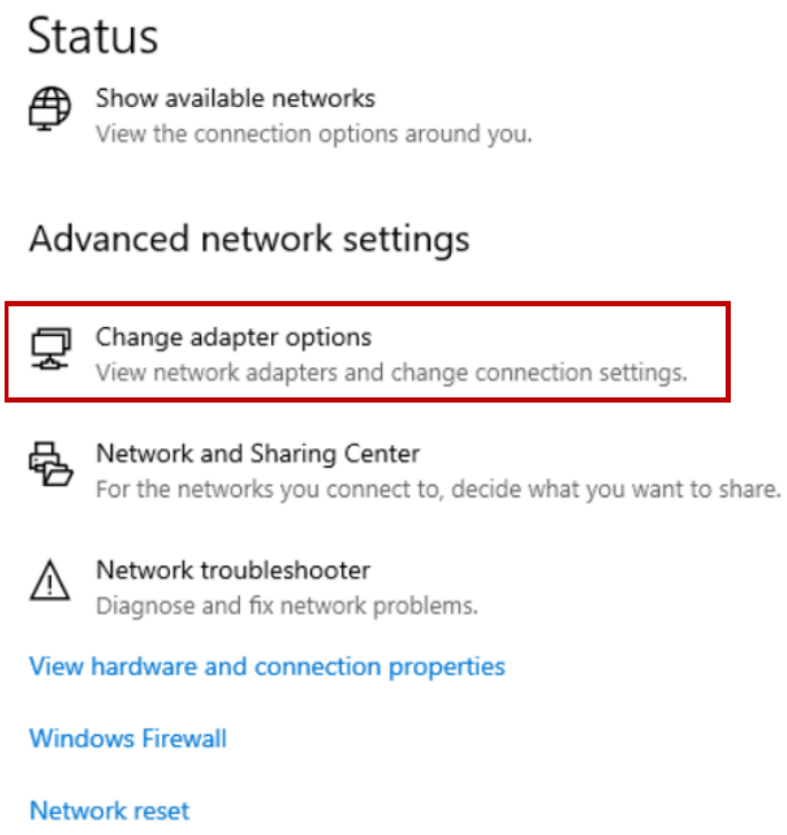
6.1 Running the Clone

Boot the cloned Windows 10 VM. The network tray icon shows "crack.local – No Internet" confirming it is still domain-connected from the clone. Open Network & Internet settings.



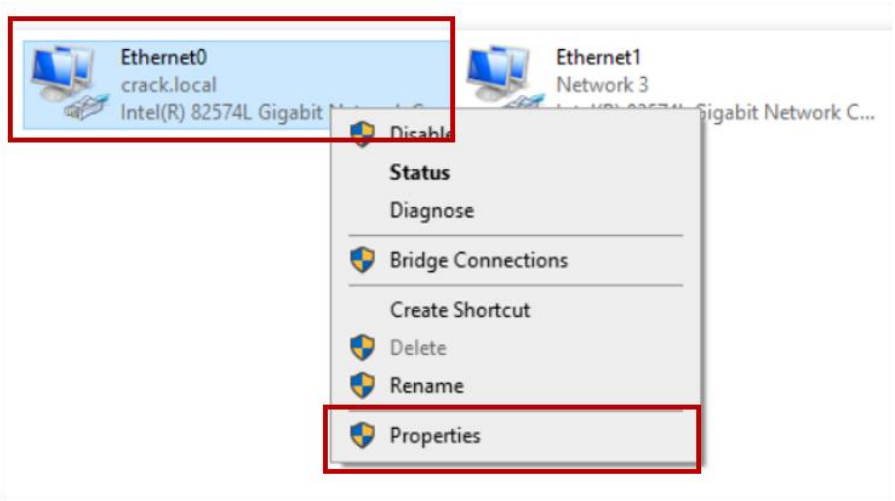
6.2 Change Adapter Options on CLI01

Navigate to Advanced network settings and click Change adapter options.



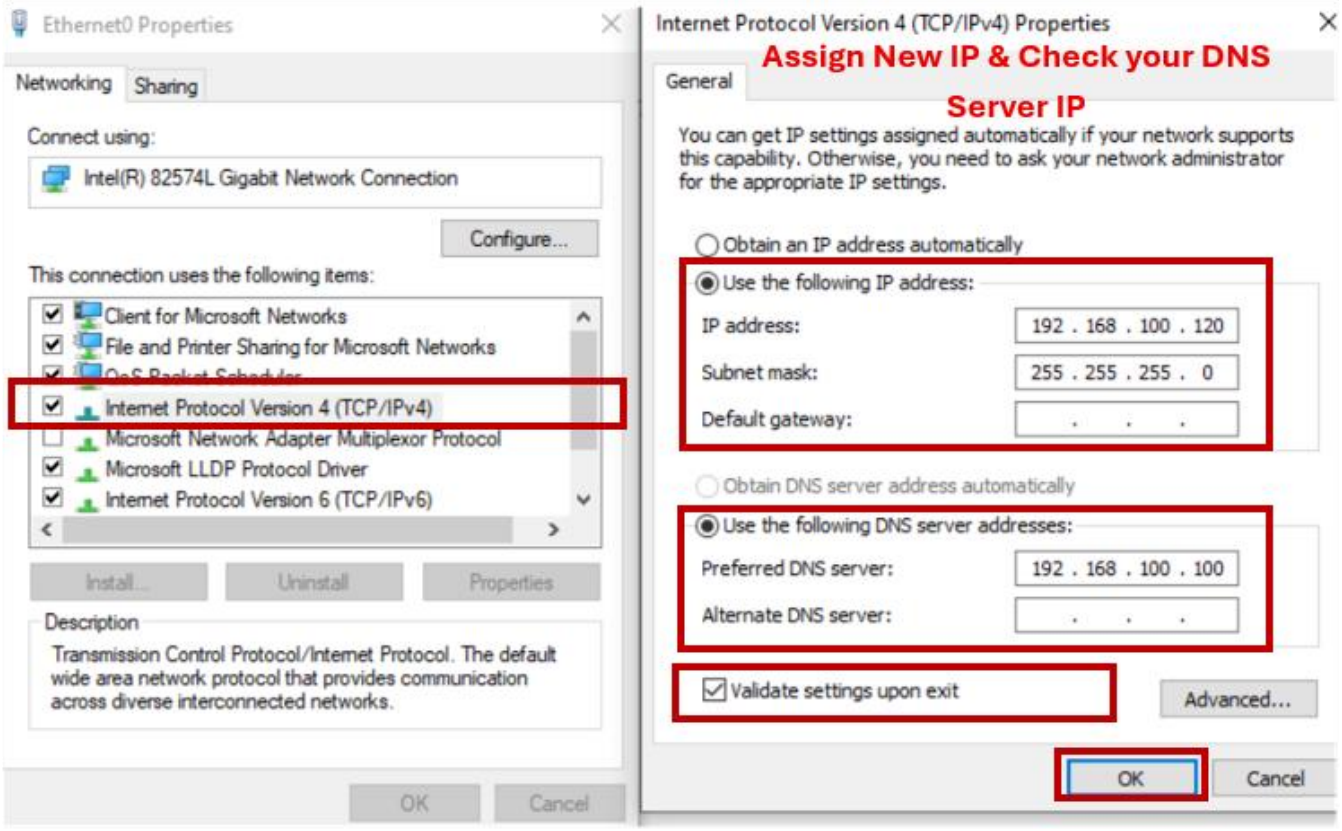
6.3 Assigning a New IP Address to CLI01

Right-click Ethernet0 (shows crack.local – the domain-connected adapter) and select Properties. Open IPv4 Properties and assign a new unique IP address.



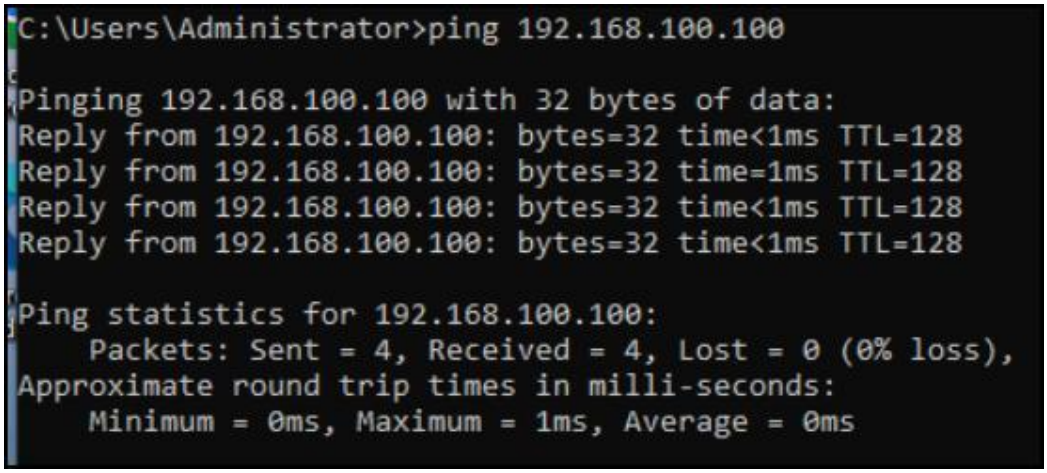
IPv4 Properties for CLI01:

- IP Address: 192.168.100.120
- Subnet Mask: 255.255.255.0
- Default Gateway: (leave blank)
- Preferred DNS Server: 192.168.100.100



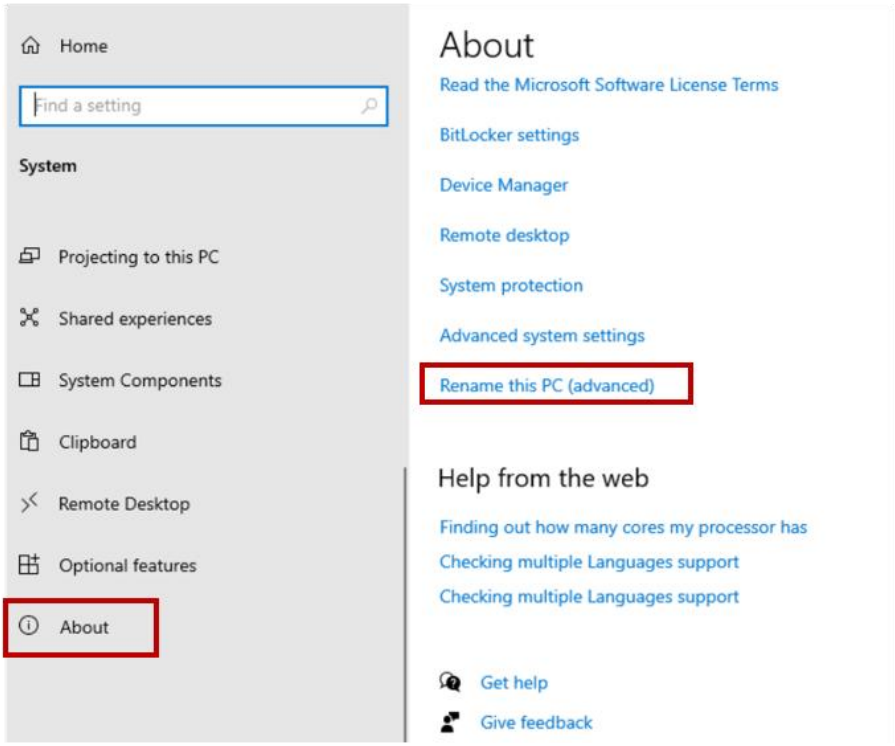
6.4 Verify Ping to Domain Controller

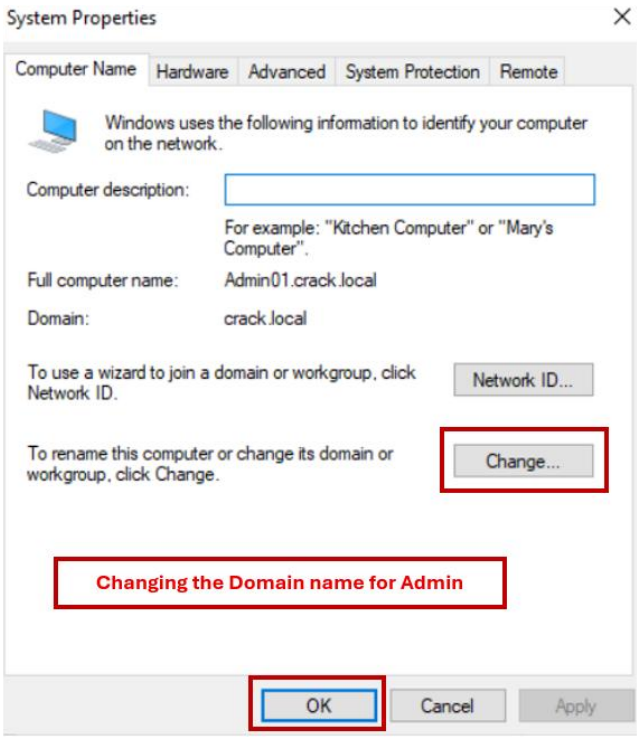
Open Command Prompt and run ping 192.168.100.100 to confirm connectivity from CLI01 to the domain controller.



6.5 Renaming CLI01 and Re-joining Domain

Navigate to Settings > System > About > Rename this PC (advanced). The System Properties dialog shows the current computer name "Admin01" from the clone. We need to rename it to CLI01 and re-join the domain with the correct unique SSID.

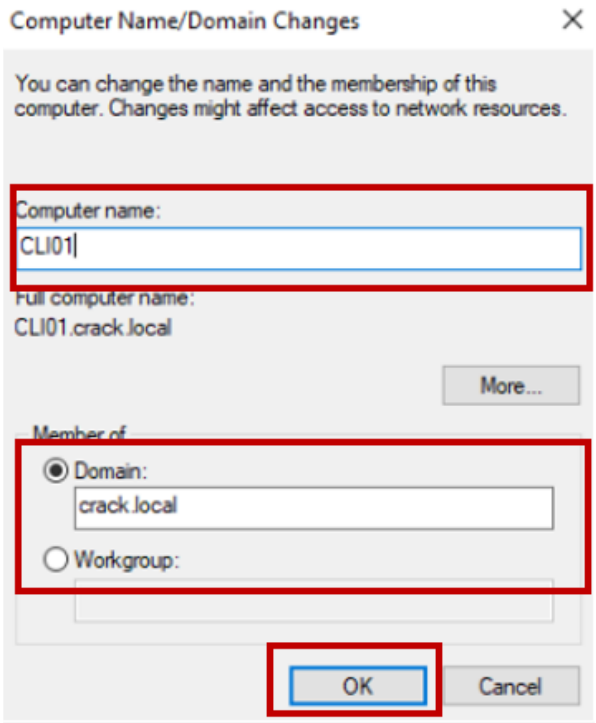




Computer Name/Domain Changes:

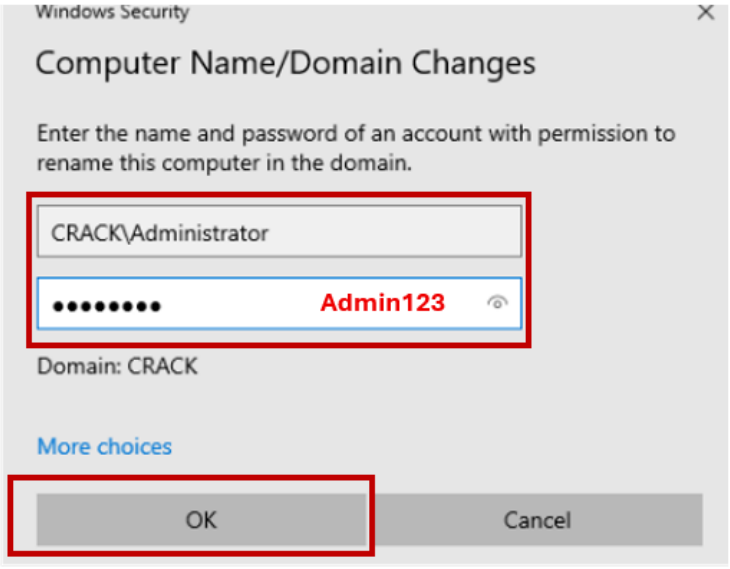
- Computer name: CLI01
- Full computer name: CLI01.crack.local
- Domain: crack.local

1. Click OK



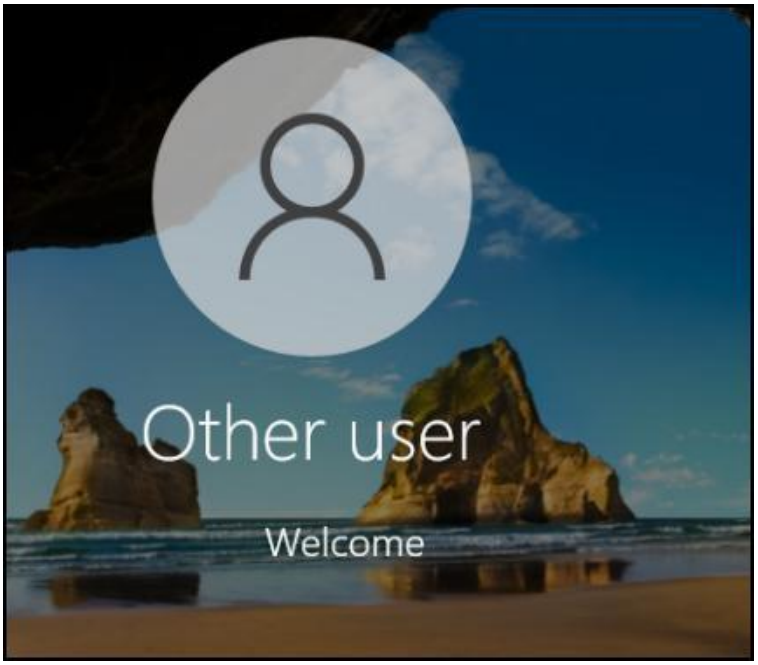
6.6 Providing Domain Credentials for CLI01

Enter CRACK\Administrator and password Admin123 to authorize the computer rename in the domain.



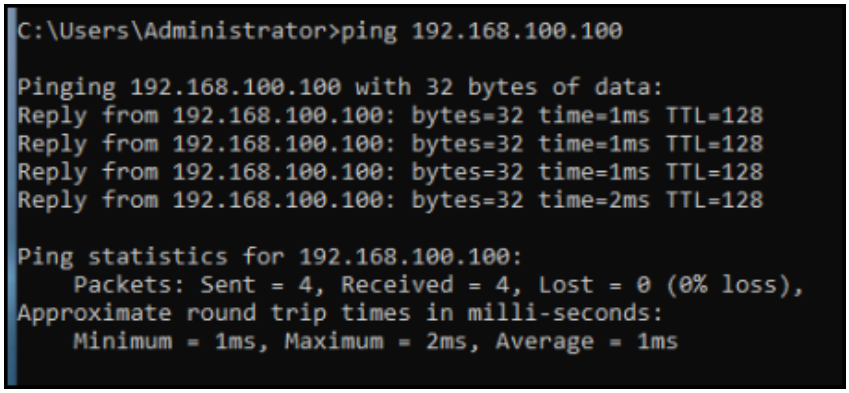
6.7 Post-Login Verification on CLI01

After restarting CLI01, log in with CRACK\Administrator. The login screen shows "Other user" with domain CRACK available.



6.8 Testing CLI01 to Server Connectivity

Ping the domain controller from CLI01 to confirm network communication is working correctly.

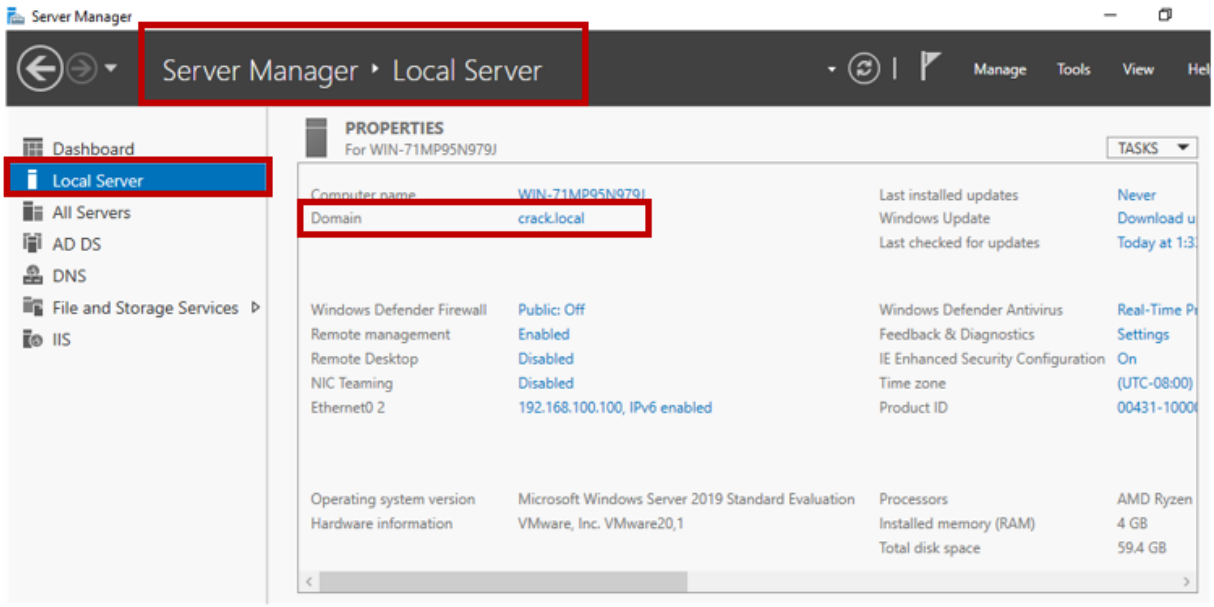


7. Renaming the Domain Controller Server

The domain controller was initially given a random name (WIN-71MP95N979J). For a professional setup, rename it to DC01 to clearly identify its role. This is done via Server Manager on the server itself.

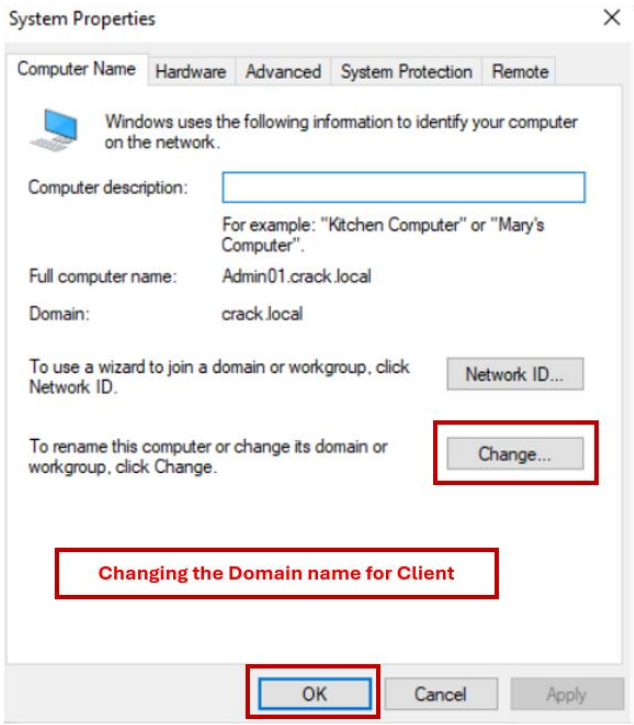
7.1 Viewing Current Computer Name in Server Manager

On the domain controller, open Server Manager > Local Server. The Properties panel shows the current computer name and domain (crack.local).



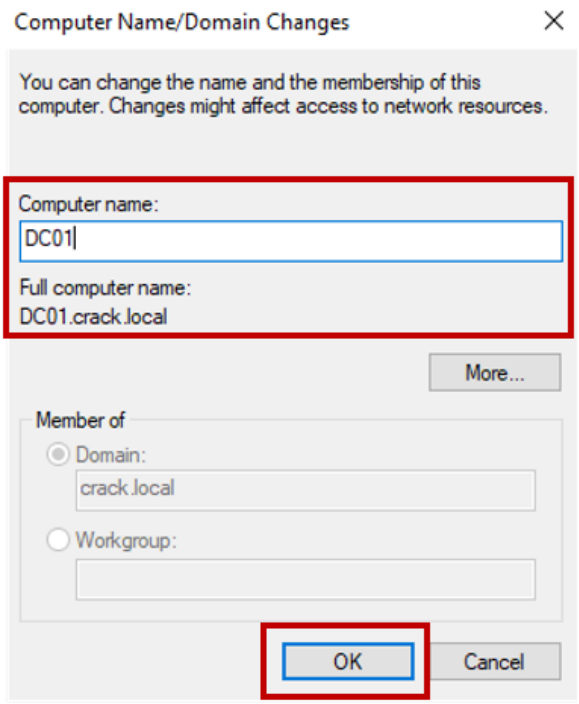
7.2 Changing the Computer Name to DC01

Click on the computer name link in Server Manager Local Server properties. The System Properties dialog opens. Click "Change…".



Computer Name/Domain Changes:

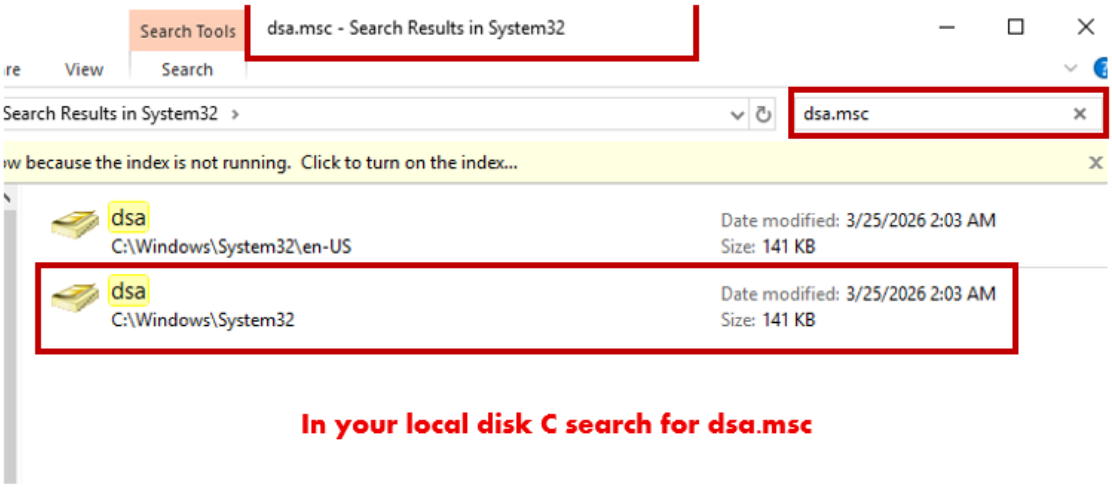
- Computer name: DC01
- Full computer name: DC01.crack.local
- Domain: crack.local (already a member)



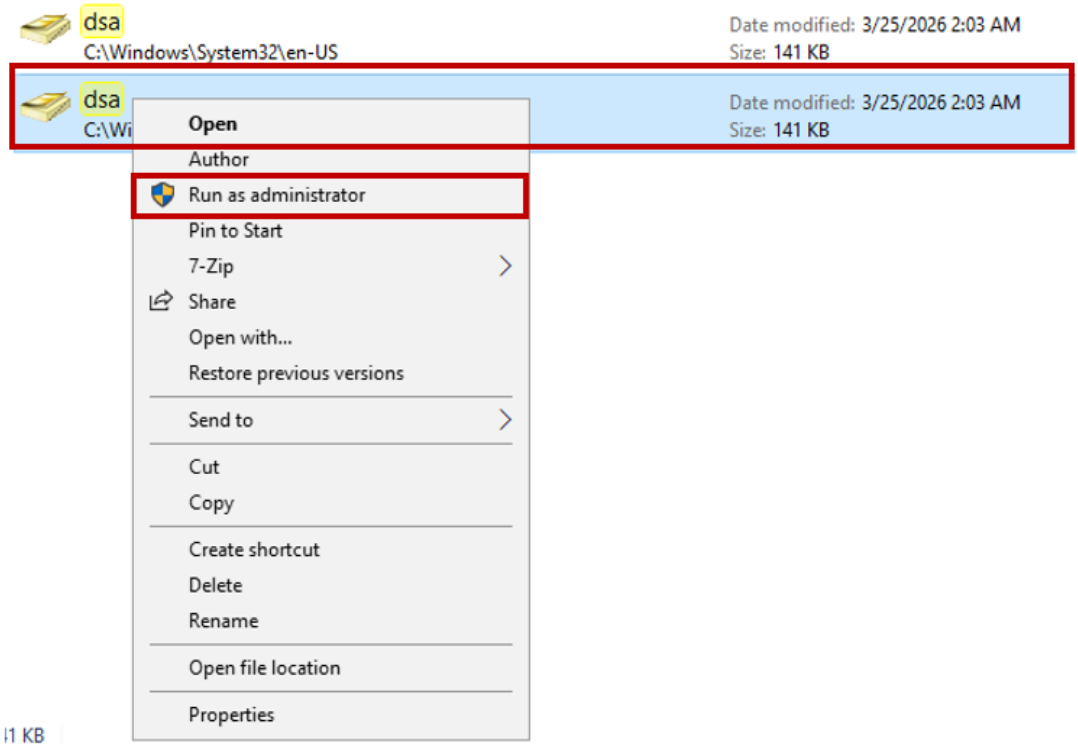
Click OK. A restart is required. After restart, the domain controller will be identifiable as DC01.crack.local throughout the domain.

7.3 Accessing dsa.msc – Active Directory Users and Computers

To manage the Active Directory, you need to launch dsa.msc (Active Directory Users and Computers) as Administrator. Navigate to C:\Windows\System32 and search for dsa.msc.



Right-click on the dsa.msc file in System32 and select "Run as administrator".

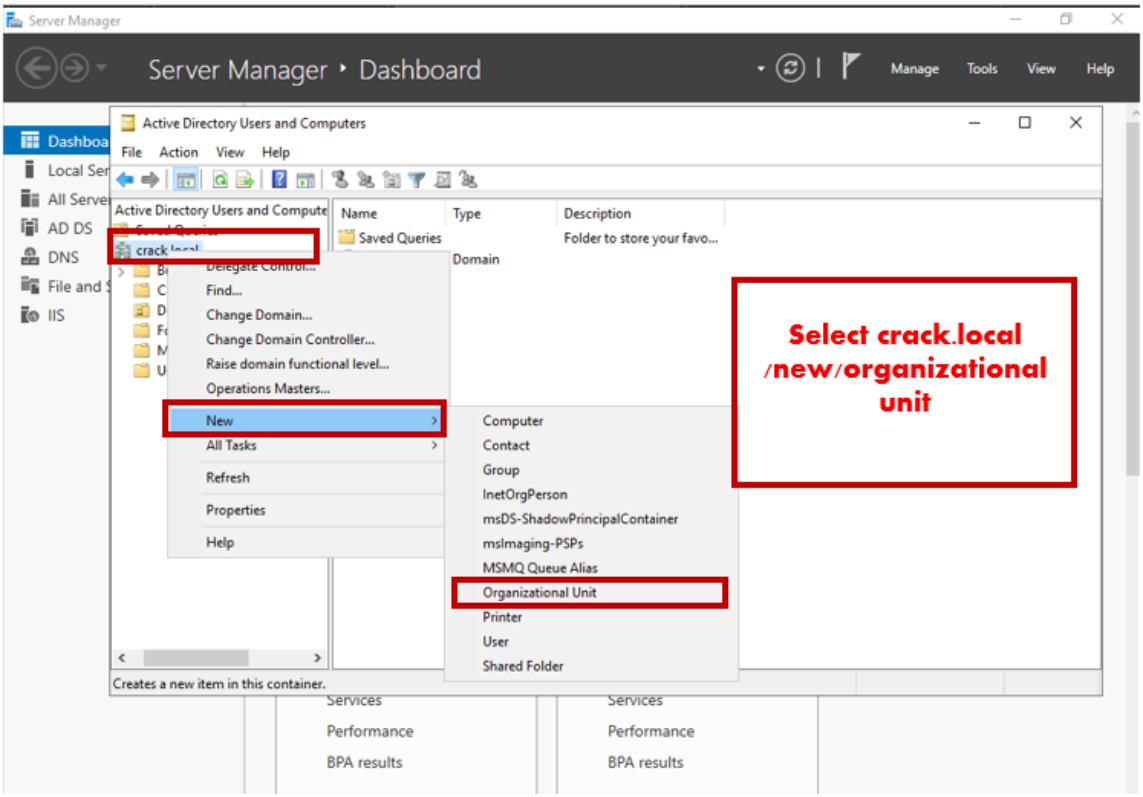


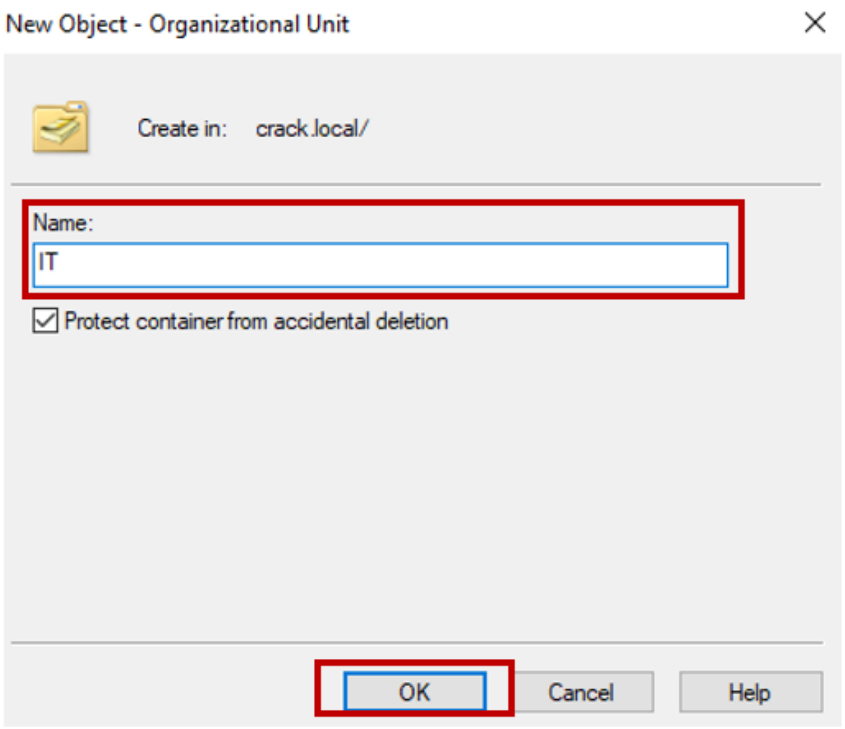
8. Organizational Units (OUs)

Organizational Units (OUs) provide a hierarchical structure for organizing domain objects such as users, computers, and groups. Group Policy Objects (GPOs) can be linked to OUs to apply different settings to different departments.

8.1 Creating Organizational Units

In Active Directory Users and Computers, right-click on "crack.local" in the left tree, select New > Organizational Unit. Create the following OUs one by one:



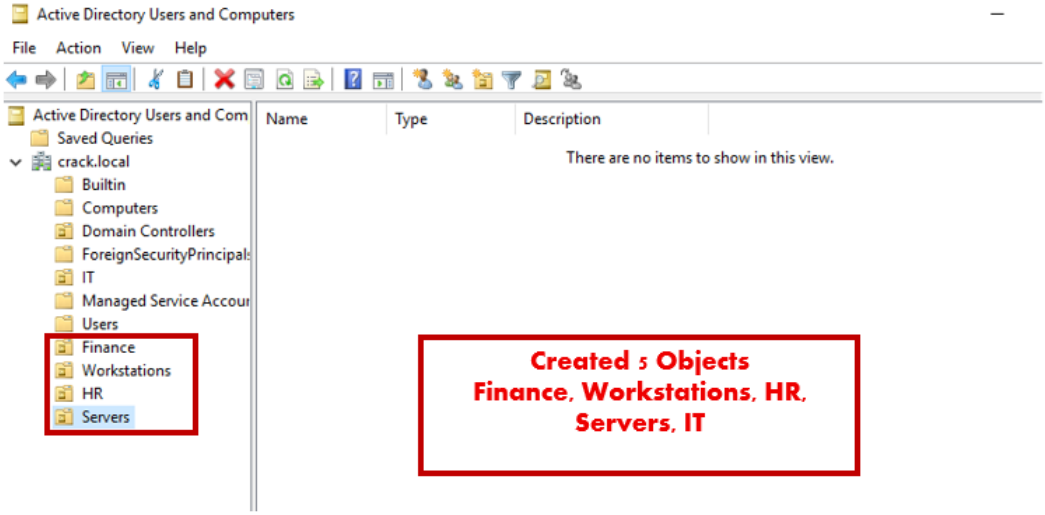


OUs to Create:

- IT — for IT administrators
- Finance — for Finance department users
- HR — for Human Resources users
- Servers — for server computer objects
- Workstations — for workstation computer objects

8.2 Viewing All Created OUs

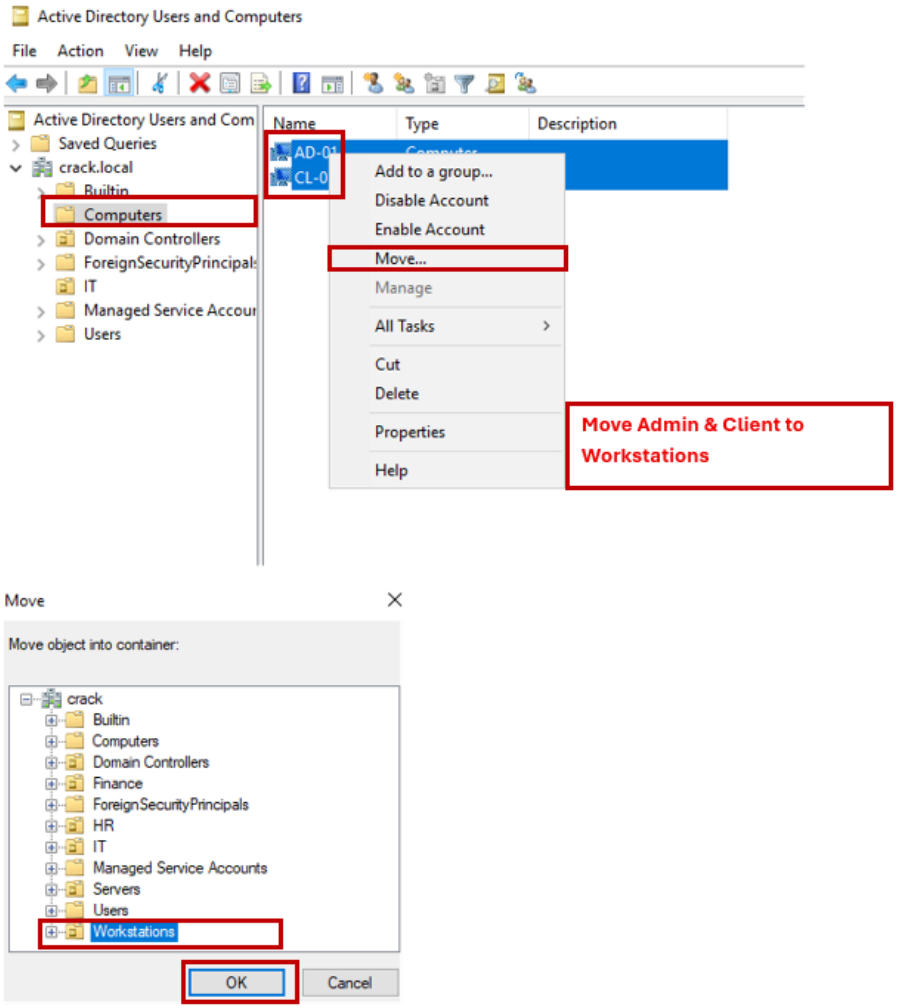
After creating all five OUs, the left panel of Active Directory Users and Computers shows Finance, Workstations, HR, Servers, and IT under crack.local.



8.3 Moving Computers to Workstations OU

The domain-joined computers (ADM01 and CLI01) currently reside in the default Computers container. Move them to the Workstations OU for proper policy management.

1. Click on the Computers container in the left panel
2. Select both AD-01 and CL-0 computer objects (Ctrl+click)
3. Right-click and select "Move…"
4. Select the Workstations OU from the tree
5. Click OK

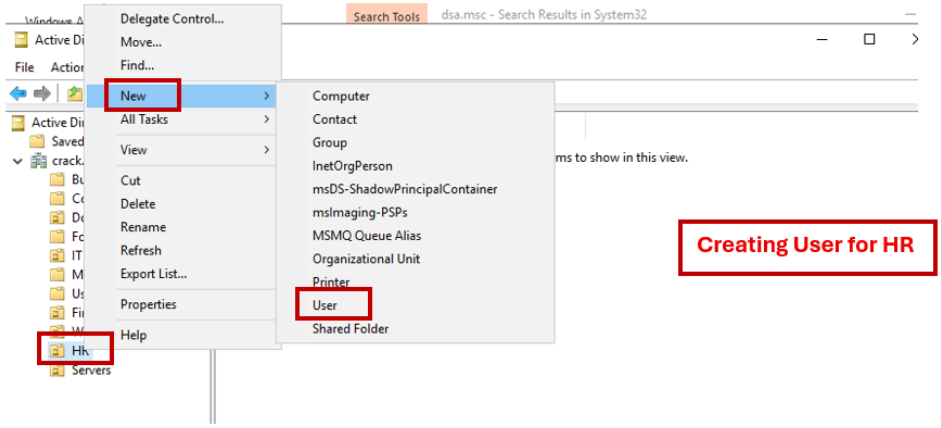


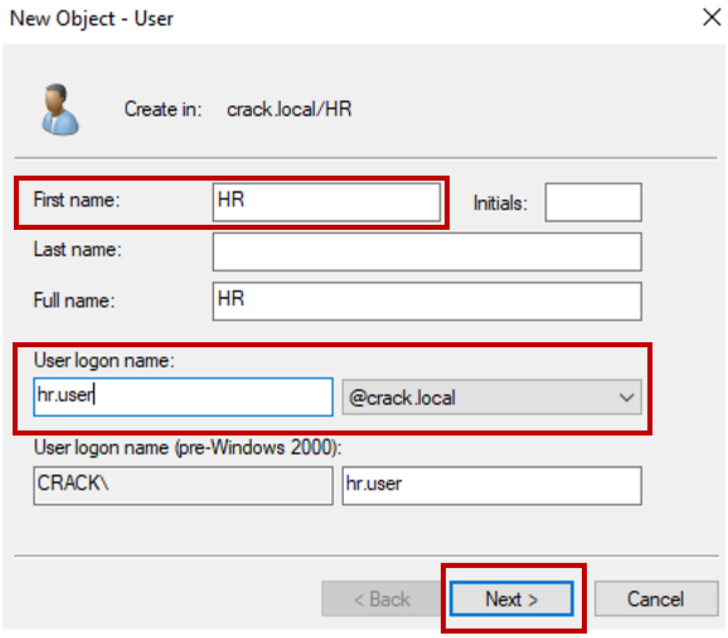
9. User Account Management

Create user accounts in each department OU. These accounts represent real employees and will be used to test Group Policy, login, and access control.

9.1 Creating HR User (hr.user)

Right-click on the HR OU, select New > User. Fill in the user details.

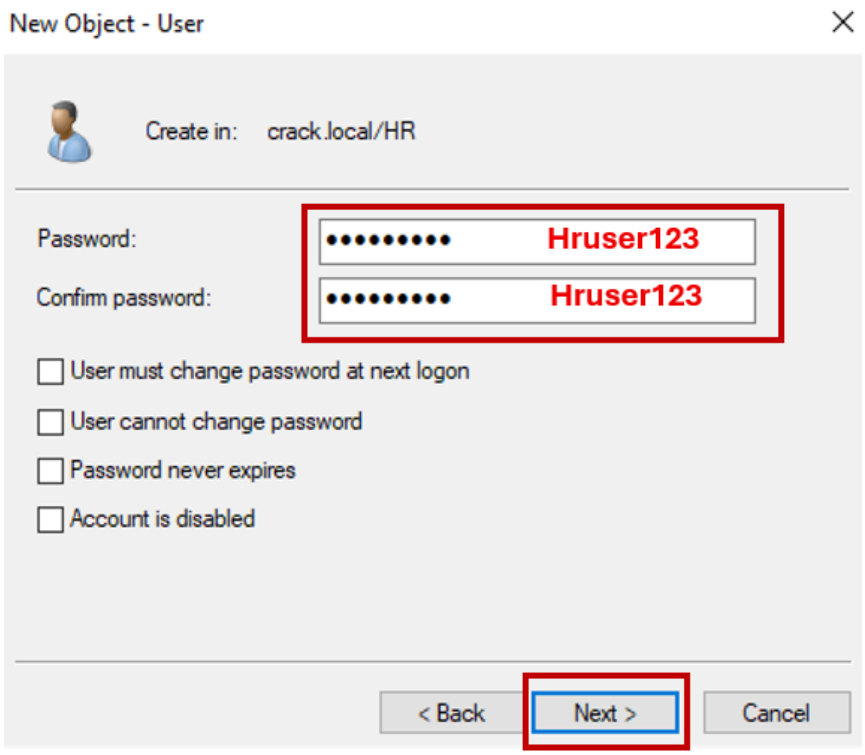


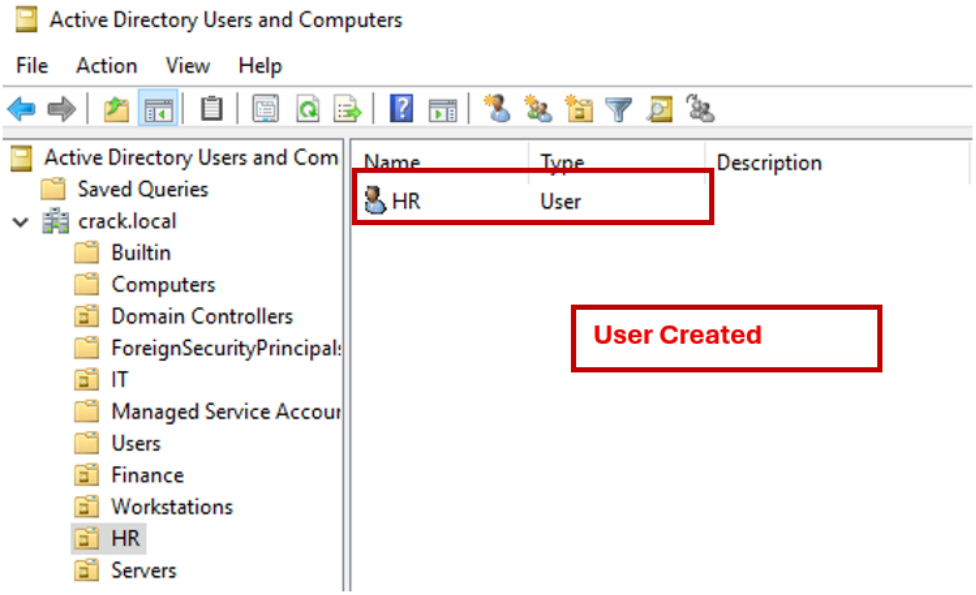


HR User Details:

- First name: HR
- User logon name: hr.user@crack.local
- Password: Hruser123
- Uncheck: User must change password at next logon

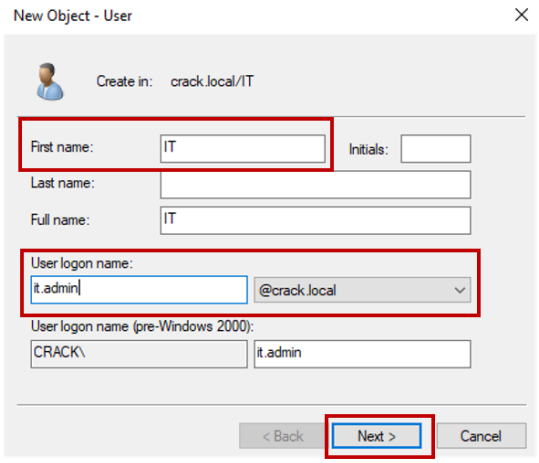
Click Next and set the password, then click Finish. The HR user is successfully created.


New Object - User
Create in: crack.local/HR
Password:
Hruser123
Confirm password:
Hruser123
User must change password at next logon
User cannot change password
Password never expires
Account is disabled
< Back
Next >
Cancel


Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
crack.local
Builtin
Computers
Domain Controllers
ForeignSecurityPrincipal:
IT
Managed Service Accour
Users
Finance
Workstations
HR
Servers
Name
Type
Description
HR
User
User Created

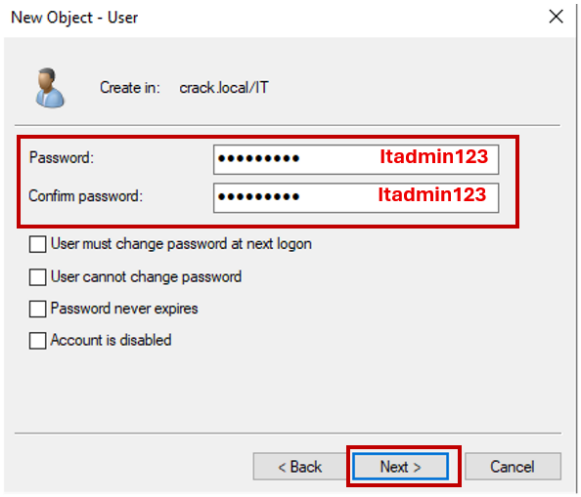
9.2 Creating IT Admin User (it.admin)

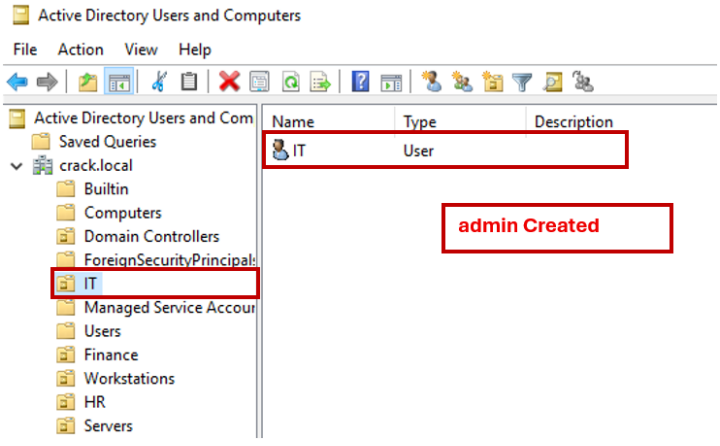
Right-click on the IT OU and select New > User.



IT User Details:

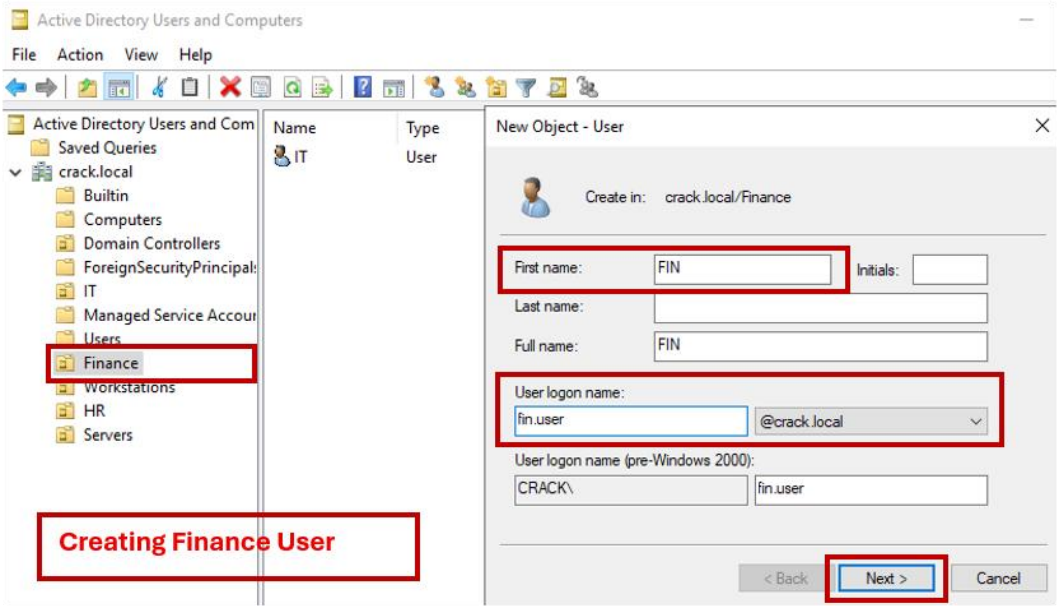
- First name: IT
- User logon name: it.admin@crack.local
- Password: Itadmin123





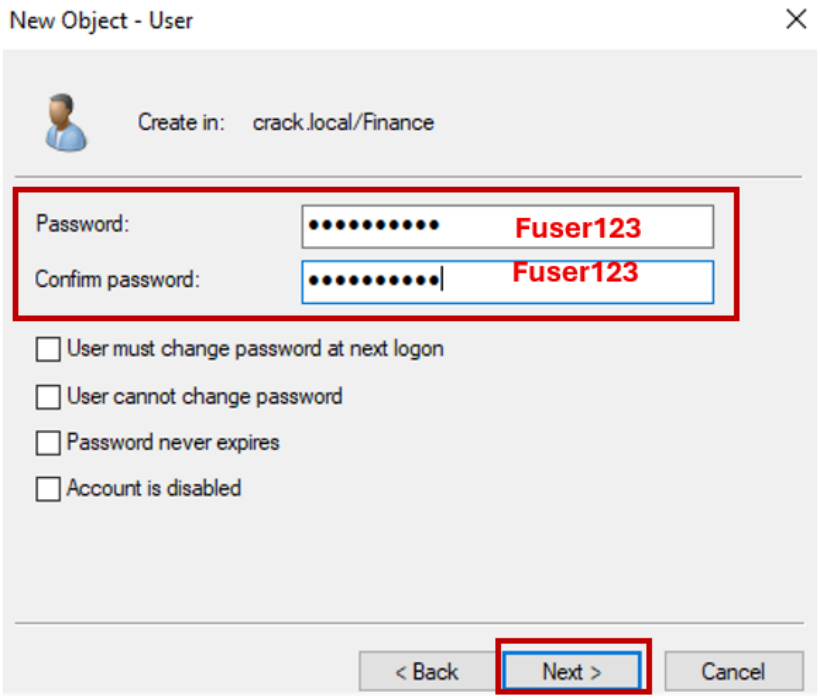
9.3 Creating Finance User (fin.user)

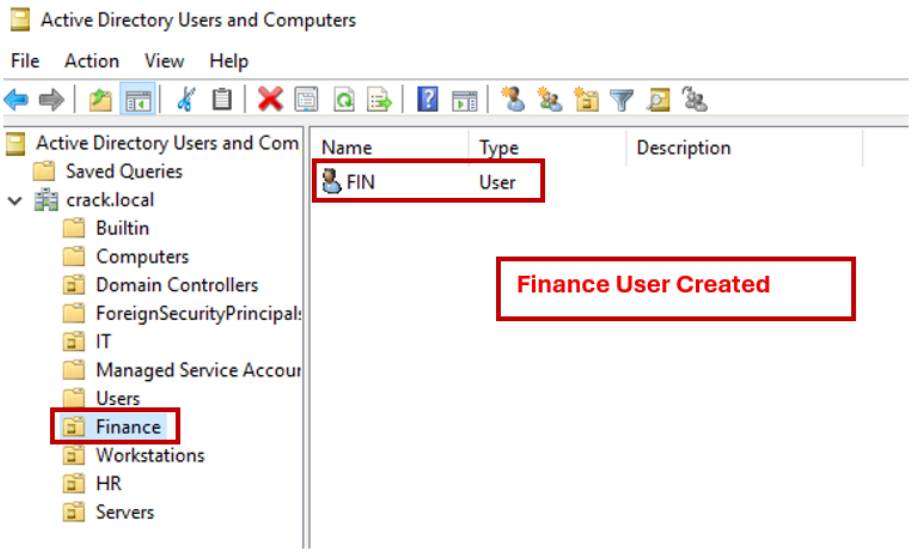
Right-click on the Finance OU and select New > User. Create the finance department account.



Finance User Details:

- First name: FIN
- User logon name: fin.user@crack.local
- Password: Fuser123

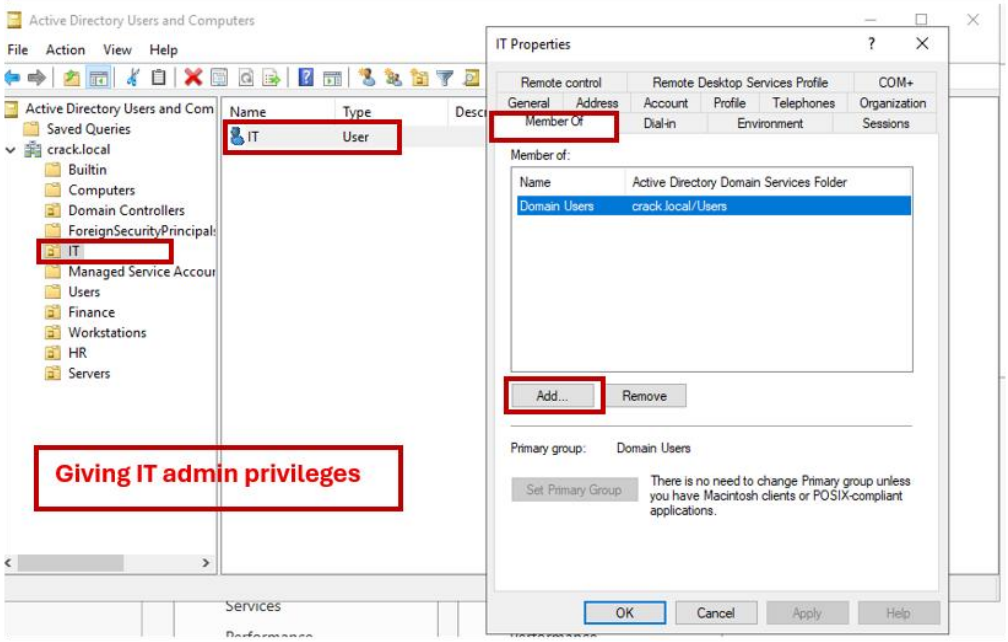


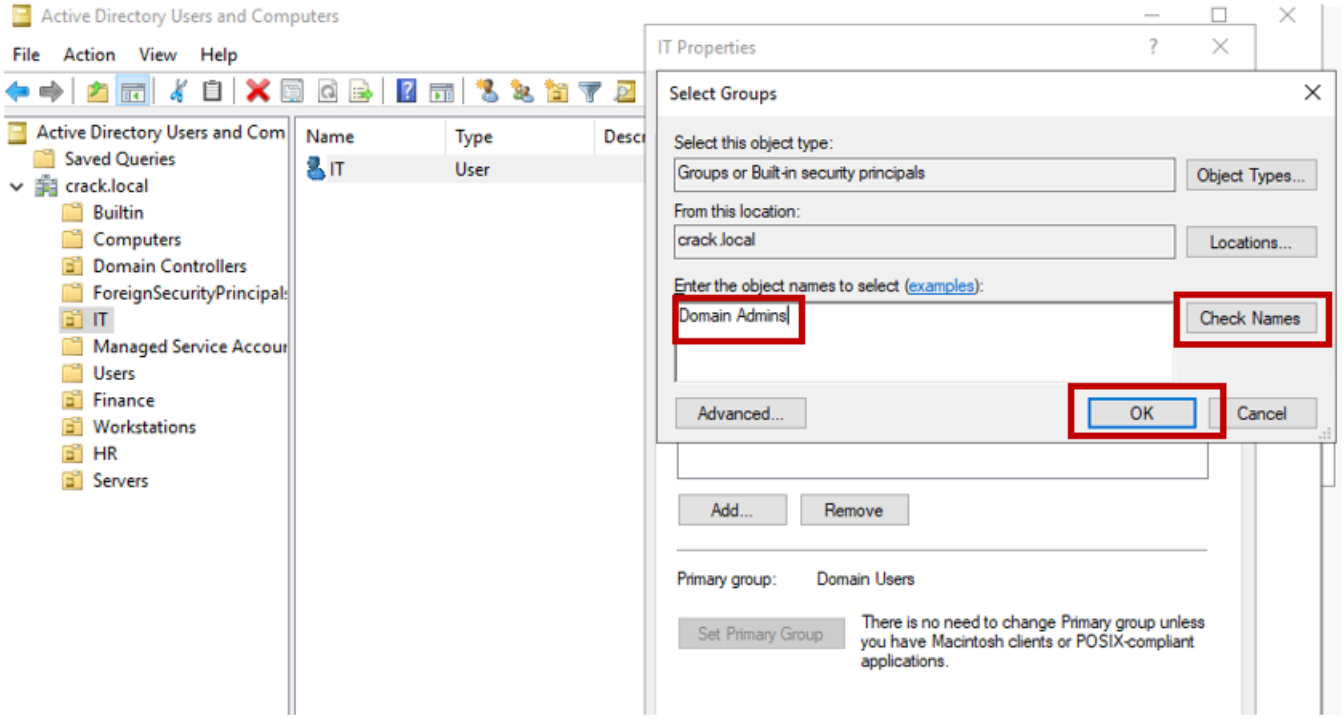


9.4 Granting it.admin Domain Admin Privileges

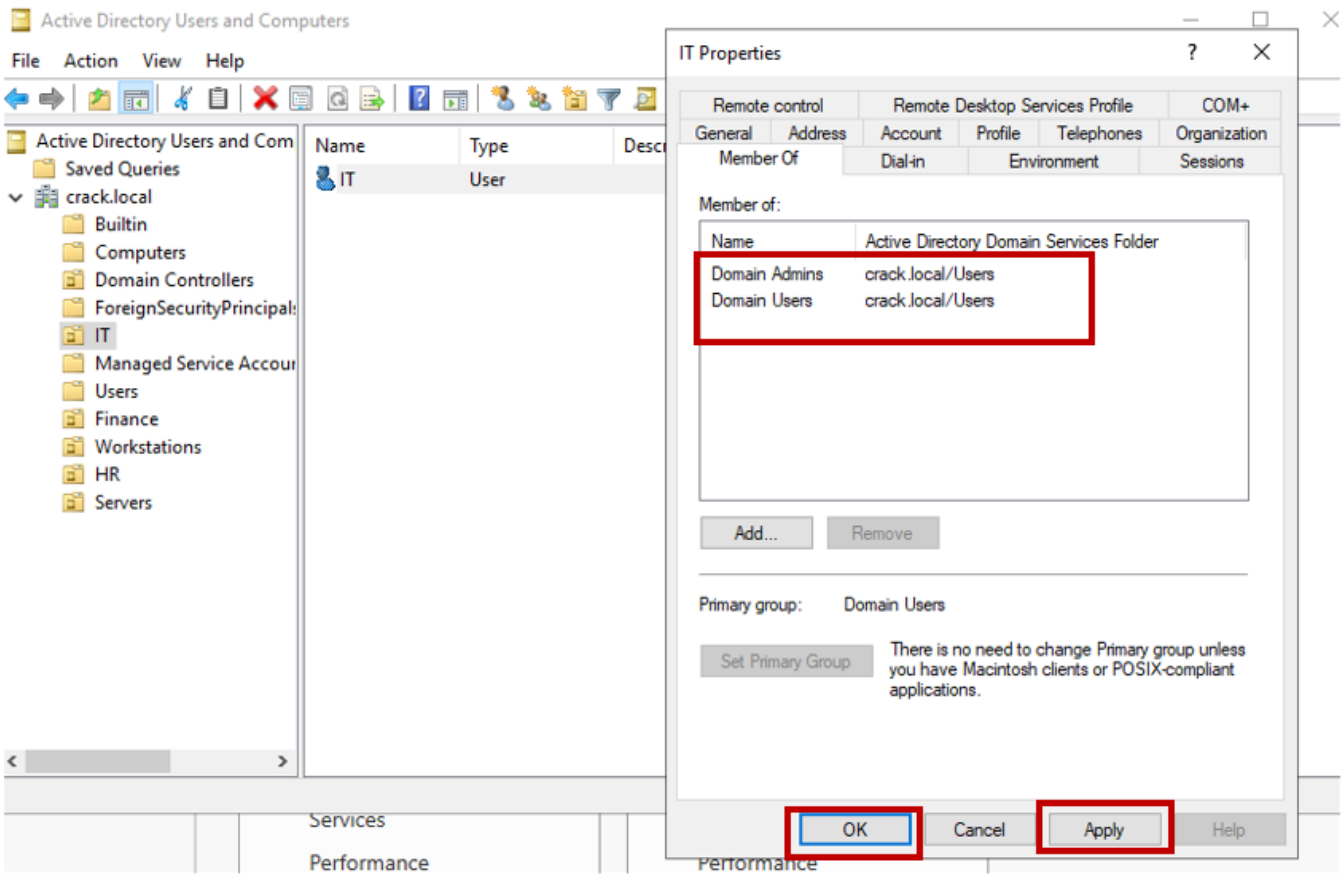
The IT administrator needs elevated permissions. Add it.admin to the Domain Admins group via the "Member Of" tab in the user's Properties.

1. Double-click on the IT user in Active Directory Users and Computers
2. Click the "Member Of" tab
3. Click "Add…"
4. Type "Domain Admins" and click "Check Names"
5. Click OK to confirm
6. Click Apply then OK





it.admin now belongs to both Domain Users and Domain Admins groups. This is role-based access control (RBAC) — IT administrators have elevated privileges while HR and Finance users have standard access.

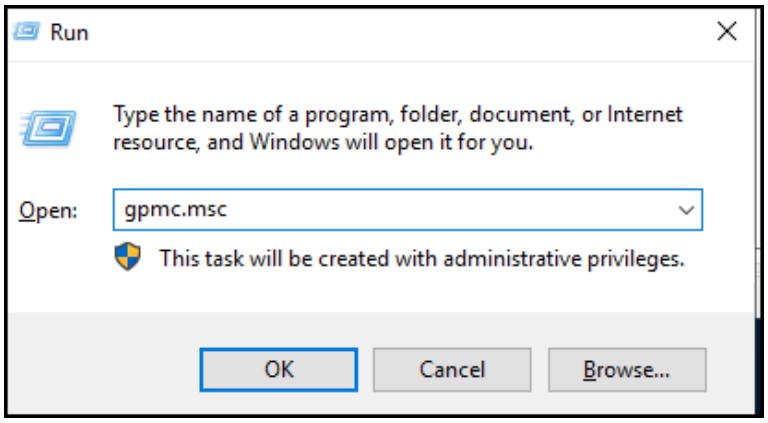


10. Group Policy Configuration

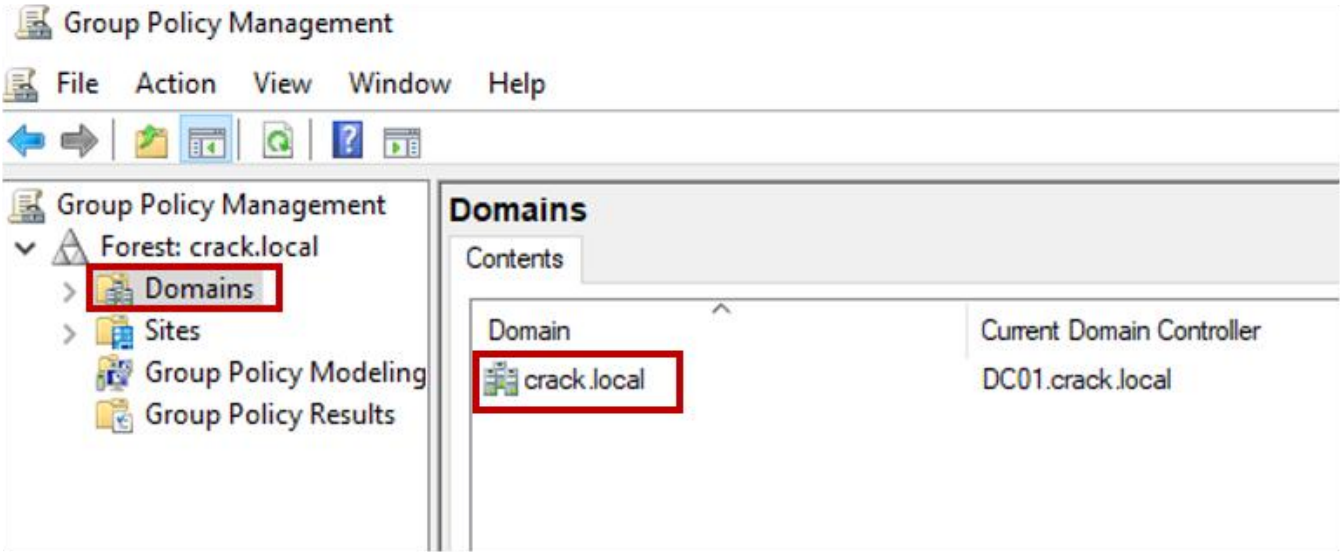
Group Policy Objects (GPOs) allow administrators to enforce security settings, password requirements, and account lockout policies across all domain computers and users.

10.1 Opening Group Policy Management Console

On the domain controller, open the Run dialog (Windows + R), type gpmc.msc and press Enter.

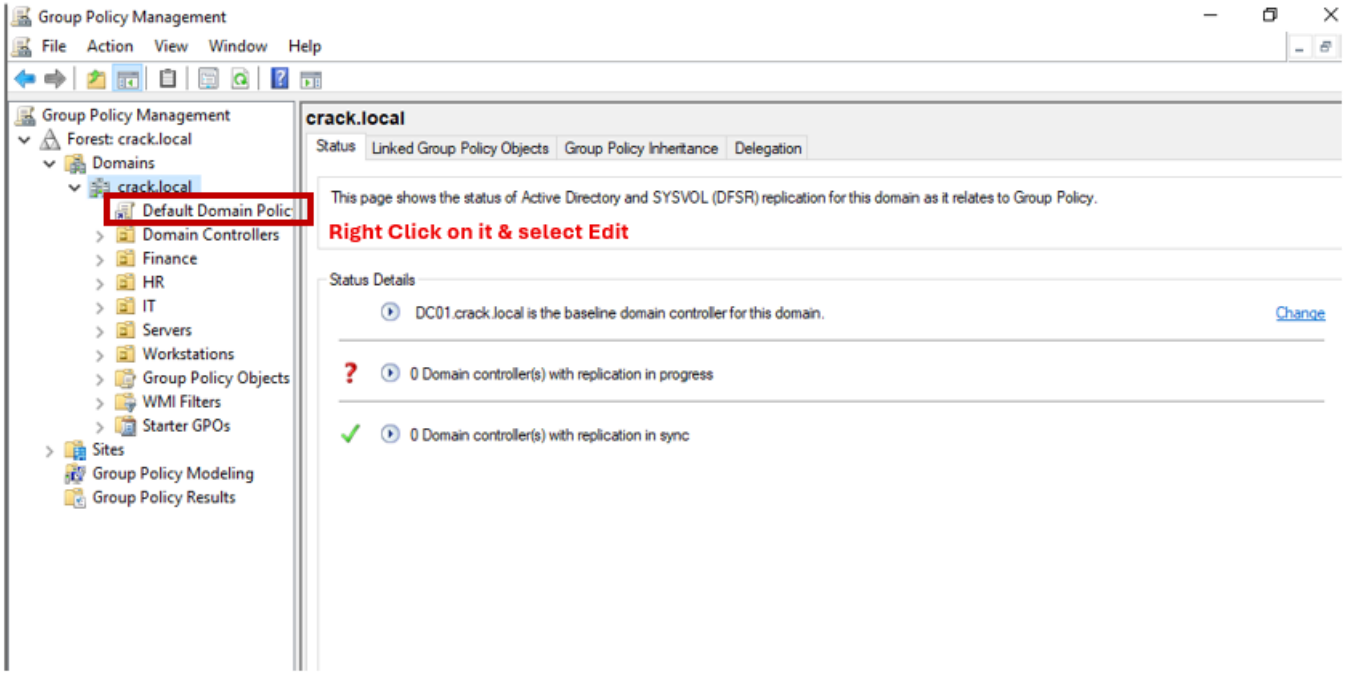


The Group Policy Management console opens, showing Forest: crack.local > Domains > crack.local. The Default Domain Policy is visible here.

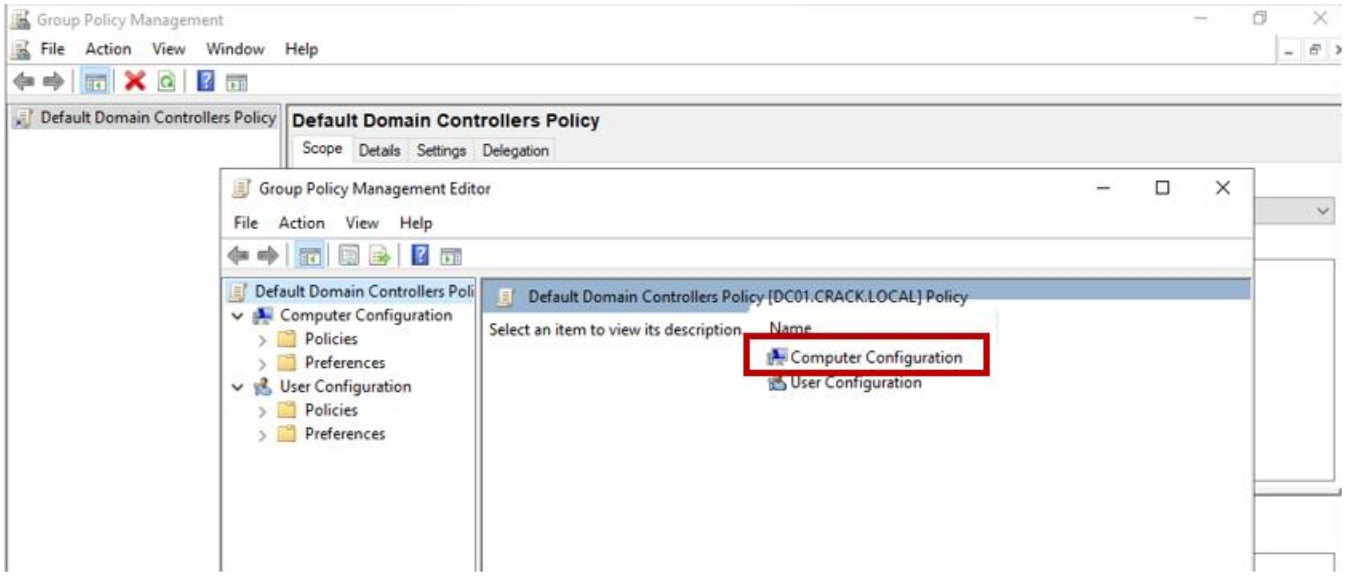


10.2 Editing Default Domain Policy

Right-click on "Default Domain Policy" under crack.local and select "Edit". The Group Policy Management Editor opens.

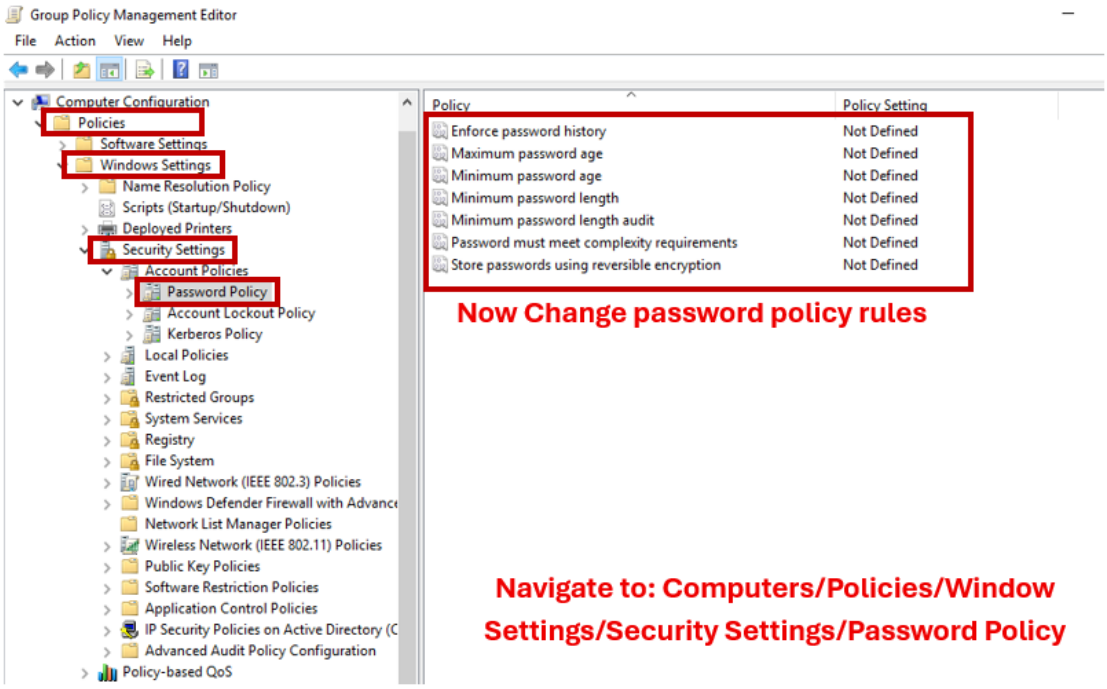


Navigate to Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy.



10.3 Configuring Password Policy

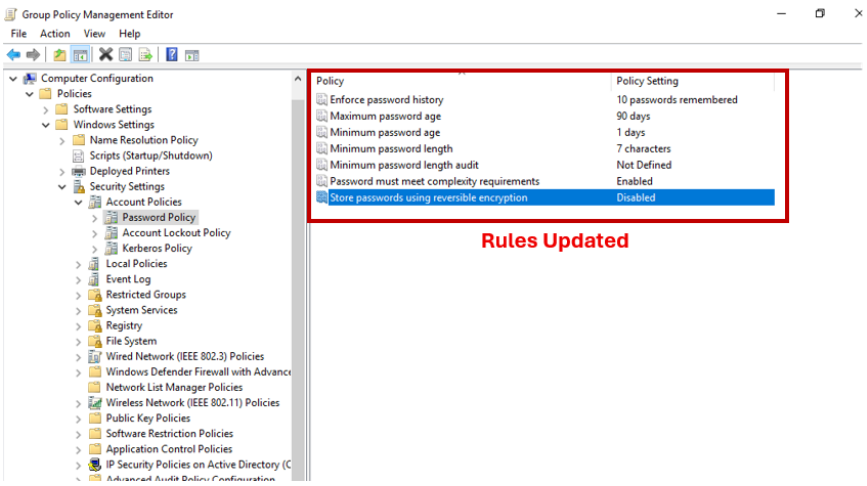
Navigate to: Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy



Password Policy Settings:

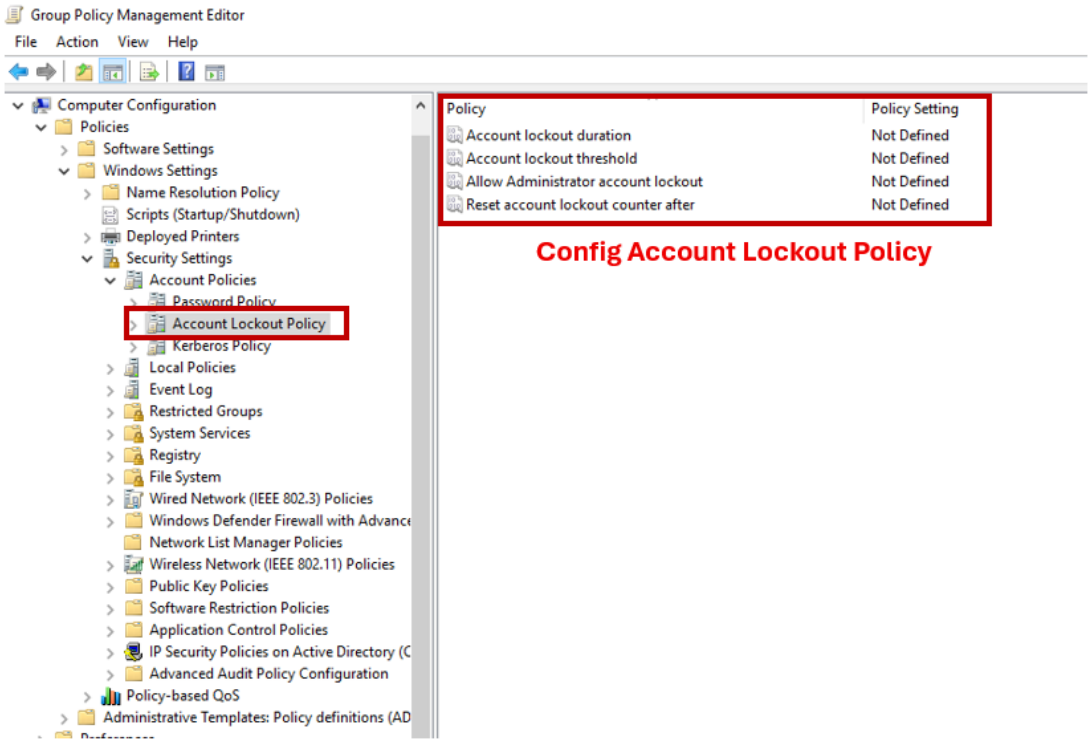
- Enforce password history: 10 passwords
- Maximum password age: 90 days
- Minimum password age: 1 day
- Minimum password length: 7 characters
- Password must meet complexity requirements: Enabled
- Store passwords using reversible encryption: Disabled

Double-click each policy item to configure it. After all settings are applied, the Rules Updated label confirms the changes.



10.4 Configuring Account Lockout Policy

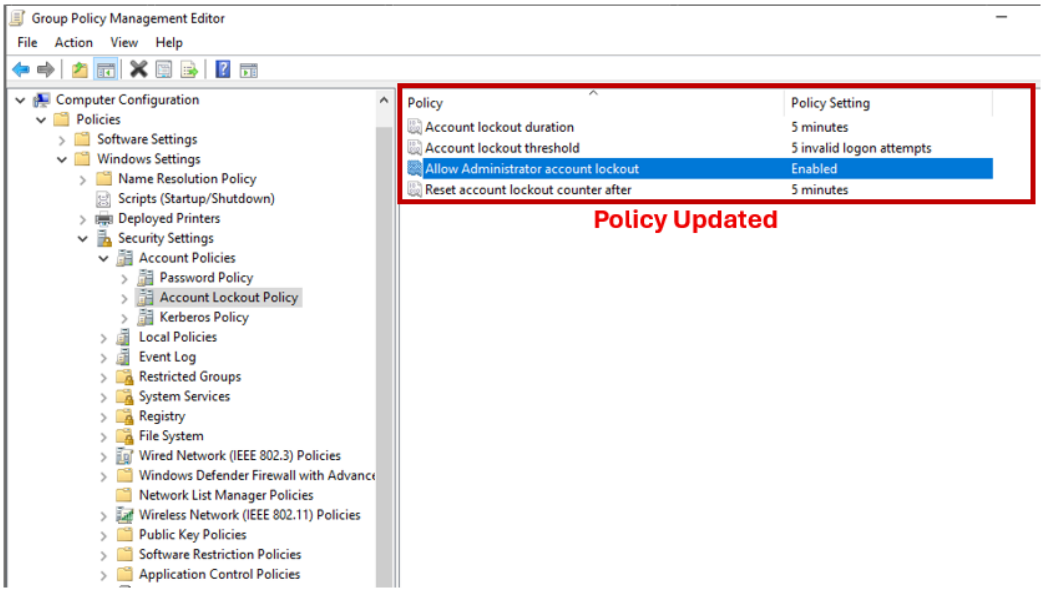
Navigate to Account Lockout Policy in the same Security Settings tree.



Account Lockout Policy Settings:

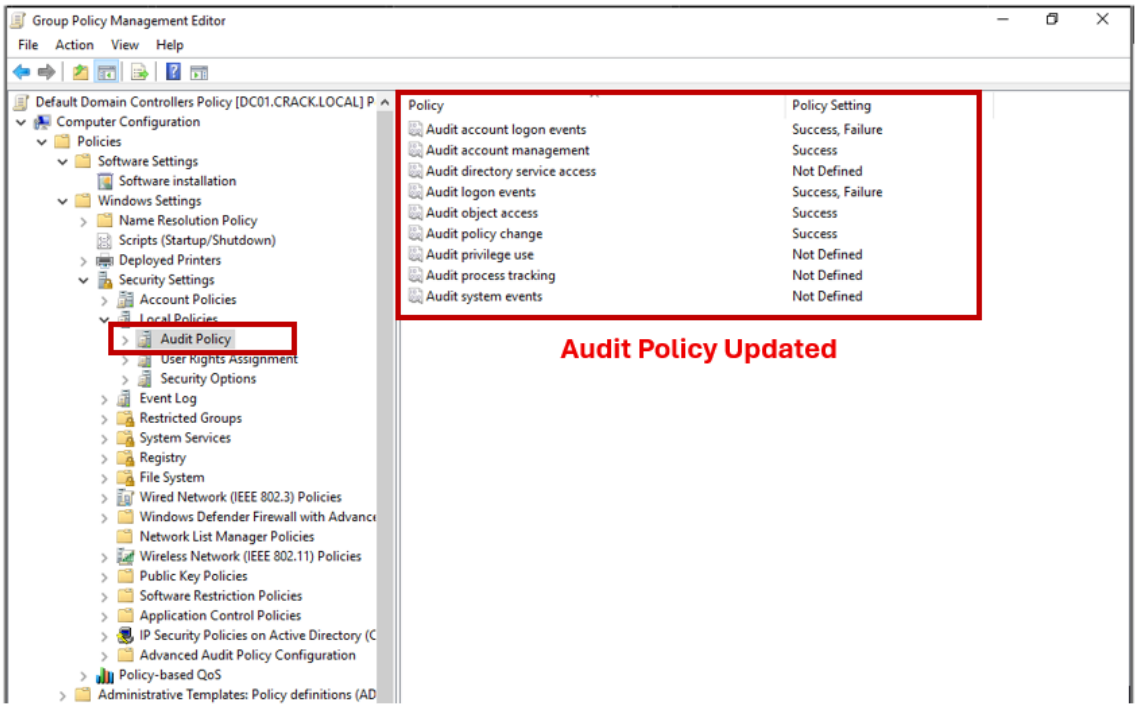
- Account lockout duration: 5 minutes
- Account lockout threshold: 5 invalid logon attempts
- Allow Administrator account lockout: Enabled
- Reset account lockout counter after: 5 minutes

After saving, the Policy Updated status confirms the lockout policy is active.



10.5 Configuring Audit Policies

Navigate to: Computer Configuration > Policies > Windows Settings > Security Settings > Local Policies > Audit Policy. Enable auditing for logon events to track login successes and failures.



Audit Policy Settings:

- Audit account logon events: Success, Failure
- Audit account management: Success
- Audit logon events: Success, Failure
- Audit object access: Success
- Audit policy change: Success

10.6 Applying Group Policy Updates

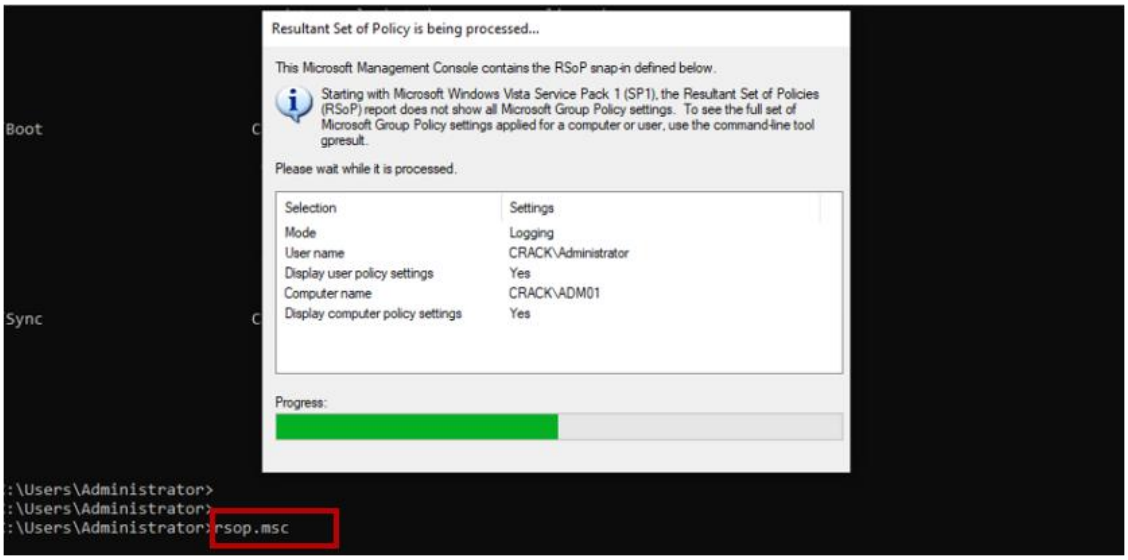
After making all Group Policy changes, force an immediate update on each machine. Run the following command in an elevated Command Prompt on the Server, ADM01, and CLI01:

Command: gpupdate /force

Also run rsop.msc (Resultant Set of Policy) to verify the applied policies. The rsop.msc console shows the effective policy settings applied to the computer and user.

```
C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```



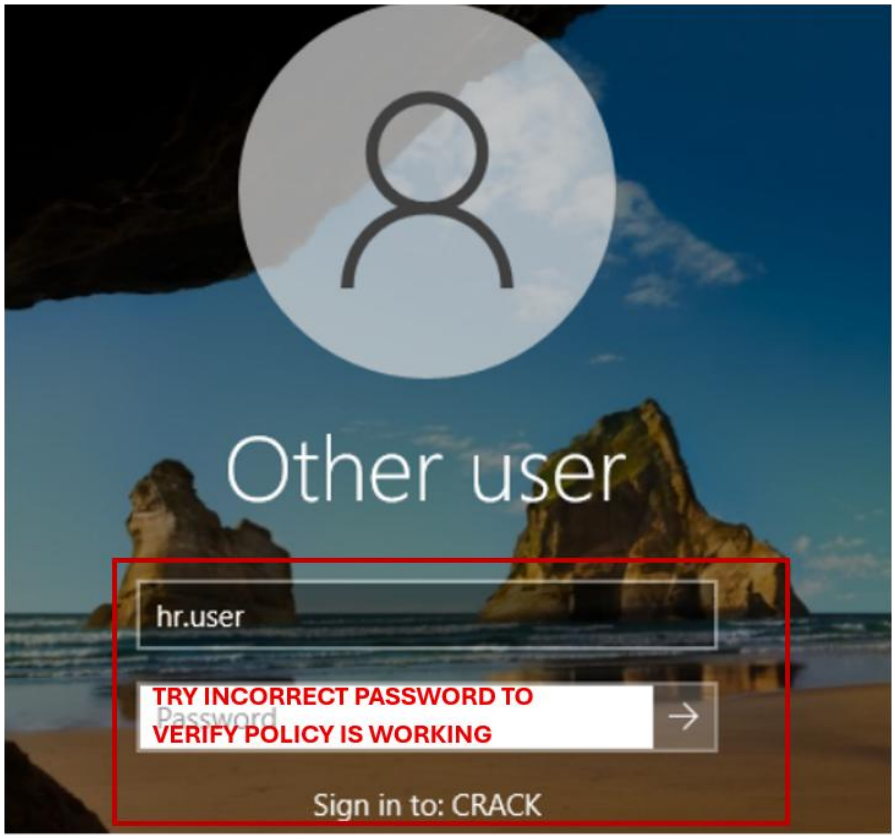
10.7 Verifying Policy is Applied (rsop.msc)

The Resultant Set of Policy confirms the Account Lockout Policy settings are in effect: 5 minutes duration, 5 invalid logon attempts threshold, 5 minutes reset counter — all sourced from Default Domain Policy.

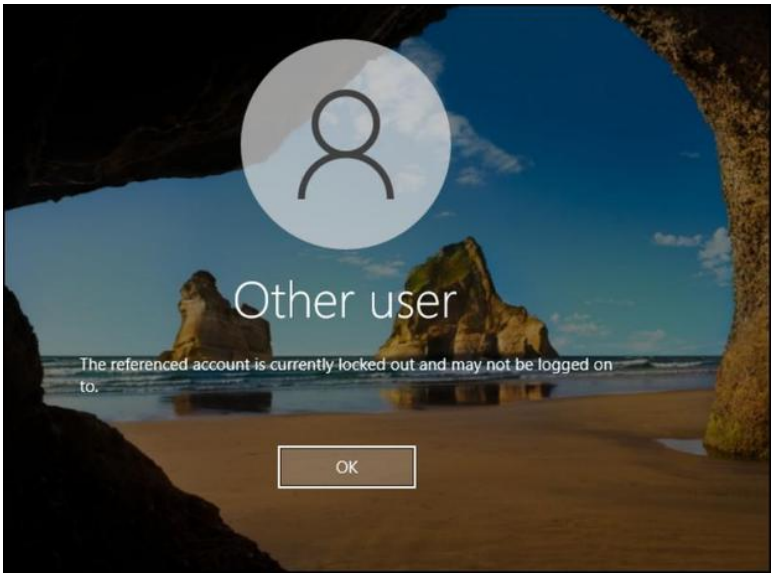
Policy	Computer Setting	Source GPO
Account lockout duration	5 minutes	Default Domai...
Account lockout threshold	5 invalid logon attempts	Default Domai...
Allow Administrator account lockout	Not Defined	
Reset account lockout counter after	5 minutes	Default Domai...

10.8 Testing Account Lockout

To verify the account lockout policy works, attempt to log in as hr.user on a domain client (CLI01 or ADM01) with an incorrect password 5 or more times.

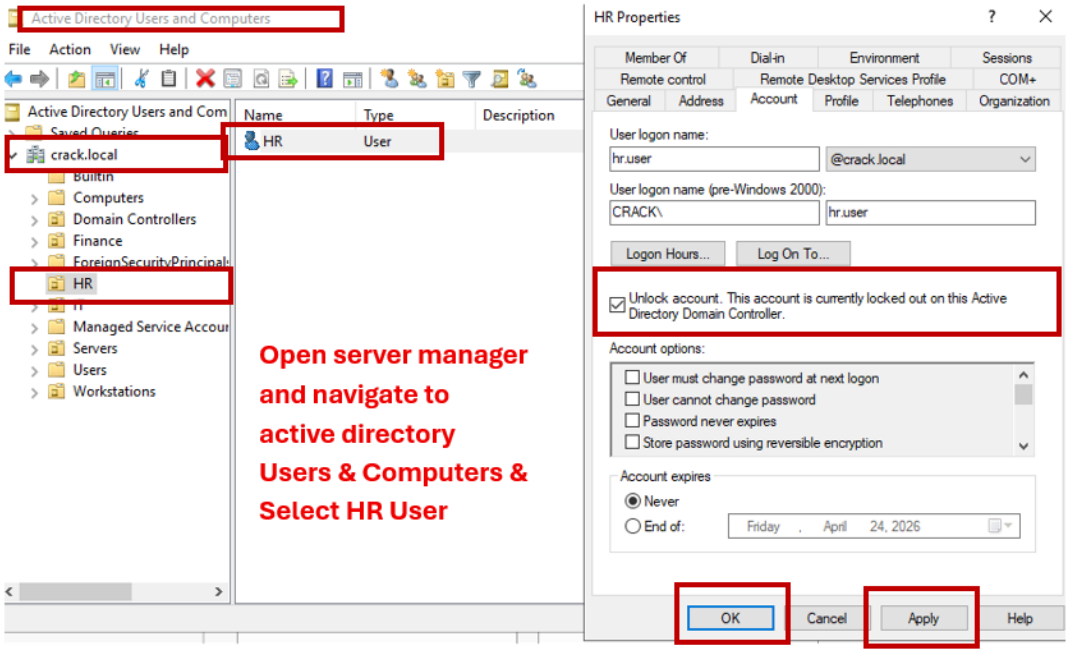


After 5 failed attempts, the account becomes locked. The error message "The referenced account is currently locked out and may not be logged on" appears.



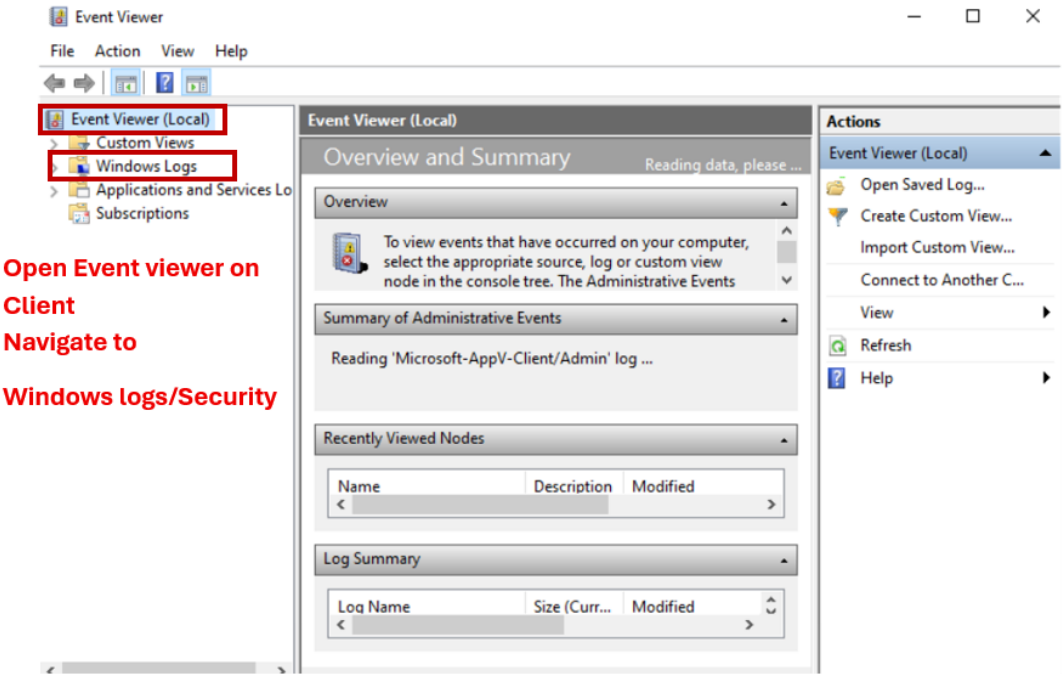
10.9 Unlocking the HR User Account

On the domain controller, open Active Directory Users and Computers. Navigate to the HR OU, double-click the HR user, and go to the Account tab. Check "Unlock account" and click OK/Apply.



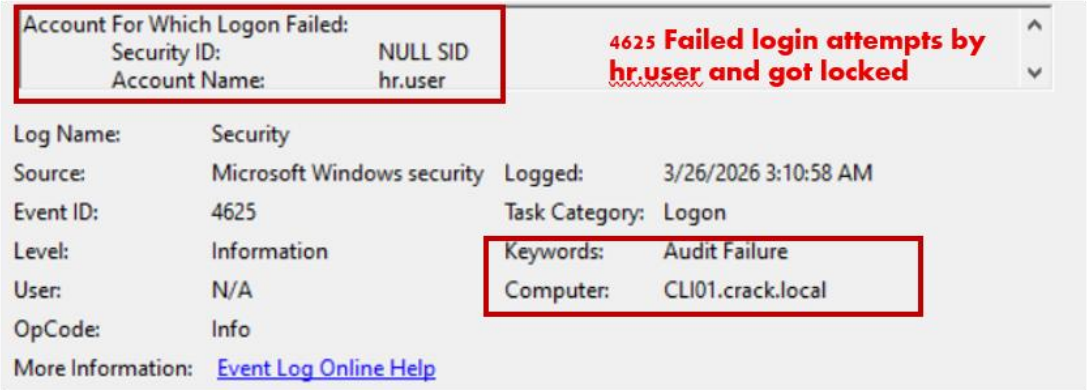
10.10 Reviewing Security Events in Event Viewer

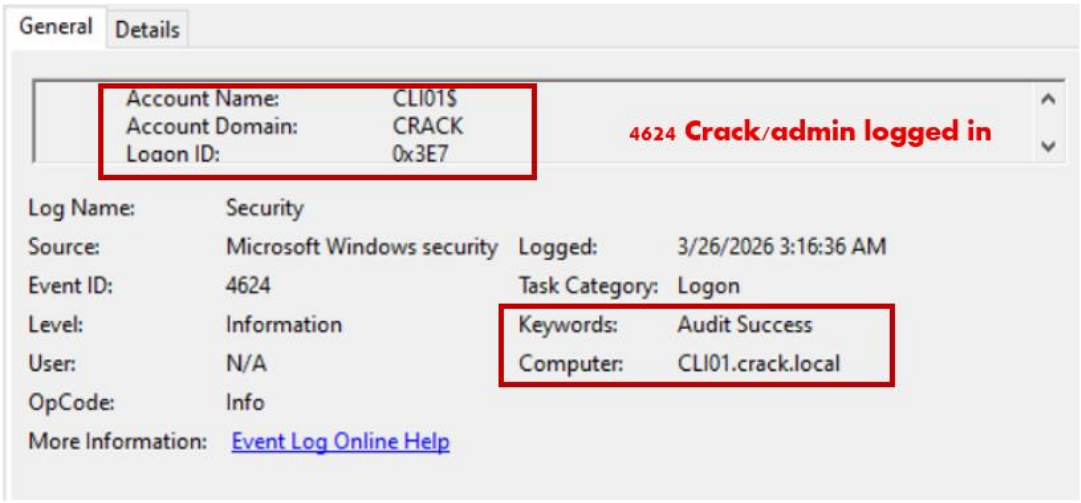
On the client machine, open Event Viewer and navigate to Windows Logs > Security. Review the audit events generated by the login attempts.



Key Event IDs:

- Event ID 4625 – Failed logon: Shows hr.user with Audit Failure keyword, Computer CLI01.crack.local
- Event ID 4624 – Successful logon: Shows CRACK/admin successful logon with Audit Success keyword



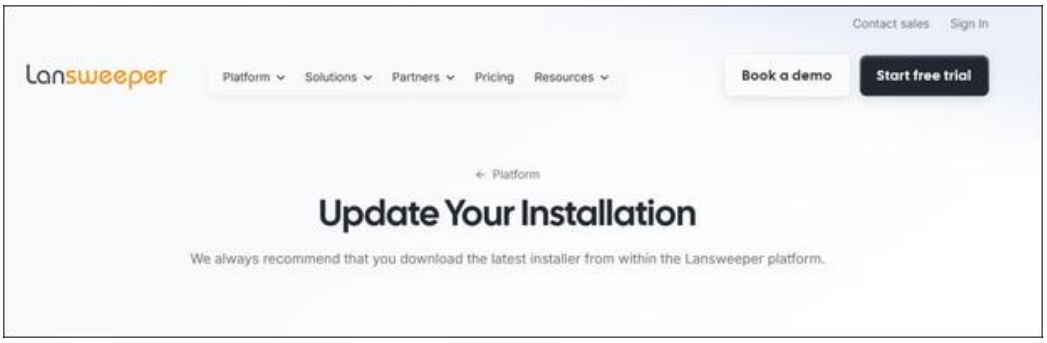


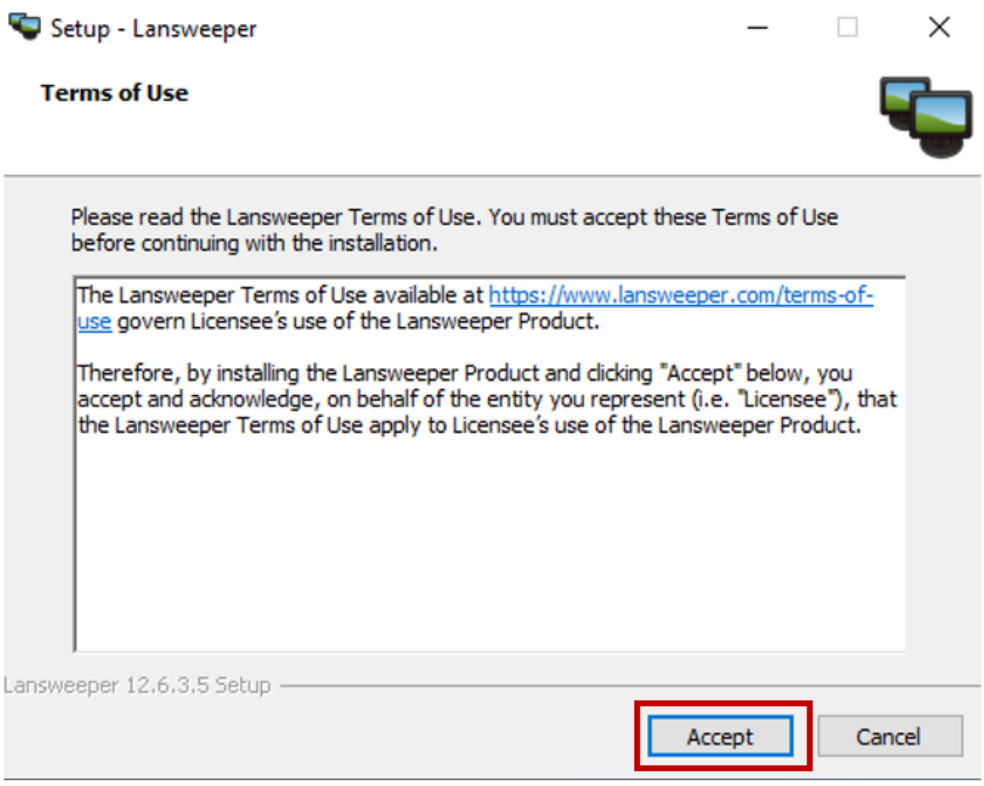
11. Lansweeper – Network Asset Management

Lansweeper is a network discovery and IT asset management platform that automatically scans your network and inventories all connected devices — computers, servers, network equipment, and more. It integrates with Active Directory to scan domain-joined machines.

11.1 Downloading Lansweeper

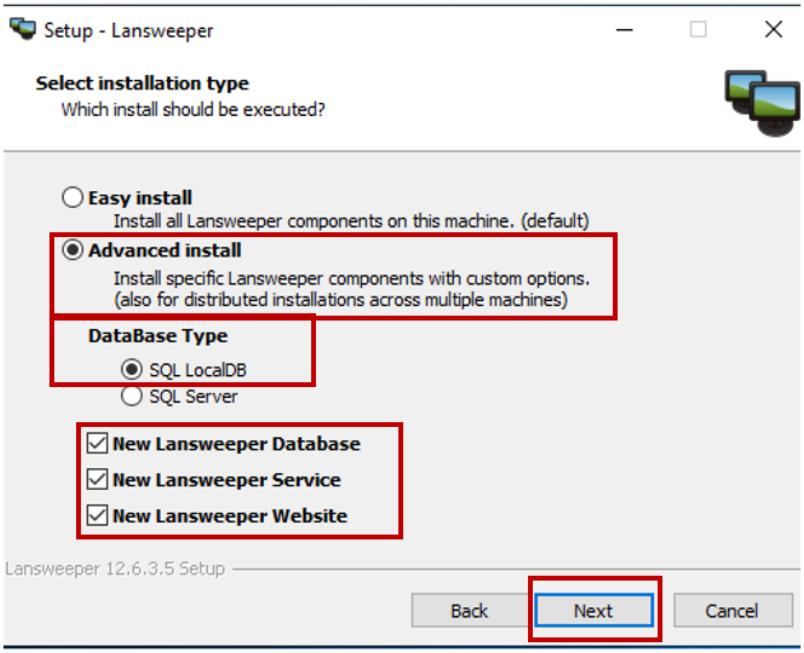
Open a browser on the domain controller and navigate to lansweeper.com. Go to Platform > Update Your Installation to download the latest installer. Run the setup file and accept the Terms of Use.





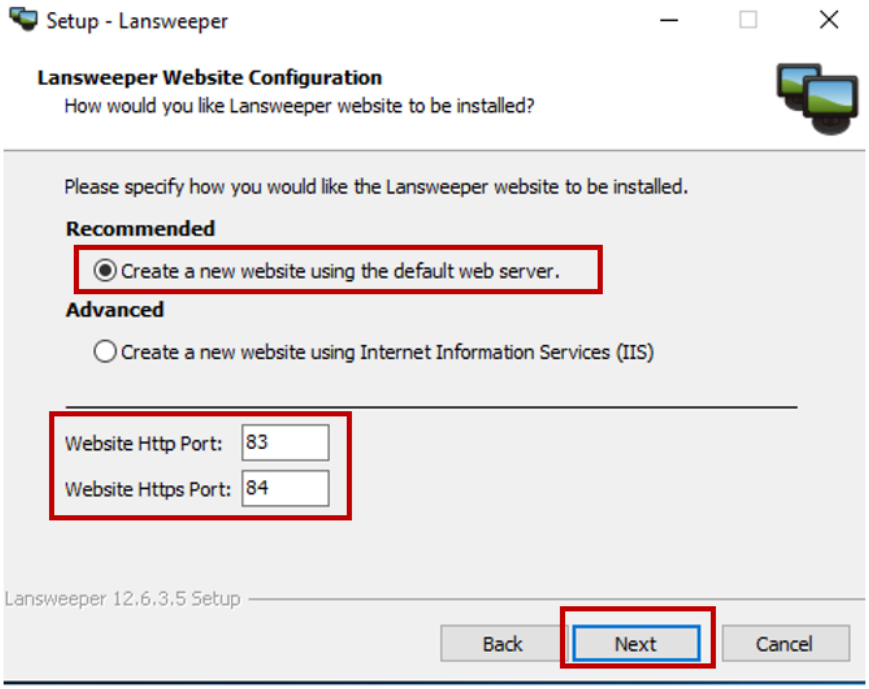
11.2 Installation Type and Database

On the installation type screen, select "Advanced install". Choose SQL LocalDB as the database type (recommended for single-server deployments). Check all three options: New Lansweeper Database, New Lansweeper Service, and New Lansweeper Website.



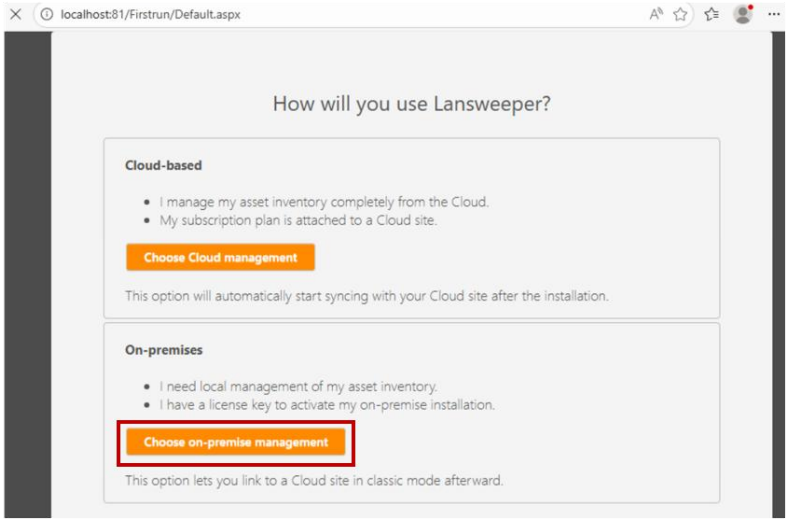
Website Configuration:

- Select: Create a new website using the default web server
- Website HTTP Port: 83
- Website HTTPS Port: 84



11.3 Initial Lansweeper Setup – Management Type

After installation, a browser opens automatically. Select "On-premises" management since we are running a local Active Directory environment.

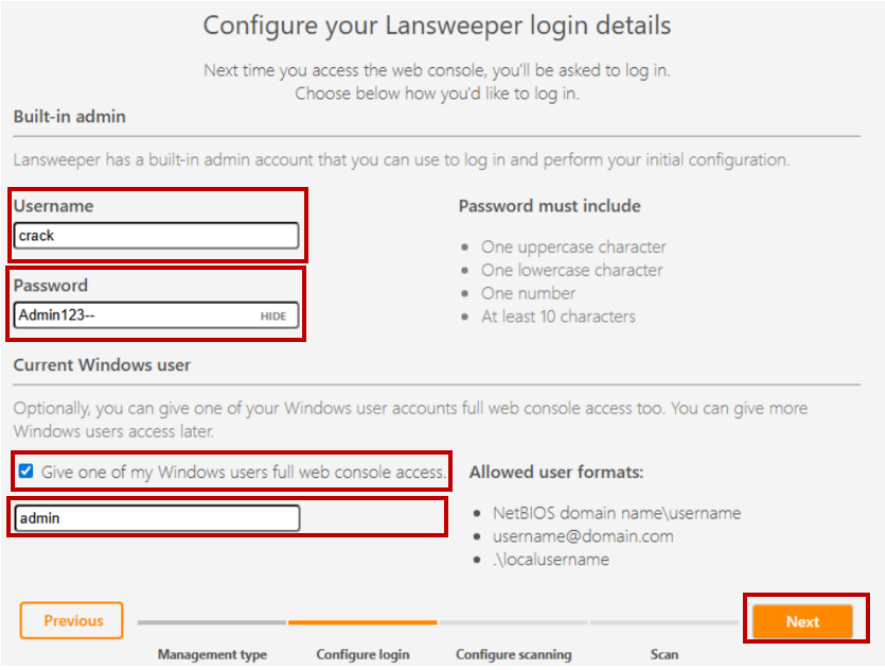


11.4 Configuring Lansweeper Login

Set up the built-in Lansweeper administrator account and optionally give Windows user access.

Built-in Admin Credentials:

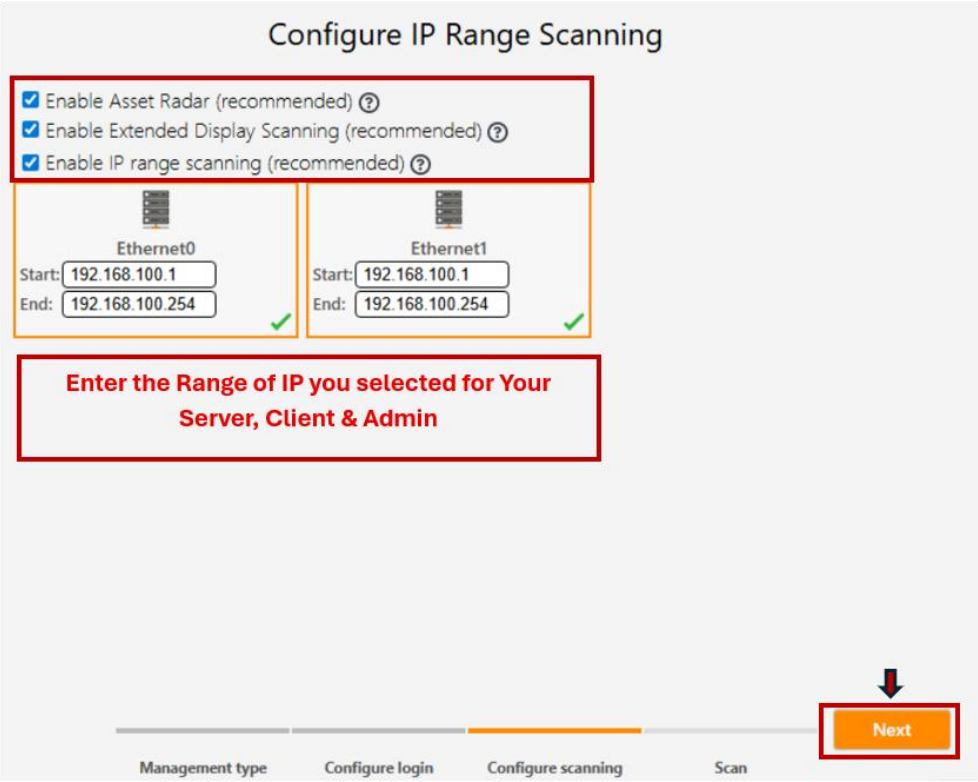
- Username: crack
- Password: Admin123-- (minimum 10 characters required)
- Check: Give one of my Windows users full web console access
- Windows user: admin



11.5 Configure IP Range Scanning

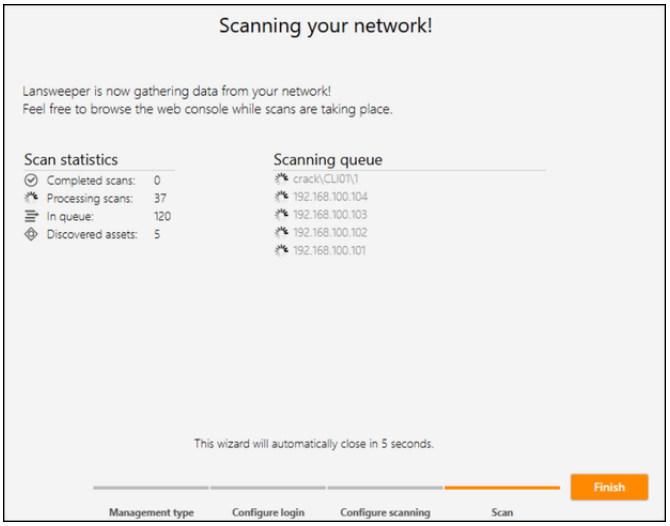
Configure the IP ranges Lansweeper will scan. Enable all scanning options and enter the range covering your server, client, and admin VMs.

- Enable Asset Radar: ✓
- Enable Extended Display Scanning: ✓
- Enable IP range scanning: ✓
- Ethernet0 – Start: 192.168.100.1 / End: 192.168.100.254
- Ethernet1 – Start: 192.168.100.1 / End: 192.168.100.254



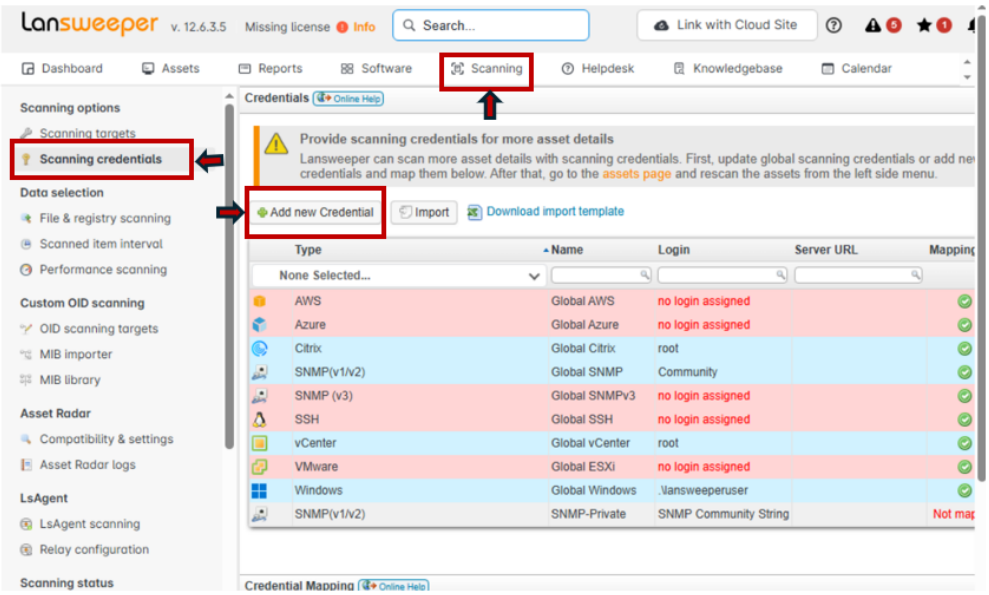
11.6 Network Scanning in Progress

Lansweeper immediately begins scanning. The Scan statistics panel shows completed scans, processing scans, queue size, and discovered assets. Click Finish when the initial scan completes.



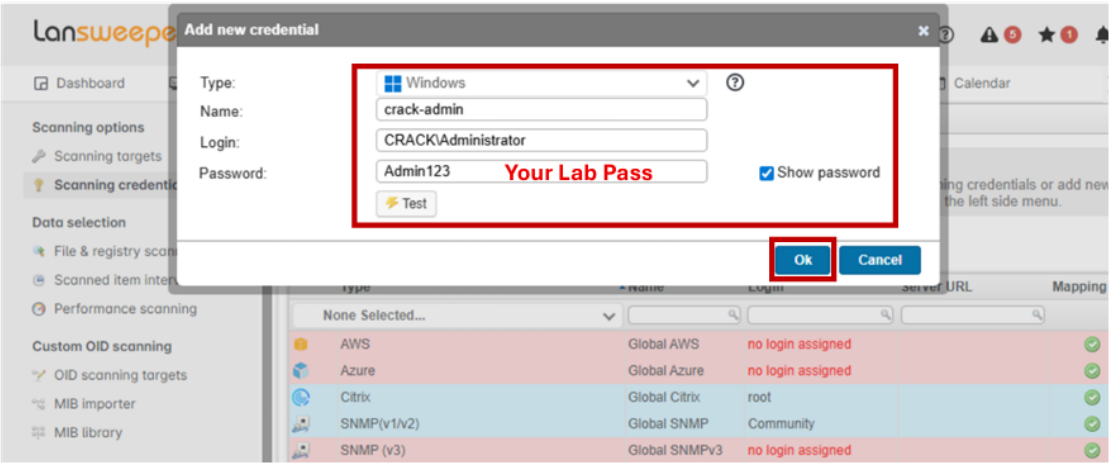
11.7 Adding Windows Scanning Credentials

Navigate to Scanning > Scanning Credentials to provide domain admin credentials. This allows Lansweeper to pull detailed hardware and software inventory from Windows machines.



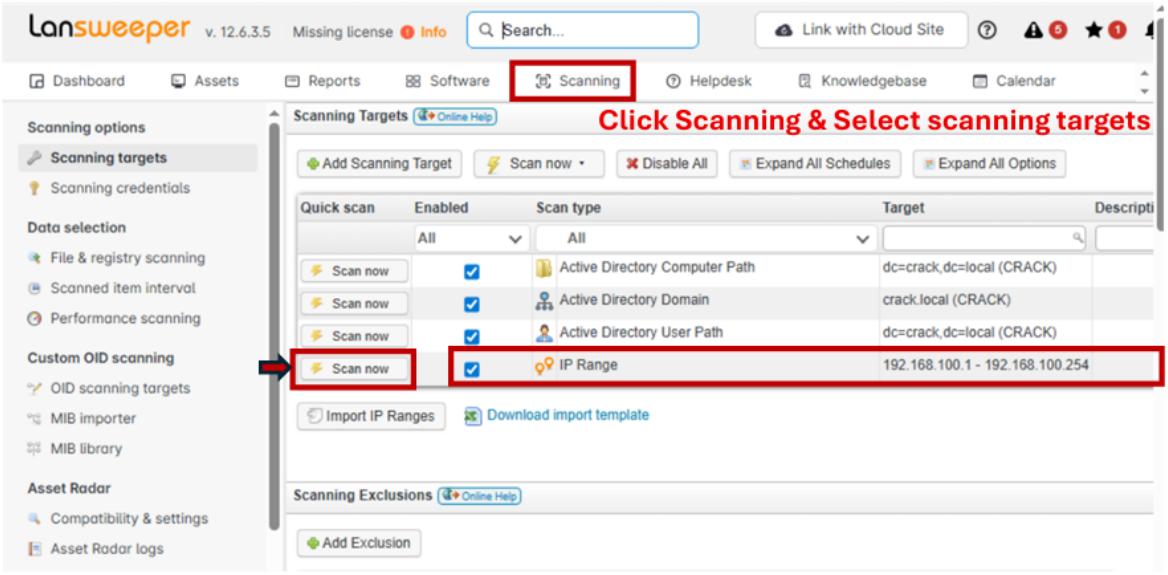
Add New Credential:

- Type: Windows
- Name: crack-admin
- Login: CRACK\Administrator
- Password: Admin123 (Your Lab Pass)



11.8 Scanning Targets and Results

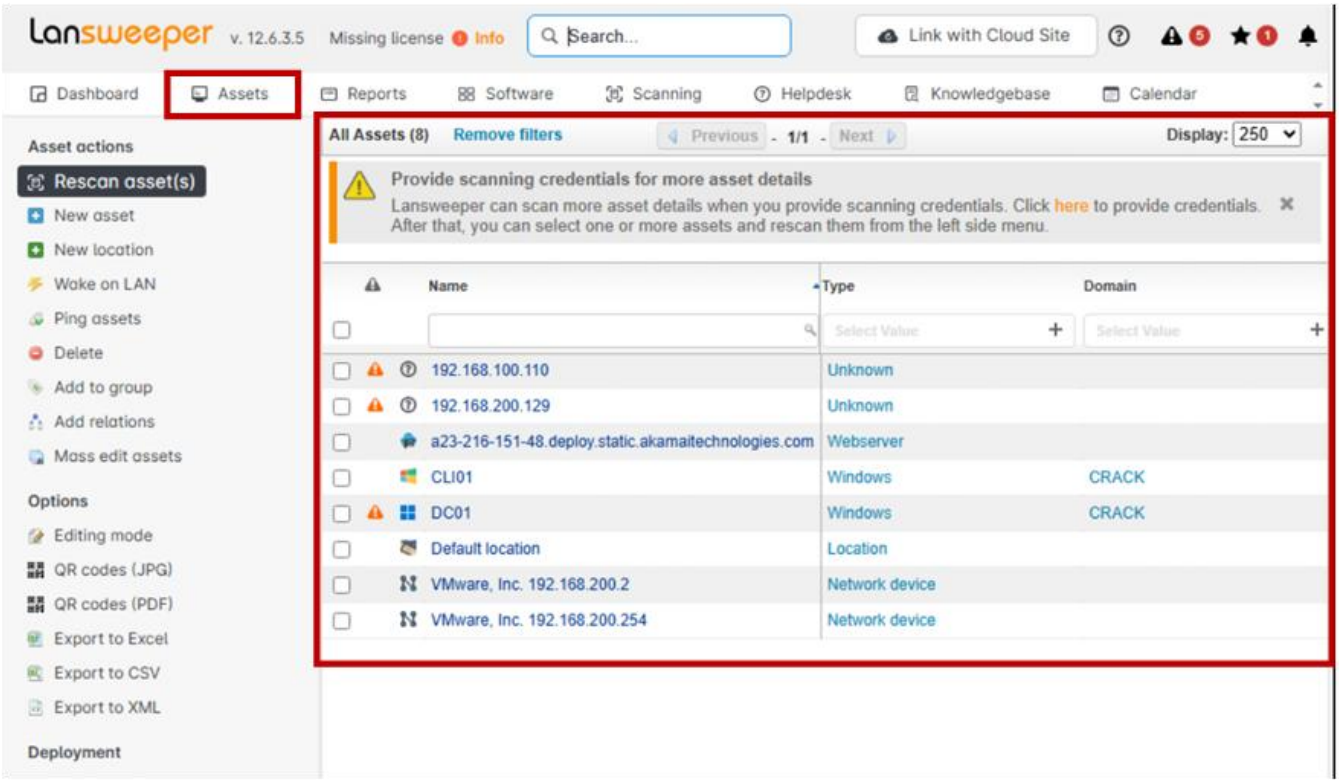
Navigate to Scanning > Scanning Targets. Click "Scan now" on the IP Range target to run a full scan. After scanning completes, review the scan statistics.





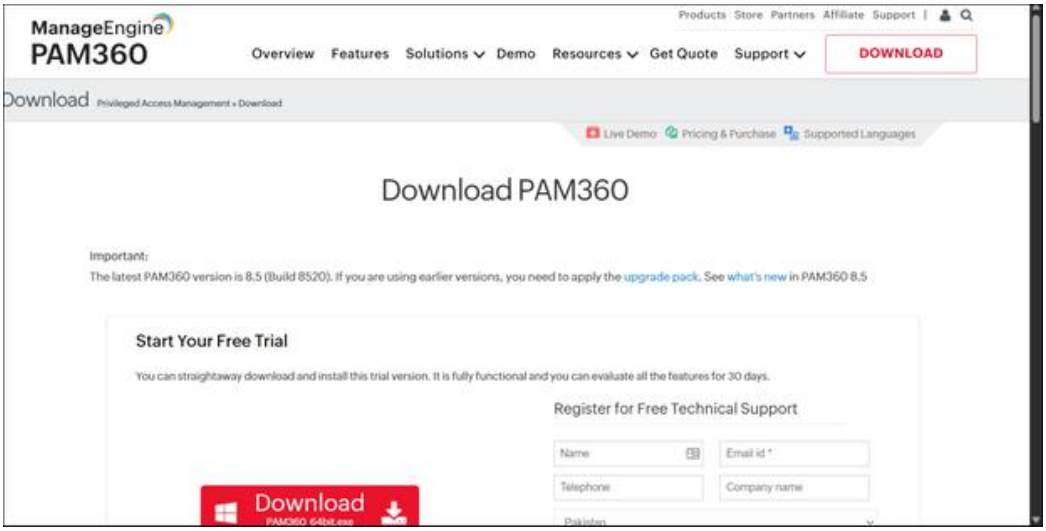
11.9 Viewing Discovered Assets

Navigate to Assets. Lansweeper displays all 8 discovered assets including CLI01 and DC01 as Windows machines on the CRACK domain, plus IP devices, VMware network adapters, and a web server.



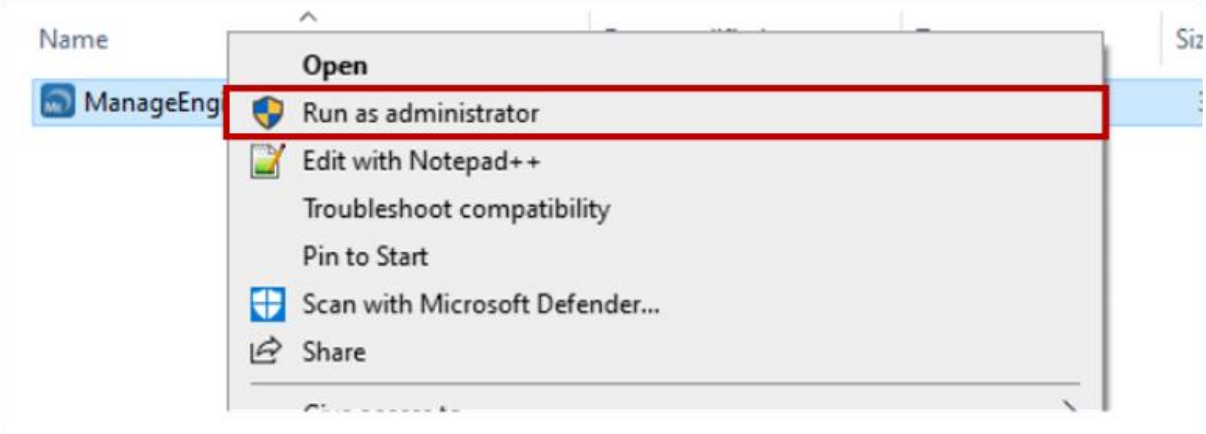
12. ManageEngine PAM360 – Privileged Access Management

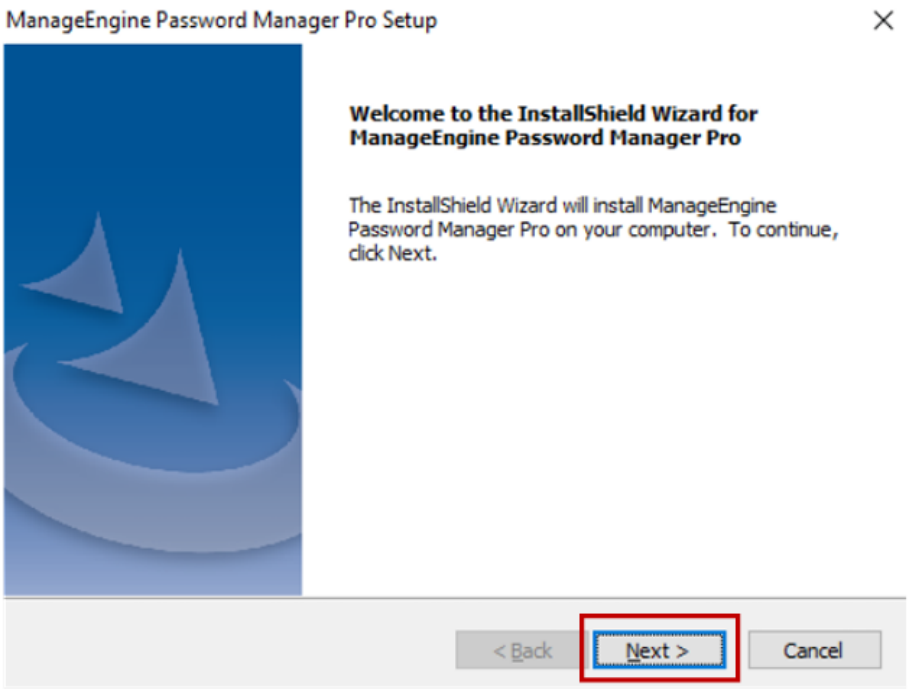
ManageEngine PAM360 (Password Manager Pro) is an enterprise privileged access management solution that centralizes the control, monitoring, and auditing of all privileged accounts in your Active Directory environment.



12.1 Downloading and Running PAM360

Download the PAM360 64-bit installer from the ManageEngine website. Run the installer as Administrator.



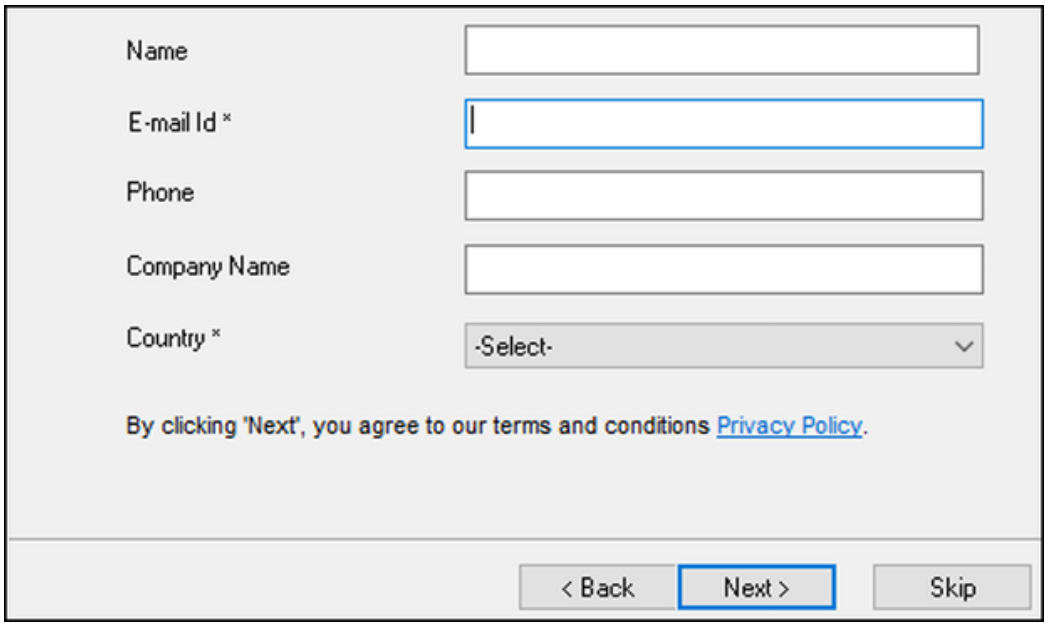


12.2 Installation Wizard

The ManageEngine Password Manager Pro InstallShield Wizard opens. Click Next to proceed through the installation steps.

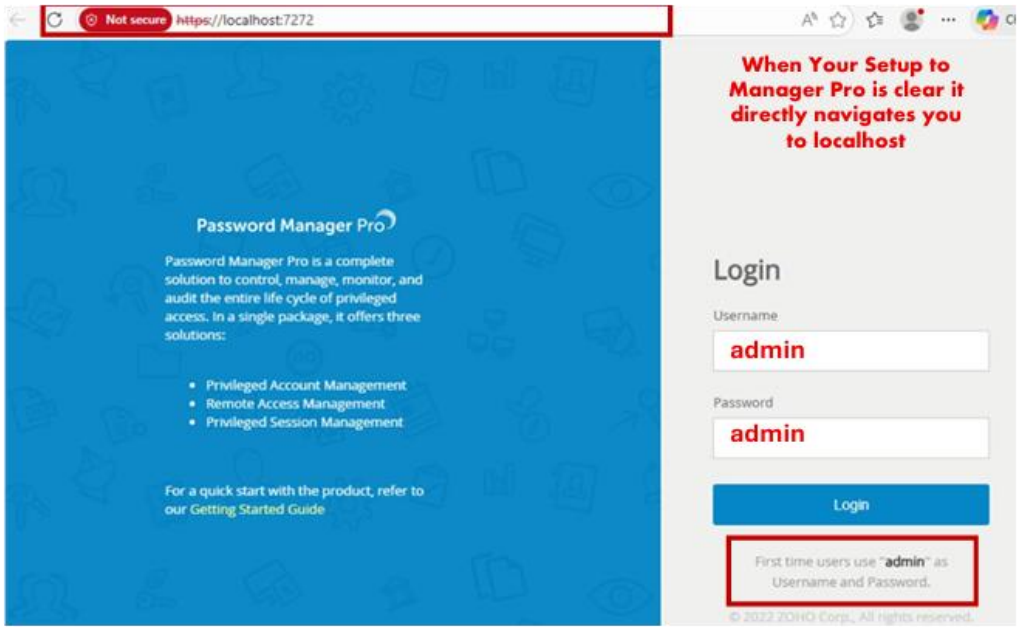
Installation Options:

- Installation Type: Standard Installation
- Database: Bundled PostgreSQL (default)
- Port: 7272



12.3 First Login and Password Reset

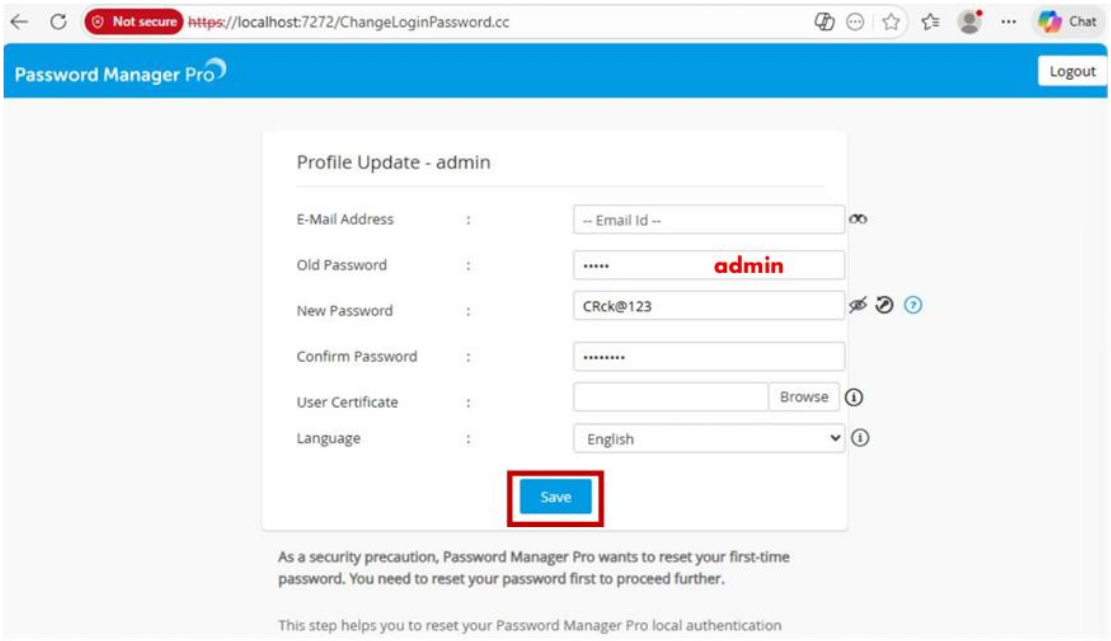
After installation, PAM360 automatically opens in your browser at https://localhost:7272. Log in with the default credentials (admin/admin). PAM360 immediately prompts a password reset for security.



Password Reset:

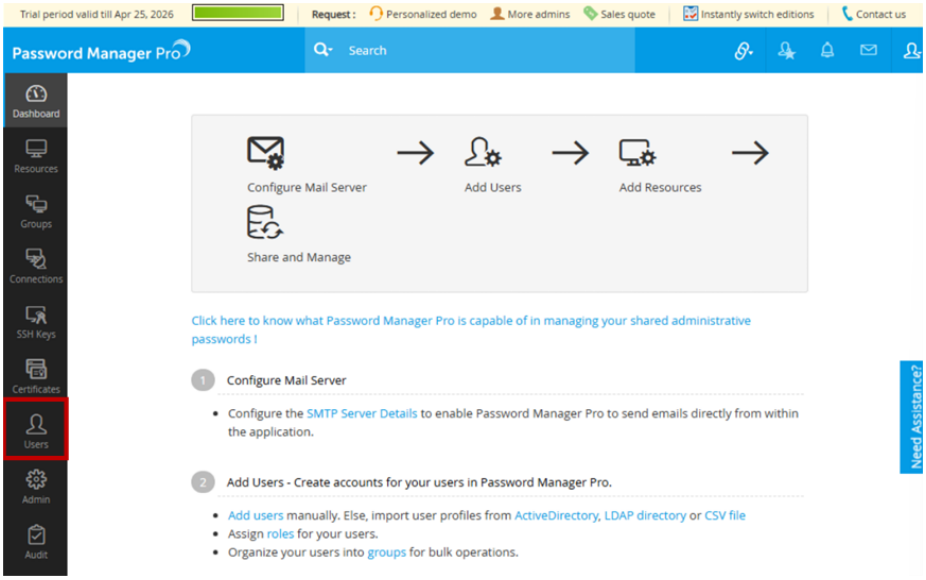
- Old Password: admin
- New Password: CRck@123
- Confirm Password: CRck@123

1. Click Save to proceed to the dashboard



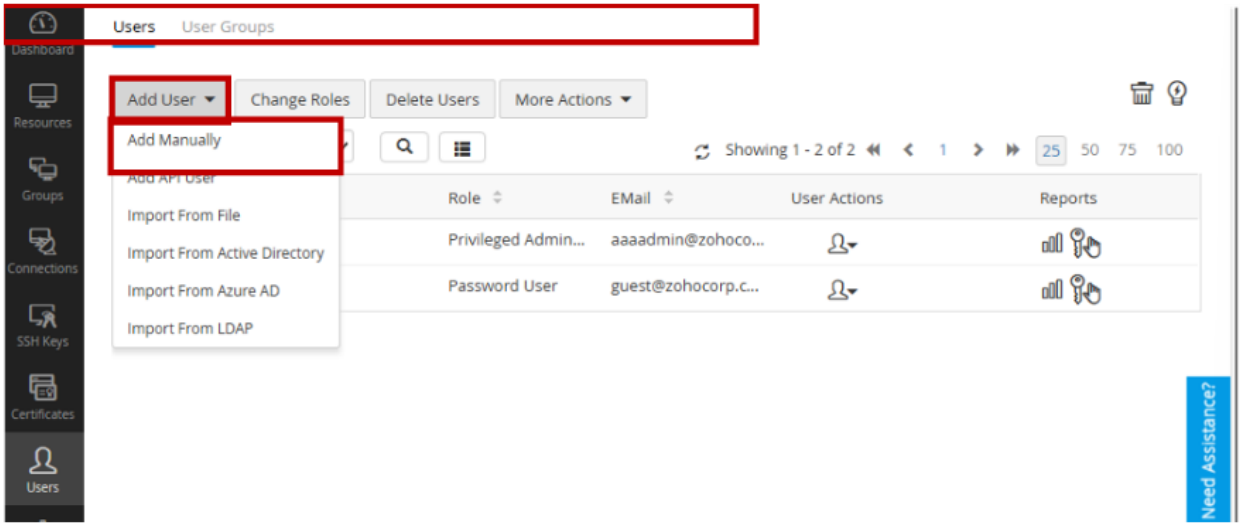
12.4 PAM360 Dashboard

The PAM360 dashboard shows the 4-step setup workflow: Configure Mail Server > Add Users > Add Resources > Share and Manage.



12.5 Adding Users to PAM360

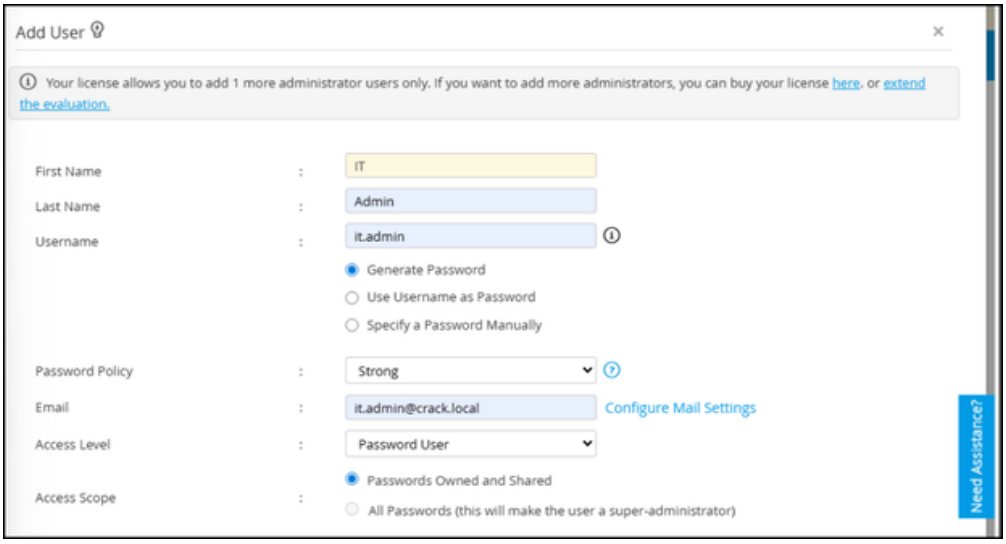
Click the Users icon in the left navigation. Click "Add User" > "Add Manually" to create a new PAM360 user.



New User – IT Admin:

- First Name: IT
- Last Name: Admin
- Username: it.admin
- Email: it.admin@crack.local

- Access Level: Password User
- Department: IT



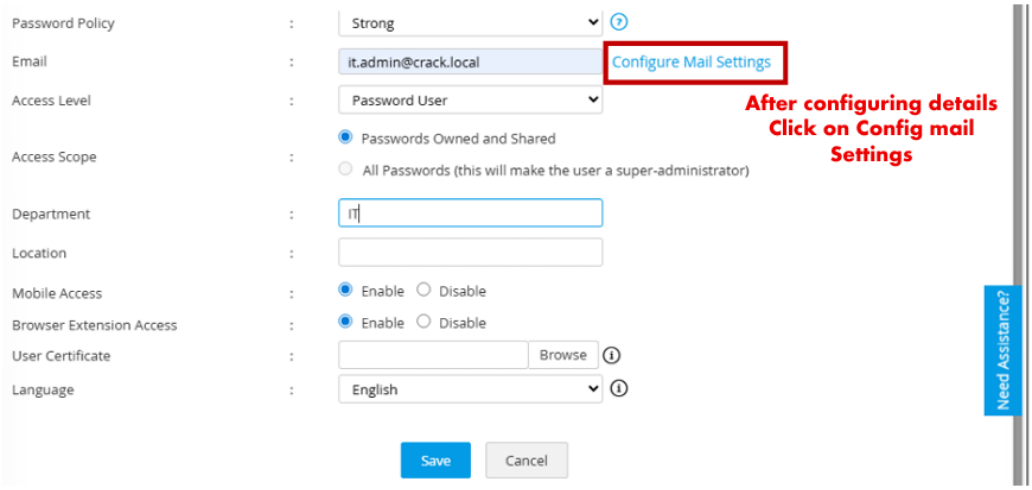
12.6 Configuring Mail Server

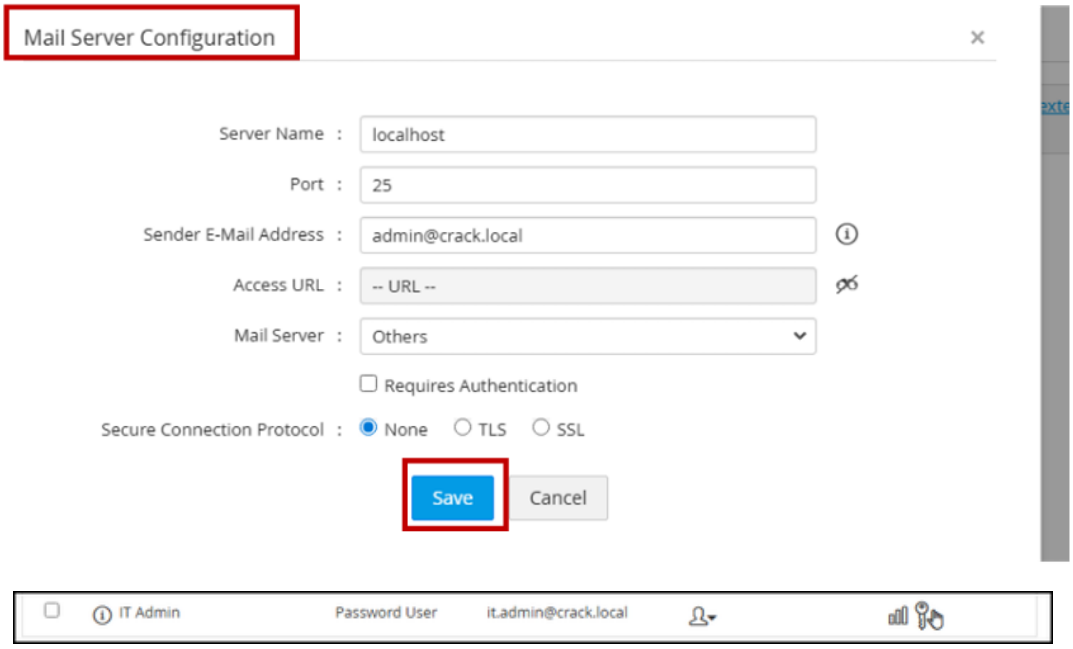
Click "Configure Mail Settings" in the Add User form. Set up the local mail server configuration.

Mail Server Settings:

- Server Name: localhost
- Port: 25
- Sender E-Mail: admin@crack.local
- Mail Server: Others
- Secure Connection: None

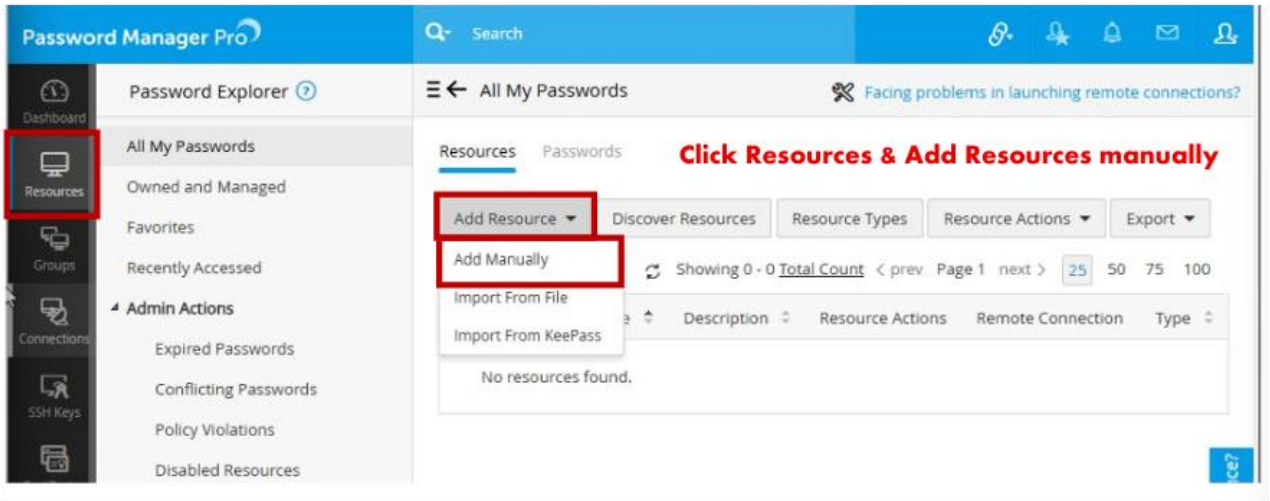
After saving mail settings, save the user. The IT Admin user now appears in the Users list.





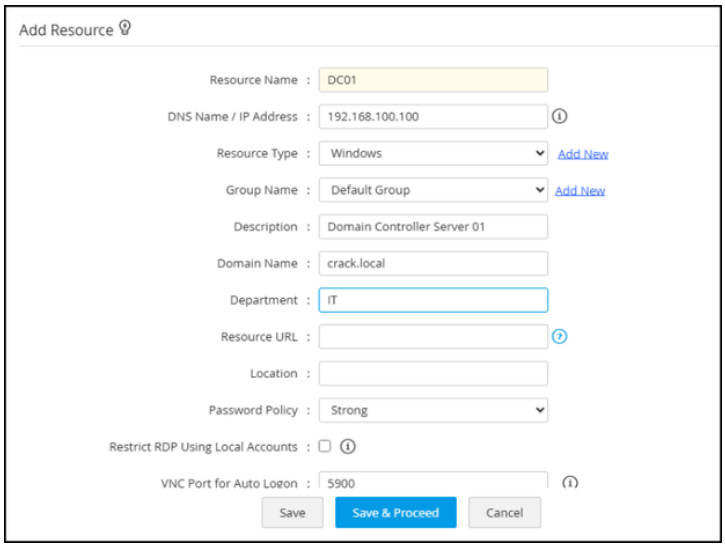
12.7 Adding Resources to PAM360

Click Resources in the left navigation. Click "Add Resource" > "Add Manually". Add the domain controller as the first managed resource.



Resource Details – DC01:

- Resource Name: DC01
- DNS Name / IP Address: 192.168.100.100
- Resource Type: Windows
- Description: Domain Controller Server 01
- Domain Name: crack.local
- Department: IT
- Password Policy: Strong



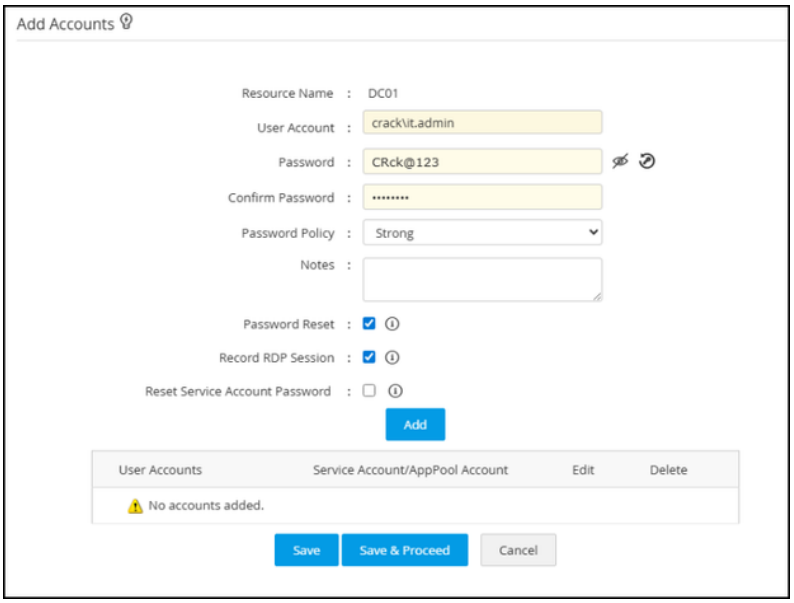
12.8 Adding Accounts to the Resource

After saving the resource, add the domain admin account credentials to manage for DC01.

Account Details:

- User Account: crack\it.admin
- Password: CRck@123
- Password Policy: Strong
- Password Reset: ✓ Checked
- Record RDP Session: ✓ Checked

Click Add, then Save & Proceed. DC01 now appears in the Resources list ready for privileged access management.





13. SpyAgent – Employee Activity Monitoring

SpyAgent by Spytech is a comprehensive computer monitoring and activity recording tool used by businesses to monitor employee activity. It captures keystrokes, websites visited, programs used, screenshots, files accessed, and more.

⚠ Note: SpyAgent should only be deployed in compliance with applicable laws and your organization's disclosure policies.

13.1 Installing SpyAgent

Run the Spytech SpyAgent installer. Click Next through the Welcome screen and complete the installation.



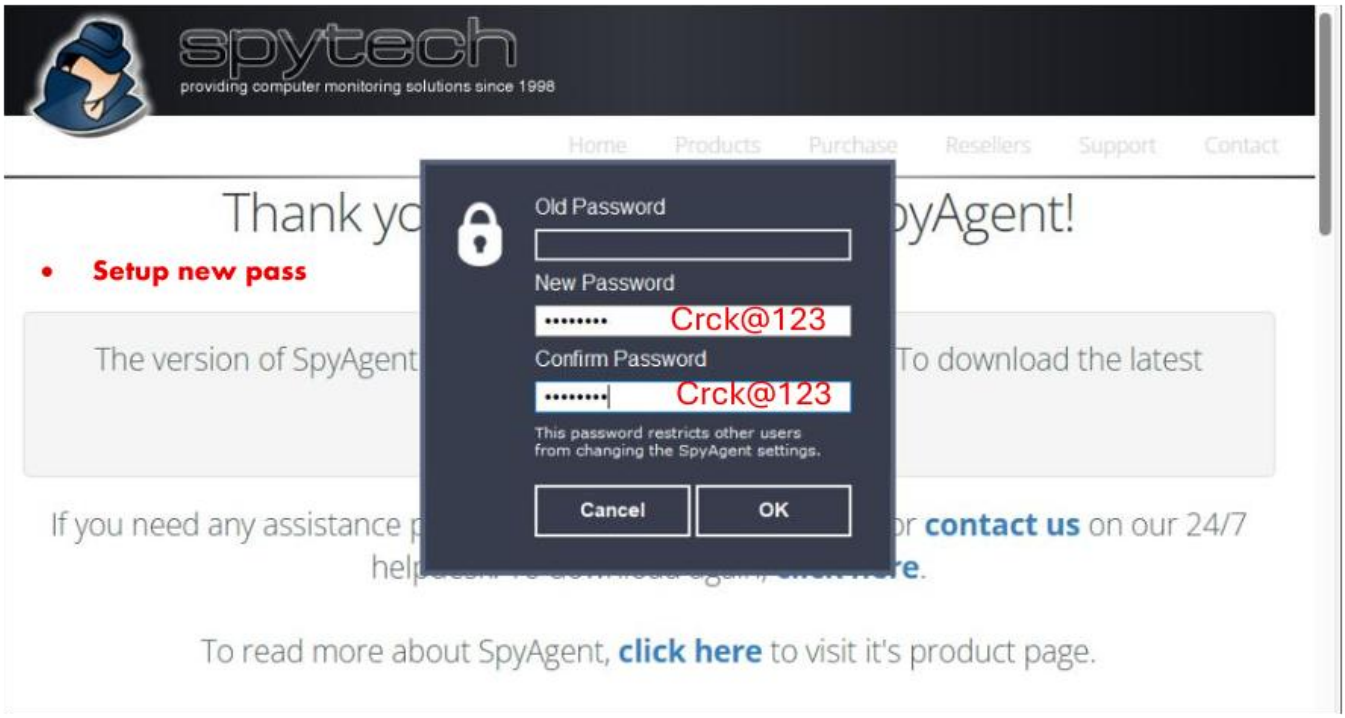
13.2 Setting the Access Password

On first launch, SpyAgent opens a browser page and immediately prompts you to set an access password to protect its settings.

Password Setup:

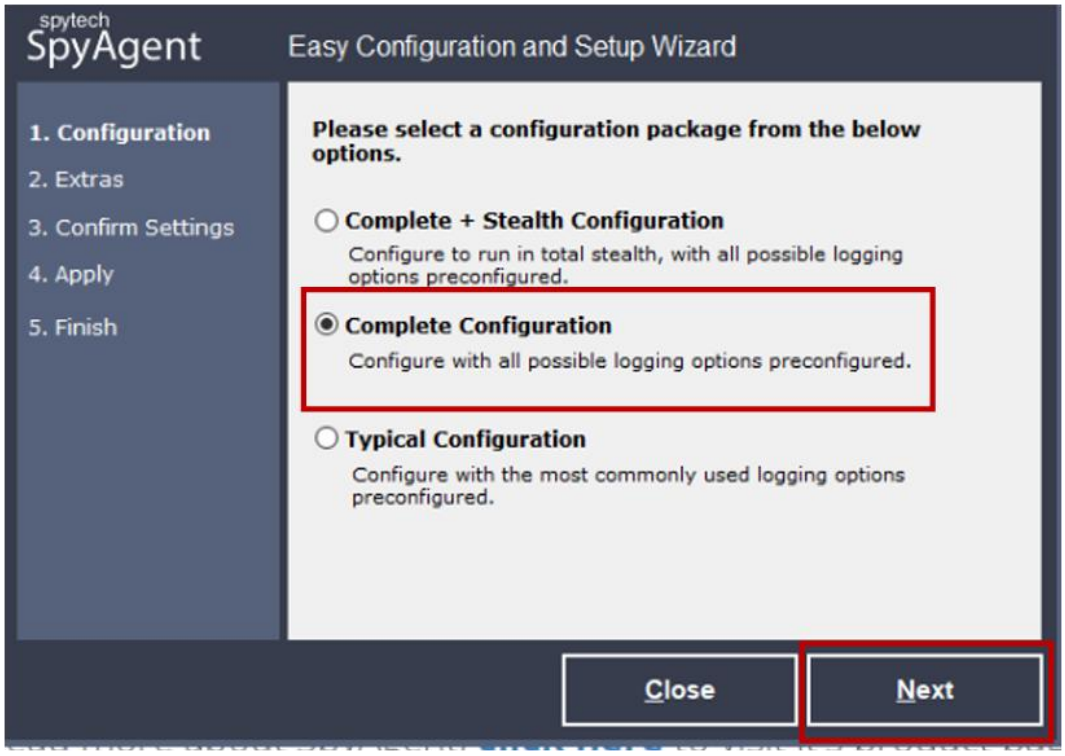
- New Password: Crck@123
- Confirm Password: Crck@123

1. Click OK to save the password

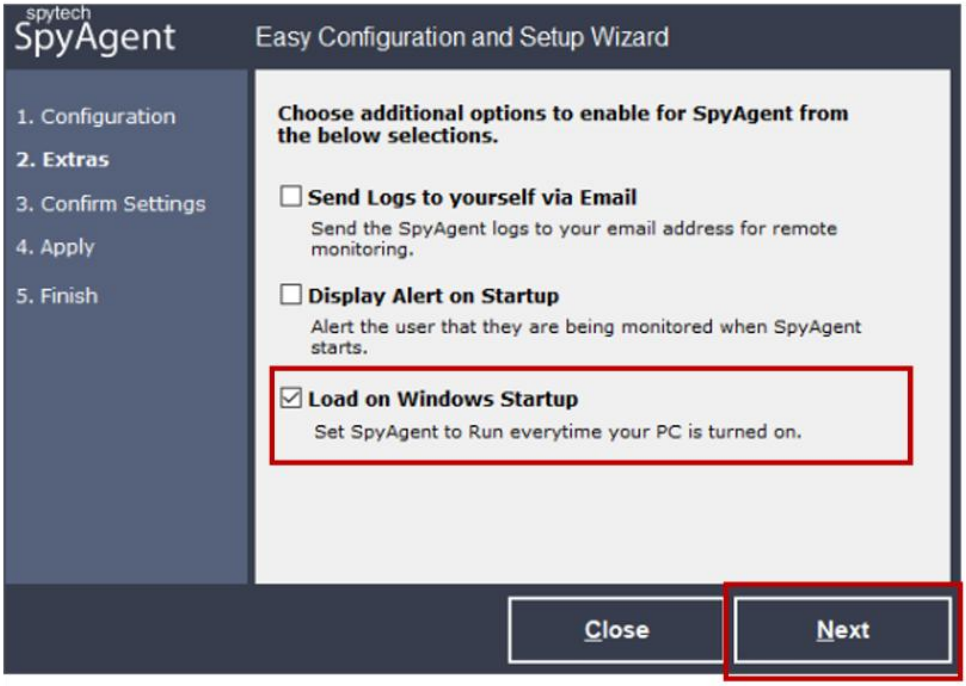


13.3 Easy Configuration Wizard

The Easy Configuration and Setup Wizard opens. Select "Complete Configuration" to enable all logging options. Click Next.

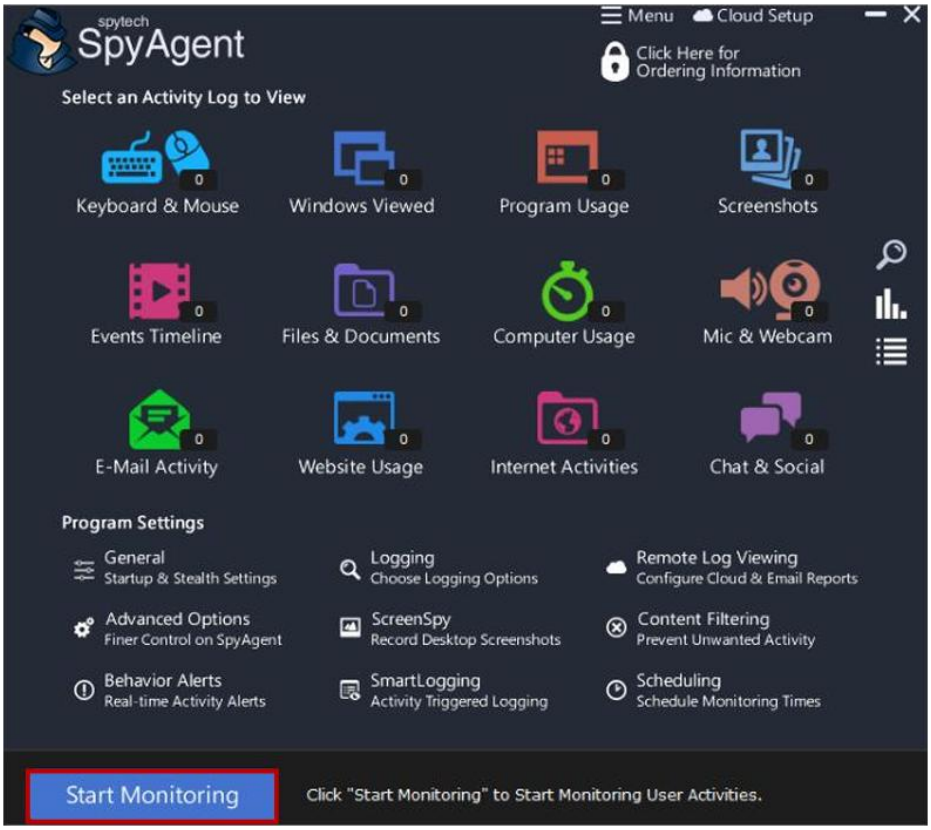


On the Extras page, check "Load on Windows Startup" to ensure monitoring continues after restarts. Click Next through Confirm Settings and Apply.



13.4 SpyAgent Dashboard – Start Monitoring

The SpyAgent main dashboard shows all activity log categories. All counters start at zero. Click "Start Monitoring" at the bottom to begin recording all user activity.



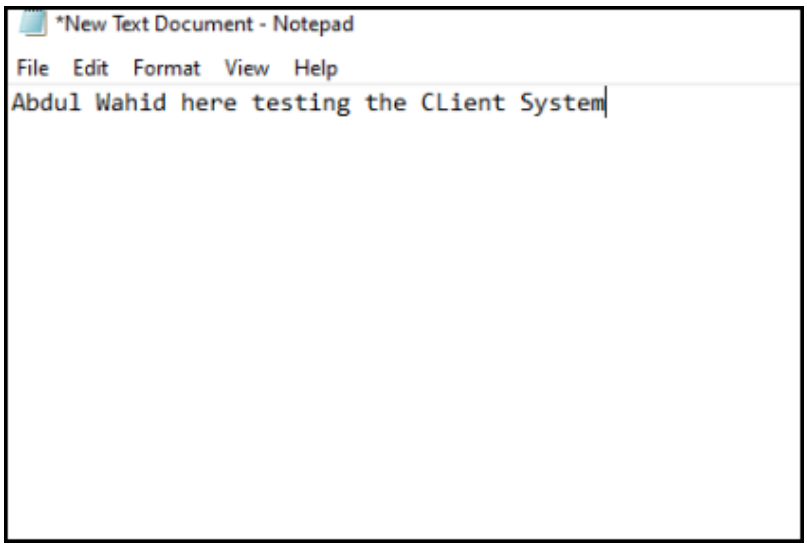

spytech
SpyAgent
Menu
Cloud Setup
Click Here for
Ordering Information
Select an Activity Log to View
Keyboard & Mouse
Windows Viewed
Program Usage
Screenshots
Events Timeline
Mic & Webcam
E-Mail Activity
Chat & Social
Enter Access Password
Crck@123
Cancel
OK
Program Settings
General
Startup & Stealth Settings
Logging
Choose Logging Options
Remote Log Viewing
Configure Cloud & Email Reports
Advanced Options
Finer Control on SpyAgent
ScreenSpy
Record Desktop Screenshots
Content Filtering
Prevent Unwanted Activity
Behavior Alerts
Real-time Activity Alerts
SmartLogging
Activity Triggered Logging
Scheduling
Schedule Monitoring Times
Start Monitoring
Click "Start Monitoring" to Start Monitoring User Activities.

13.5 Testing Phase – Generating Activity

Perform test activities while SpyAgent is actively monitoring to verify it captures everything correctly.

13.5.1 Testing Phase 1 – Notepad Keystrokes

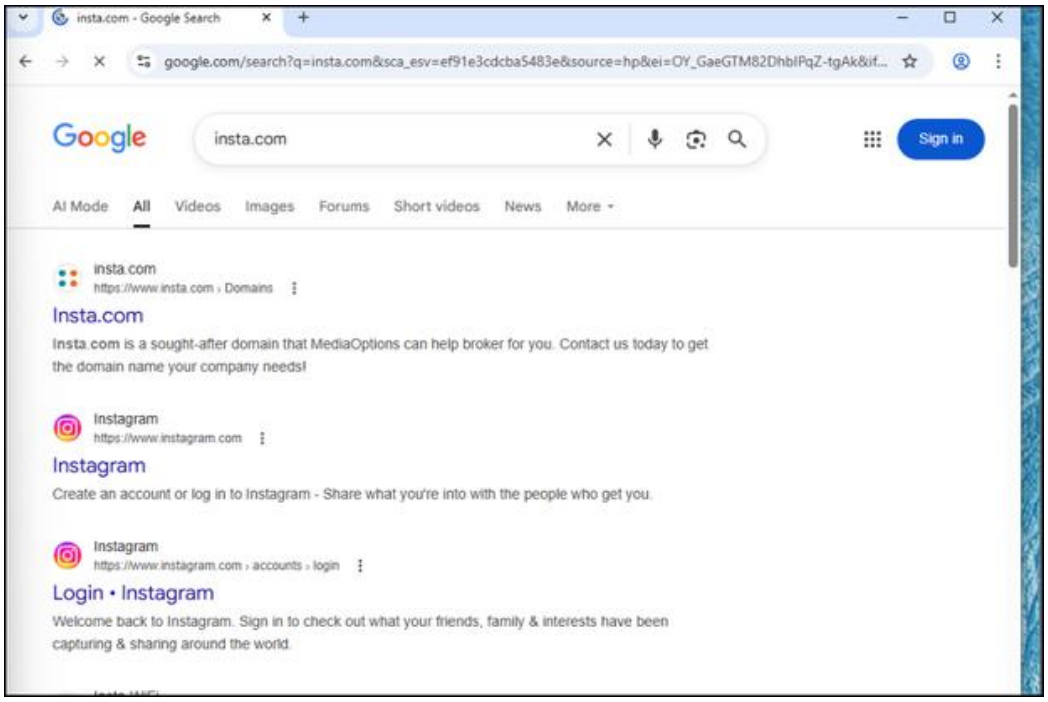
Open Notepad and type some test text. SpyAgent will capture every keystroke made in Notepad.



13.5.2 Testing Phase 2 – Website Browsing

Open Google Chrome and navigate to google.com. Search for "insta.com" — this will be captured as a website visit and online search keyword.





13.5.3 Testing Phase 3 – Calculator Usage

Open the Windows Calculator and perform some calculations. SpyAgent logs all programs opened and their usage duration.



13.6 Stopping Monitoring and Accessing SpyAgent

To review logs, Stop Monitoring



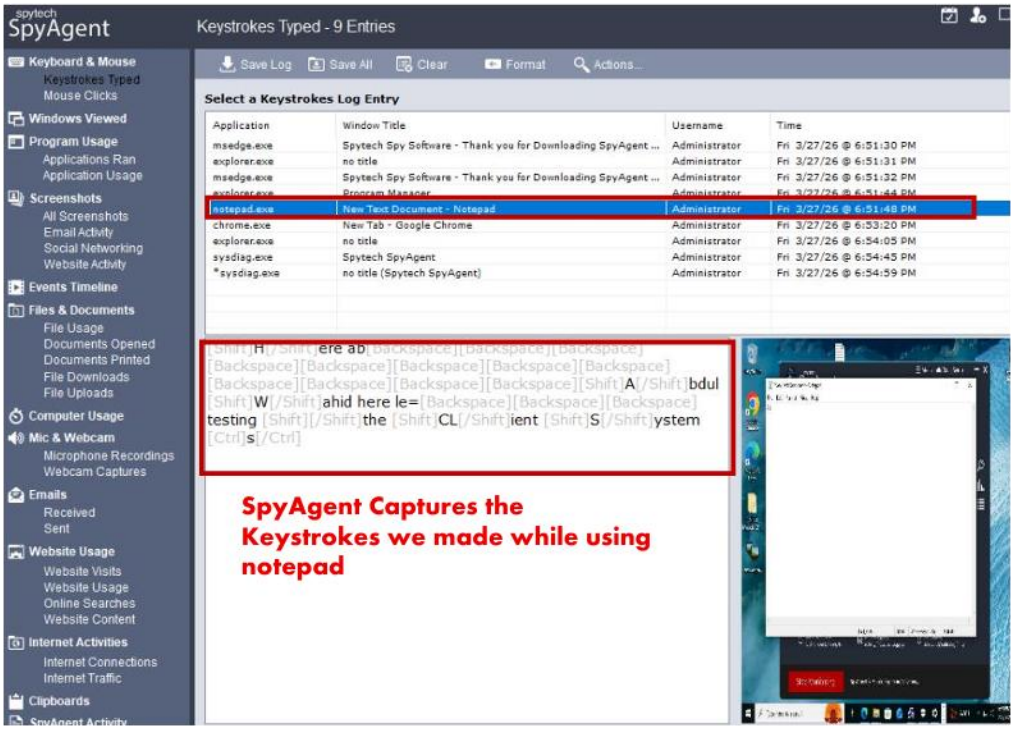
After stopping monitoring, the dashboard shows all recorded activity counts: Keyboard & Mouse (150), Windows Viewed (20), Program Usage (71), Screenshots (7), Events Timeline (400), Files & Documents (990), Website Usage (5), Internet Activities (42).



13.7 Reviewing Keystroke Logs

Click "Keyboard & Mouse" > "View Keystrokes Log". Select the notepad.exe entry to see the exact text typed during the test session. SpyAgent captures every keystroke including backspaces and modifier keys.

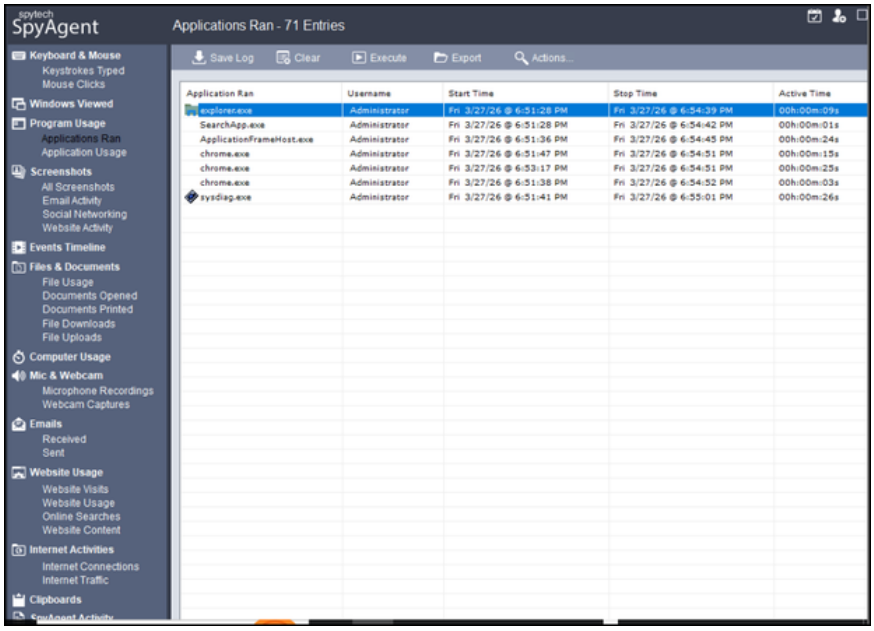




13.8 Reviewing Program Usage

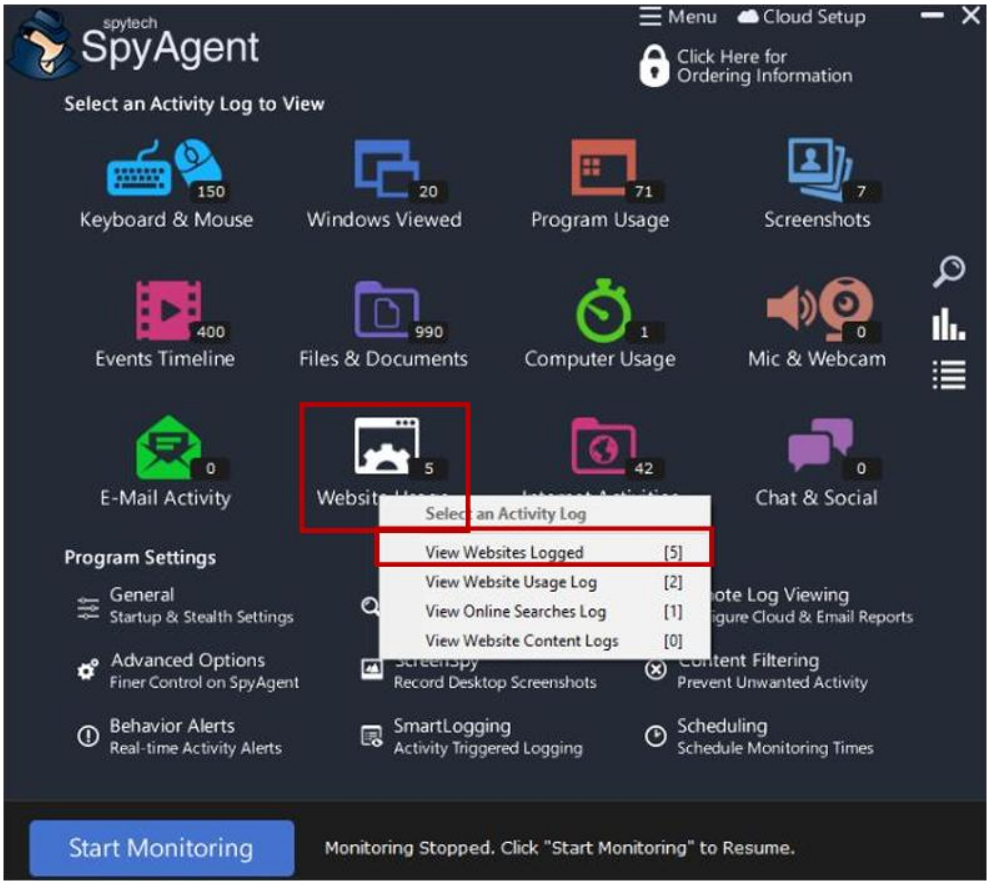
Click "Program Usage" > "View Applications Log". The log shows all executables that ran — explorer.exe, chrome.exe, notepad.exe — with username, start time, stop time, and active duration.



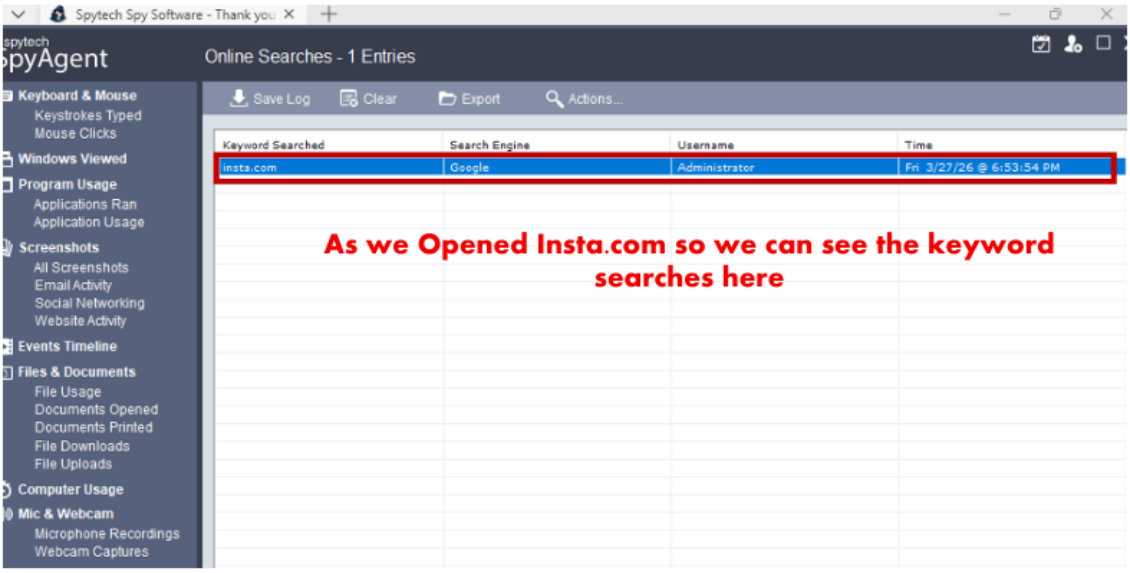


13.9 Reviewing Website and Search Activity

Click "Website Usage" > "View Websites Logged" to see all websites visited. Click "View Online Searches Log" to see search keywords entered.



The Online Searches log shows "insta.com" was searched via Google by Administrator on CLI01.crack.local — exactly matching the test activity performed during Phase 2.



14. Lab Summary & Credentials Reference

14.1 Network Architecture

Virtual Networks:

- VMnet3 (Host-Only) — Internal domain: 192.168.100.0/24 — No internet, DHCP disabled
- VMnet2 (NAT) — Internet access: 192.168.200.0/24 — DHCP enabled

Virtual Machines:

- DC01 (Windows Server 2019) — IP: 192.168.100.100 — Domain Controller, DNS, AD DS, Lansweeper, PAM360
- ADM01 (Windows 10 x64) — IP: 192.168.100.110 — Admin workstation, SpyAgent
- CLI01 (Windows 10 clone) — IP: 192.168.100.120 — Client workstation

14.2 Active Directory Summary

Domain:

- Domain Name: crack.local
- NetBIOS: CRACK
- Forest / Domain Level: Windows Server 2016
- Domain Controller: DC01.crack.local

Organizational Units:

- IT / Finance / HR / Servers / Workstations

User Accounts:

- CRACK\hr.user — Password: Hruser123
- CRACK\it.admin — Password: Itadmin123 — Domain Admin
- CRACK\fin.user — Password: Fuser123

14.3 Group Policy Settings

Password Policy (Default Domain Policy):

- History: 10 | Max age: 90 days | Min length: 7 chars | Complexity: Enabled

Account Lockout Policy:

- Threshold: 5 attempts | Duration: 5 min | Reset: 5 min

Audit Policy:

- Account logon events: Success + Failure | Logon events: Success + Failure

14.4 Application Credentials

⚠ Note: These credentials are for this lab environment only. Use strong unique passwords in production.

Domain Administrator:

- Username: CRACK\Administrator | Password: Admin123

DSRM Password: Crack123

Lansweeper:

- URL: http://localhost:83 | Username: crack | Password: Admin123--

PAM360:

- URL: https://localhost:7272 | Username: admin | Password: CRck@123

SpyAgent:

- Access Password: Crck@123

14.5 Tools Summary

Lansweeper v12.6.3.5

- Network asset discovery — auto-scans 192.168.100.0/24
- Windows scanning credential: CRACK\Administrator
- Discovered: CLI01, DC01, IP devices, VMware adapters

ManageEngine PAM360 (Password Manager Pro)

- Privileged Access Management — manages domain admin credentials
- Resource DC01 added with crack\it.admin account
- Records RDP sessions for audit compliance

SpyAgent

- Employee monitoring — keystrokes, websites, programs, screenshots
- Deployed on workstations for activity auditing
- Verified: captures Notepad keystrokes, Google searches, application usage

— End of Document —